

Cylindrical digester:

$$\text{Inner diameter} = \sqrt[3]{\frac{4.6V_d}{\pi r}}$$

Where: V_d = Effective digester volume, m³

r = height/diameter ratio

Digester height = rD_d

Square digester:

$$\text{Inner side of the square digester, } S_d = \sqrt[3]{\frac{1.15V_d}{r}}$$

Digester height = rS_d

Total height = $1.2\sqrt[3]{V_d r^2}$

Rectangular digester:

$$\text{Inner width of rectangular digester, } W_d = \sqrt[3]{\frac{1.15V_d}{rp^2}}$$

Digester height = rV_d

p = desired width and length proportion

Total height = $1.2\sqrt[3]{pV_d r^2}$

Cylindrical digester:

$$\text{Total height} = 1.2\sqrt[3]{\frac{4r^2V_d}{\pi}}$$

$$\text{Number of fixtures} = \frac{\text{Illumination x area}}{(\text{lamp per fixture})(\text{lumens})(\text{cu})(\text{mf})}$$

$$\text{initial footcandle} = \frac{(\text{footcandle})(\text{cu})}{\text{Area}}$$

$$\text{Maintenance illumination} = \frac{(\text{lamp lm})(\text{cu x mf})}{\text{Area}}$$

$$\text{Grashof Number (Gr)} = \frac{N_{Gr} = \frac{(D^3 \rho^2 g \beta \Delta T)}{\mu^2}}$$

$$\text{Nusselt Number} = Nu = hD/k; \text{ Reynolds Number} = Re = \rho v D / \mu$$

$$\text{Prandtl Number} = Pr = \mu c_p / k$$

PART 3 SOLVED PROBLEMS

1. If a wind mill rotor has a tip-speed ratio of 1:1, what windspeed is required for its rotor tip to move at 3.3 m/s?

Solution:

Wind Speed = tip speed (since tip speed ratio is 1: 1)

$$\text{Wind speed} = \frac{3.3\text{m}}{\text{s}}$$

2. What is the electrical energy consumption of a 30-kw vermicast production shredder plugged in a 240-volt source if it requires 1 hour to shred 2 tonnes of farm waste?

Solution:

$$E_c = \frac{P_c T_o}{W_i} = \frac{(30\text{kW})(1\text{hr})}{2,000\text{kg}} = \frac{0.015\text{kWh}}{\text{kg}}$$

3. A 2.8-tonne capacity Jatropha oil expeller was loaded full with dried Jatropha seeds in 3 minutes. It produced 900 kg of crude oil in 2 hours. What is its crude oil production rate?

Solution:

$$\text{Oil Production} = \frac{\text{Weight of crude oil collected}}{\text{Total time}} = \frac{900\text{kg}}{2\text{hrs}} = \frac{450\text{kg}}{\text{hr}}$$

4. A Jatropha oil expeller was loaded with 2.2 tons dried Jatropha seeds. It produced 600 kg of crude oil in 4 hours. What is its Jatropha cake production rate?

Solution:

$$\text{cake Production} = \frac{\text{Weight of resulting Jatropha cake produced}}{\text{Total time}} = \frac{2200\text{kg} - 600\text{kg}}{4\text{hrs}} = \frac{400\text{kg}}{\text{hr}}$$

5. A 21-tonne capacity coconut oil expeller was loaded full with grated copra. It produced 7,000kg of crude oil in 20 hours. What is the copra cake production rate of the machine at 95% cake collection efficiency?

Solution:

$$\begin{aligned} \text{Production} &= \frac{(\text{Wt. of copra input} - \text{Wt. of crude oil collected})(\text{Eff})}{\text{Total time}} \\ &= \frac{(21,000 \text{ kg} - 7,000 \text{ kg})(0.95)}{20\text{hrs}} = \frac{665\text{kg}}{\text{hr}} \end{aligned}$$

6. How many tons of agricultural waste is needed annually to optimally operate a 25-ampere 5-kw vermicast production shredder having a capacity of 0.2 ton of waste per hour? It optimally operates 16 hours daily during the 3-month peak sunny months and 8 h/day at 26days/month for the rest of the year.

Solution:

Wt = (Input Rate)(Peak Operating Time + Lean Operating Time)

$$Wt = \left(\frac{0.2\text{tonne}}{\text{h}} \right) \left[\left(\left(\frac{16\text{hr}}{\text{day}} \right) \left(\frac{30\text{days}}{\text{month}} \right) \left(\frac{3\text{month}}{\text{yr}} \right) \right) + \left(\left(\frac{8\text{hrs}}{\text{day}} \right) \left(\frac{26\text{day}}{\text{month}} \right) \left(\frac{9\text{months}}{\text{yr}} \right) \right) \right]$$

$$Wt = \frac{662\text{tonnes}}{\text{yr}}$$

7. In open channel construction, how many 40-kg bags of cement are needed for a 20-km lateral channel if it requires 0.6 cubic meter concrete per linear meter? Use 5% safety factor.

Solution:

$$\text{Bags} = (\text{Safety factor}) \left(\frac{\text{Bags cement}}{\text{cu. m. concrete}} \right) \left(\frac{\text{Concrete volume}}{\text{m}} \right) (\text{m length})$$

$$\text{Bags} = (1.05) \left(\frac{9\text{bags}}{\text{m}^3} \right) \left(\frac{0.6\text{m}^3}{\text{m}} \right) (20,000\text{m}) = 113,400 \text{ bags}$$

8. How many truck-loads of sand are needed for 50-km main irrigation canal requiring 3.5 cubic meters concrete per linear meter? 1 truck load is 2.5 cubic meters. Use 5% safety factor.

Solution:

$$\text{Truck load} = (\text{Safety factor}) \left(\frac{\text{m}^3 \text{ sand}}{\text{m}^3 \text{ concrete}} \right) \left(\frac{\text{trucks}}{\text{m}^3} \right) (\text{Concrete Volume})$$

$$\text{Truck loads} = (1.05) \left(\frac{0.5\text{m}^3}{\text{m}^3} \right) \left(\frac{1 \text{ truck load}}{2.5\text{m}^3} \right) \left(\frac{3.5\text{m}^3}{\text{m}} \right) (50,000\text{m}) = 36,750 \text{ truck loads}$$

9. Estimate the truck loads of gravel needed for an 8-km main irrigation canal requiring 4 cubic meters concrete per linear meter. 1 truck load is 2.5 cubic meters. Use 10% safety factor.

Solution:

$$\text{truck loads} = (\text{safety factor}) \left(\frac{\text{m}^3 \text{ gravel}}{\text{m}^3 \text{ concrete}} \right) \left(\frac{\text{trucks}}{\text{m}^3} \right) (\text{Concrete volume})$$

$$\text{Truck loads} = (1.1) \left(\frac{1\text{m}^3}{\text{m}^3} \right) \left(\frac{1\text{truck}}{2.5\text{m}^3} \right) \left(\frac{4\text{m}^3}{\text{m}} \right) (8000\text{m}) = 14,080 \text{ truck loads}$$

10. How much should be budgeted for cement for 20-km irrigation canal if it requires half cubic meter concrete per linear meter and cement costs Php210/bag? Use 5% safety factor.

Solution:

$$\text{Budget} = (\text{Safety factor}) \left(\frac{\text{Bags cement}}{\text{m}^3 \text{ concrete}} \right) (\text{concrete volume})(\text{Unit cost})$$

$$\text{Budget} = (1.05) \left(\frac{9\text{bags}}{\text{m}^3} \right) \left(\frac{0.5\text{m}^3}{\text{m}} \right) (20,000\text{m}) \left(\frac{210}{\text{bag}} \right) = \text{P}19,845,000$$

11. Give cement budget estimate for 10-km canal requiring 2/3 cubic meter concrete per linear meter. Cement costs Php200/bag while transport costs Php 10/bag. Use 5% safety factor.

Solution:

$$\text{Budget} = (\text{Safety factor}) \left(\frac{\text{Bags cement}}{\text{m}^3 \text{ concrete}} \right) (\text{concrete volume})(\text{Unit cost})$$

$$\text{Budget} = (1.05) \left(\frac{9\text{bags}}{\text{m}^3} \right) \left(\frac{0.667\text{m}^3}{\text{m}} \right) (10,000\text{m}) \left(\frac{210}{\text{bag}} \right) = \text{P}13,236,615$$

12. A gravity dam weighing 75,000 tons is to be constructed with length along dam axis of 1km and base of 100 meters. What should be the minimum soil bearing stress? Use factor of safety against sinking of 2.

Solution:

$$\text{Sinking Factor of safety} = \frac{\text{Min. defensive bearing stress}}{\text{Offensive bearing stress}}$$

$$\text{Min. def. stress} = (\text{Sinking FS})(\text{Offensive Stress})$$

$$\text{Min. def. stress} = (2) \left(\frac{75,000\text{tons}}{(1,000\text{m})(100\text{m})} \right) = \frac{1.5\text{ton}}{\text{m}^2}$$

$$\text{Note: Offensive stress} = \frac{\text{Weight}}{\text{Area}}$$

13. A 2-meter thick trapezoidal section of a concrete dam has resisting moment about the dam toe of 350,000 tonne-meters. The water exerts 4,050 tonnes horizontally at 30 meters above the dam toe. What is the factor of safety against overturning?

Solution:

$$\text{Overturing Factor of Safety} = \frac{\text{Resisting moment}}{\text{Overturning moment}} = \frac{350,000\text{ton} - \text{m}}{(4,050\text{ton})(30\text{m})} = 2.88$$

14. A concrete dam with $60,000\text{m}^2$ submerged vertical upstream wall is to be constructed against a 546 tons/m^2 average water pressure. Coefficient of friction at dam base is 0.8. What should be the minimum dam weight? Use factor of safety against sliding of 2.

Solution:

$$\text{Sliding Factor of Safety} = \frac{\text{Min. defensive horizontal force}}{\text{Offensive horizontal force}}$$

$$\text{Min. defensive force} = (\text{Coef. of friction})(\text{Weight}) = (\text{FS})(\text{Offensive force})$$

$$\text{Weight} = \frac{(\text{FS})(\text{Offensive force})}{\text{Coef. of friction}} = \frac{(2)\left(\frac{546\text{ton}}{\text{m}^2}\right)(60,000\text{m}^2)}{0.8} = 81,900,000 \text{ tonnes}$$

$$\text{Note: Min. defensive force} = (\text{Coef. of friction})(\text{Weight})$$

15. Determine the daily biogas consumption in cubic meters of an industrial boiler using 5 units of 50-cm gas burner 8 hours a day. The consumption of each gas burner is 1.2m^3 hour.

Solution:

$$C = \text{NBT}$$

where C = Consumption, N = No. of units, B = biogas consumption of device, T = Time

$$C = (5 \text{ units})\left(\frac{1.2\text{m}^3}{\text{hr}}\right)(8 \text{ hrs}) = 48\text{m}^3$$

16. Estimate the biogas production in cubic meters per day from a large balloon type digester using 1:1 water-manure ratio. The source farm has 2,000 heads of porkers of mixed ages. Use a retention period of 30 days. The mean daily manure production of porkers is 2.2 kg/head while the specific gas production for 30-day manure retention period is $0.063\text{ m}^3/\text{kg}$.

Solution:

$$P = \text{NMG}$$

where P = biogas production potential, N = No. of heads, M = manure production,

G = Specific gas production of the manure for a specific retention period

$$P = (2000 \text{ heads})\left(\frac{2.2\text{kg}}{\text{head}}\right)\left(\frac{0.063\text{m}^3}{\text{kg}}\right) = 277.2\text{m}^3$$

17. How many heads of breeding cattle are needed as source of manure to generate 85 m^3 of biogas a day if the retention period is 25 days and water-manure ratio is 1:1? The mean daily manure production of breeding cattle is 13 kg/head while its specific gas production for 25day manure retention period is half of that of chicken dung.

Solution:

$$N = \frac{P}{MG}$$

where N = No. of heads, P = biogas production, M = manure production,

G = Specific gas production of the manure for a specific retention period

$$G = (0.5)(G \text{ for chicken dung at 25day retention period})$$

$$G = (0.5)\left(\frac{0.06\text{m}^3}{\text{kg}}\right) = \frac{0.03\text{m}^3}{\text{kg}}$$

$$N = \frac{P}{MG} = \frac{85\text{m}^3}{\left(\frac{13\text{kg}}{\text{head}}\right)\left(\frac{0.03\text{m}^3}{\text{kg}}\right)} = 217.9 \text{ heads or } 218 \text{ heads}$$

18. When approved as Clean Development Mechanism (CDM) project, how much will the project owner of a new covered lagoon bio-digester receive annually from CDM fund for the first 10 years

if the digester produces 7,000 tons of methane plus 3,000 tons of carbon dioxide annually? Assume current carbon credit costs US\$12 per ton of carbon equivalent.

Solution:

Considering that methane is 21 times more destructive than carbon,

$$\text{Amount} = (\text{Carbon cost per tonne}) \left[\left(\frac{\text{tons carbon}}{\text{year}} \right) + 21 \left(\frac{\text{tons of methane}}{\text{year}} \right) \right]$$

$$\text{Amount} = (\$12)(3,000 + 21(7,000)) = \frac{\$800,000}{\text{year}}$$

19. How many grams of ethanol (C₂H₅OH) can be theoretically derived from 100 g of glucose (C₂H₁₂O₆) if the reaction yields 2 moles of ethanol and 2 moles of carbon dioxide per mole of glucose feedstock.

Solution:



$$181 \text{ g} \qquad 92.1 \text{ g} \qquad 88 \text{ g}$$

Therefore by ratio 100/181 = x/92.1;

$$x = 51.4 \text{ g}$$

20. Determine the volume of manure produced in 30 days by a facility that houses 200 50-lb animals if the daily excreted manure is 3 kg/animal/day.

Solution:

By looking at the choices, only c and d have volumetric units, so you can concentrate on the two. However, 9000 m³ is quite large for this facility. Leaving c as a practical answer.

Alternately, you can solve the problem by:

$$30 \text{ days} (200 \text{ animals}) (3 \text{ kg/animal/day}) = 18000 \text{ kg}$$

Since the density of the waste is not given, you can make assumptions. Usually, the density is similar to soil. Say, 2000 kg/m³, thus:

$$18000 \text{ kg} / 2000 \text{ kg/m}^3 = 9 \text{ m}^3 = 9000 \text{ liters}$$

21. A 12 amperes electric fan and blower with a power factor of 0.85 was connected to a 240 volts convenient outlet. Calculate the power in the circuit.

Solution:

$$\text{Power (watts)} = \text{volts} \times \text{amperes} \times \text{power factor}$$

$$P = (240\text{V})(12\text{amp})(0.85)$$

$$P = 2,448 \text{ watts or } 2.448\text{kW}$$

22. An electric motor has a nameplate rating of 2 horsepower, 240 volts, and 15 amperes. Calculate the motor power factor assuming a motor efficiency of 85%. Also compute for the motor power input

Solution:

$$\text{Power factor } (\cos\theta) = \frac{P}{VI} = \frac{(2\text{hp})(746\text{W})/0.85}{(240\text{V})(15\text{amp})} = 0.4875$$

$$\text{Power input} = (240\text{V})(15\text{amp}) = 3600\text{Volt} - \text{ampere}$$

23. Determine the current flow in a circuit having a resistance of 5 ohms on a 120 volt and 240 volt current supply.

Solution:

$$I = \frac{V}{R} = \frac{120\text{V}}{5 \text{ ohms}} = 24\text{amp}$$

$$I = \frac{V}{R} = \frac{240V}{5\text{ohms}} = 48\text{amp}$$

24. A circuit has a resistance of 20 ohms and the current flows at 12 amperes. Determine the voltage.

Solution:

$$V = IR = (12\text{amp})(20\text{ohms}) = 240\text{Volts}$$

25. Two automobile headlights are connected in series to a 12 volt battery each having a resistance of 1 ohm. What is the current flowing in the circuit?

Solution:

$$R_T = R_1 + R_2 = 2\text{ohms}$$

$$I = \frac{V}{R} = \frac{12V}{2\text{ohms}} = 6\text{amp}$$

26. A mercury lamp having a hot resistance of 50 ohms is connected to a socket with 240-volt current supply. How much current flows through the lamp? Calculate the power drawn.

Solution:

$$I = \frac{V}{R} = \frac{240\text{Volt}}{50\text{ ohms}} = 4.8\text{amp}$$

$$P = VI(PF) = (240V)(4.8\text{amp})(1) = 1,152\text{Watts}$$

or

$$P = I^2R = (4.8\text{amp})^2(50\text{amp}) = 1,152\text{Watts}$$

27. An office room with general dimensions of 8 x 20 meters is to be lighted at an average maintained foot candle of 50. How many 3-lamp fixtures of 120 centimeters long F40 T12 WW rapid start fluorescent lamps are required assuming 0.38 cu and 0.75 mf?

Solution:

$$\text{Lamp lumens} = \frac{(\text{maintained footcandle})(\text{area})}{(\text{cu})(\text{mf})} = \frac{(50\text{fc})(8\text{m} \times 20\text{m})}{(0.38)(0.75)} = 28,070 \text{ lumens}$$

Each 40 watt fluorescent lamp has an output of 3,200 lumen, the number of lamps will be:

$$\text{Number of lumen} = \frac{28,070}{3200} = 8.77 \text{ lamps}$$

Since there are 3 lamps for each fixture, divide:

$$\frac{8.77}{3} = 2.93 \text{ or } 3 \text{ lamps in each fixture}$$

28. A school classroom with general dimensions of 24 x 30 feet is lighted with 10 fluorescent lamp 4F 40 T12 WW rapid start lamp. Calculate the initial and maintained illumination in foot candle (English) and Lux (Metric), Assuming that the coefficient of utilization (cu), is 0.35 and the maintenance factor (mf) is 0.70.

Solution:

$$\text{Lamp lumens} = (10\text{fixtures}) \left(4 \frac{\text{lamps}}{\text{fixture}} \right) \left(\frac{3,200\text{lumens}}{\text{lamp}} \right) = 128,000\text{lumens}$$

$$\text{Initial footcandle} = \frac{(\text{Lumens})(\text{cu})(\text{mf})}{\text{Area}} = \frac{(128,000)(0.35)(0.70)}{(24\text{ft})(30\text{ft})} = 43.55 \text{ footcandle}$$

$$\text{Lux} = \frac{(\text{Lumens})(\text{cu})(\text{mf})}{\text{area}} = \frac{(128,000)(0.35)(0.70)}{(7.32\text{m})(9.14\text{m})} = 468.73\text{lux}$$

TABLE 5-3 FLUORESCENT LAMP DATA

Lamp Code ^a	Watts	Lamp Current mil-amp.	Ballast Watts ^{bc}	Total Watts	Lamp Life hr ^d	Initial Output Lumens ^e	Actual Efficacy m/w ^f	Actual Efficacy lm/w ^g
Preheat Lamp								
F15 T8 CW	15	425	8	23	7500	870	38	58
F20 T12 CW	20	425	10	30	9000	1300	43	65
Rapid Start - Preheat Lamps^h								
F40 T12 CW	40	425	6	46	18000	3150	68	79
F40 T12 WW	40	425	6	46	18000	3200	70	80
F40 T12 CWX	40	425	6	46	18000	2200	48	55
F40 T12 D	40	425	6	46	18000	2600	57	65
Rapid Start - High Output								
F48 T12 CW/HO	60	800	15	75	12000	4300	57	72
F60 T12 CW/HO	75	800	15	90	12000	5400	60	72
F72 T12 CW/HO	85	800	15	100	12000	6650	67	78
F96 T12 CW/HO	105	800	15	121	12000	9200	76	88
Rapid Start - Very High Output								
F48 T12 CWVHO	110	1500	8	118	9000	6250	53	57
F72 T12 CWVHO	165	1500	8	173	9000	9900	57	60
F96 T12 CWVHO	215	1500	13	228	9000	14500	64	67
Instant Start (Slimline) Lamps								
F42 T6 CW	25	200	15	40	7400	1750	44	70
F64 T6 CW	40	200	10	50	7500	2800	56	70
F48 T12 CW	40	430	16	56	9000	3000	54	75
F64 T12 CW	55	430	16	71	12000	3600	51	65
F96 T12 CW	75	430	17	92	12000	6300	68	84

a Standard ordering abbreviation
CW, cool white; WW, warm white;
CWX, cool white de luxe; D, daylight
b Figures are for a two-lamp circuit
per start
d Life figures are for 3 hours burning
per start

e After 100 hours burning
f Includes ballast loss
g Excludes ballast loss

h Data given for lamps in a
rapid start circuit

29. How much liter of kerosene is needed to dry 1 ton of palay from 30% to 14%? Assume latent heat of vaporization is 2500 kJ/kg; heating value of kerosene is 43 MJ/kg; specific gravity is 0.8 and efficiency of 80%.

Solution:

$$W_f = \frac{W_i(1 - MC_i)}{1 - MC_f} = \frac{(1000\text{kg})(1 - 0.30)}{(1 - 0.14)} = 813.95\text{kg}$$

$$\text{Weight of Water removed} = 1000 - 813.95 = 186.0456\text{kg}$$

$$X = \frac{(186.0456\text{kg})\left(\frac{2500\text{kJ}}{\text{kg}}\right)}{\left(\frac{43,000\text{kJ}}{\text{kg}}\right)(0.8)(0.8)} = 17\text{kg}$$

30. In a flour mill, wheat is to be adjusted to a moisture content of 15% on a dry basis. If the whole grain received at the mill is found to contain 11.4% of water initially, how much water must the miller add per 100 kg of input grain as received, to produce the desired moisture content?

Solution:

Initially the grain received has the following:

Initial weight = 100kg

Initial moisture content = 11.4%

In the mill, we want to add water in order for it to have moisture content of 15% dry basis.

$$MC_{wb} = \frac{MC_{db}}{1 + MC_{db}} = \frac{0.15}{1 + 0.15} = 0.13 \text{ or } 13\% \text{ wb}$$

Total mass balance:

Whole grain + water added = flour mill

$$100\text{kg} + W = F$$

Total moisture balance:

$$100(0.114) + W = F(0.13)$$

$$F = F$$

$$\frac{100(0.114) + W}{0.13} = 100 + W$$

$$W = 1.8\text{kg}$$

31. A 3 m diameter stainless steel tank contains wine. In the tank, the wine is filled to 5 cm depth. A discharge port, 10 cm diameter, is opened to drain the wine. Calculate the discharge velocity of wine, assuming the flow is steady and frictionless and the time required in emptying it

Solution:

$$\cancel{\frac{P_1}{\rho g}} + \cancel{\frac{v_1^2}{2g}} + z_1 = \cancel{\frac{P_2}{\rho g}} + \frac{v_2^2}{2g} + z_2$$

$$v_2 = \sqrt{2g(z_2 - z_1)} = \sqrt{2(9.81)(5\text{m} - 0)} = \frac{9.9\text{m}}{\text{s}}$$

$$Q = AV = \left(\frac{\pi(0.1\text{m})^2}{4} \right) \left(\frac{9.9\text{m}}{\text{s}} \right) = \frac{0.078\text{m}^3}{\text{s}}$$

Time to empty the tank:

$$t = \frac{\frac{\pi(3\text{m})^2(5\text{m})}{4}}{\frac{0.078\text{m}^3}{\text{s}}} = 453 \text{ seconds or } 7.55 \text{ mins}$$

32. Water flows at the rate of 0.4 m³ / min in a 7.5 cm diameter pipe at a pressure of 70kPa. If the pipe reduces to 5 cm diameter calculate the new pressure in the pipe

Solution:

$$v_1 = \frac{\frac{0.4\text{m}^3}{\text{min}}}{\left(\frac{\pi D^2}{4} \right) (60\text{s})} = \frac{1.51\text{m}}{\text{s}}$$

$$v_2 =$$

$$P_2 = P_1 + \frac{\rho_w}{2} (v_1^2 - v_2^2) = 70\text{kPa} + \frac{\left(\frac{1000\text{kg}}{\text{m}^3} \right)}{2} (1.51)^2 - (3.40)^2 = 65.36\text{kPa}$$

33. A tubular water blancher is being used to process lima bean. The product mass flow rate is 860 kg/h. It is found that the theoretical energy consumed for the blanching process amounts to 1.19 GJ/h. The energy lost due to lack of insulation around the blancher is estimated to be 0.24 GJ/h. If the total energy input to the blancher is 2.71 GJ/h. Calculate the energy required to reheat water.

Solution:

$$\begin{aligned} E_{in} &= E_{out} \\ \frac{2.71\text{GJ}}{\text{h}} &= \frac{1.19\text{GJ}}{\text{h}} + \frac{0.24\text{GJ}}{\text{hr}} + E_w \\ E_w &= \frac{1.43\text{GJ}}{\text{hr}} \end{aligned}$$

34. Processing water is to be heated in a direct fired heater, which burns natural gas with a calorific value of 20.2 MJ m⁻³. If 5,000 kg h⁻¹ of this water has to be heated from 15°C to 80°C and the heater is estimated to be 45% efficient, estimate the hourly consumption of gas.

Solution:

Energy for heating water:

$$Q = \dot{m} C_p \Delta T = \left(\frac{5,000 \text{ kg}}{\text{hr}} \right) \left(\frac{4.187 \text{ kJ}}{\text{kg}^\circ\text{K}} \right) (80^\circ\text{C} - 15^\circ\text{C}) = \frac{1,360,775 \text{ kJ}}{\text{hr}}$$

Thermal Energy:

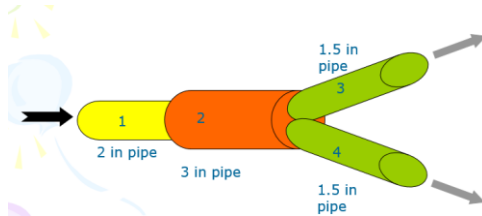
$$Q_h = \left(\frac{20.2 \text{ MJ}}{\text{m}^3} \right) (0.45) = \frac{9090 \text{ kJ}}{\text{m}^3}$$

Hourly consumption of gas:

$$X = \frac{Q}{Q_h} = \frac{\frac{1,360,775 \text{ kJ}}{\text{hr}}}{\frac{9090 \text{ kJ}}{\text{m}^3}} = \frac{149.70 \text{ m}^3}{\text{hr}}$$

35. A 20%wt sucrose (sugar) solution having a density of 1074 kg/m³ is flowing through the piping system as seen in following picture. The flow rate entering pipe 1 is 1.892 m³/h. The flow divides equally in pipe 3. Calculate (a) the velocity in m/s in pipe 2 and 3, (b) The mass velocity kg/m².s in pipe 2 and 3.

Solution:



$$Q = Q_1 = Q_2 = Q_3 + Q_4 = \frac{1.892 \text{ m}^3}{\text{hr}}$$

Discharges:

$$Q_{1\&2} = \left(\frac{1.892 \text{ m}^3}{\text{hr}} \right) \left(\frac{1 \text{ hr}}{3600 \text{ s}} \right) = \frac{5.26 \times 10^{-4} \text{ m}^3}{\text{s}}$$

$$Q_{3\&4} = \frac{\left(\frac{5.26 \times 10^{-4} \text{ m}^3}{\text{s}} \right)}{2} = \frac{2.63 \times 10^{-4} \text{ m}^3}{\text{s}}$$

Areas of each tubes:

$$A_1 = \frac{\pi D^2}{4} = \frac{\pi (2 \times 0.0254)^2}{4} = 2.03 \times 10^{-3} \text{ m}^2$$

$$A_2 = \frac{\pi D^2}{4} = \frac{\pi (3 \times 0.0254)^2}{4} = 4.56 \times 10^{-3} \text{ m}^2$$

$$A_3 = \frac{\pi D^2}{4} = \frac{\pi (1.5 \times 0.0254)^2}{4} = 1.14 \times 10^{-3} \text{ m}^2$$

Velocities:

$$v_2 = \frac{\left(\frac{5.26 \times 10^{-4} \text{ m}^3}{\text{s}} \right)}{4.56 \times 10^{-3} \text{ m}^2} = \frac{0.115 \text{ m}}{\text{s}}$$

$$v_3 = \frac{\left(\frac{2.63 \times 10^{-4} \text{ m}^3}{\text{s}} \right)}{1.14 \times 10^{-3} \text{ m}^2} = \frac{0.23 \text{ m}}{\text{s}}$$

Mass flow rates:

$$\dot{m}_2 = \rho_w Q_2 = \left(\frac{1,074 \text{ kg}}{\text{m}^3} \right) \left(\frac{5.26 \times 10^{-4} \text{ m}^3}{\text{s}} \right) = \frac{0.56 \text{ kg}}{\text{s}}$$

$$\dot{m}_2 = \rho_w Q_2 = \left(\frac{1,074 \text{ kg}}{\text{m}^3} \right) \left(\frac{2.63 \times 10^{-4} \text{ m}^3}{\text{s}} \right) = \frac{0.282 \text{ kg}}{\text{s}}$$

Mass velocities:

$$\dot{m}_2 = \frac{\left(\frac{0.56 \text{ kg}}{\text{s}} \right)}{4.56 \times 10^{-3} \text{ m}^2} = \frac{122.81 \text{ kg}}{\text{m}^2 \cdot \text{s}}$$

$$\dot{m}_3 = \frac{\left(\frac{0.282 \text{ kg}}{\text{s}} \right)}{1.14 \times 10^{-3} \text{ m}^2} = \frac{247.37 \text{ kg}}{\text{m}^2 \cdot \text{s}}$$

36. Calculate the pressure in psia and kN/m² in a spherical tank at the bottom of the tank filled with oil having a diameter of 0.8 ft. The top of the tank is vented to the atmosphere having a pressure of 14.72 psia. The density of oil is 0.922 g/cm³.

Given:

$$P_{\text{atm}} = 14.72 \text{ psia}$$

$$D = 8 \text{ ft}$$

$$\rho = \frac{0.922 \text{ g}}{\text{cm}^3}$$

Solution:

$$P = P_{\text{gauge}} + P_{\text{atm}}$$

$$P = \left(\frac{9.22 \times 10^2 \text{ kg}}{\text{m}^3} \right) \left(\frac{9.81 \text{ m}}{\text{s}^2} \right) (8 \text{ ft} \times 0.3048 \text{ m}) + \frac{(14.72 \text{ psia})(101325 \text{ Pa})}{14.696 \text{ psia}} = 123.54 \text{ kPa}$$

$$P = 123.54 \text{ kPa} \left(\frac{14.696 \text{ psia}}{101.325 \text{ kPa}} \right) = 17.92 \text{ psia}$$

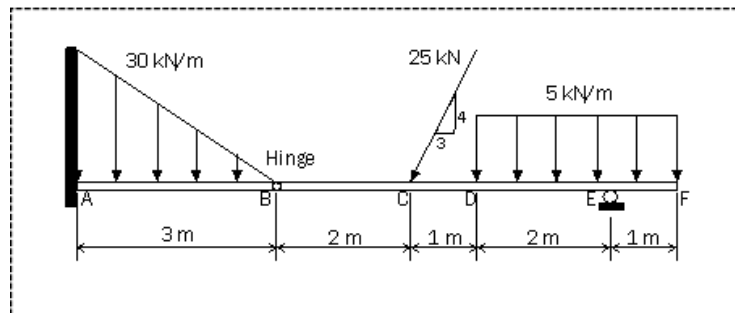
37. The pressure at the top of tank of jet fuel is 180.6 kN/m². The depth of liquid in the tank is 6.4 m. The density of the fuel is 825 kg/m³. Calculate the head of the liquid in m which corresponds to the absolute pressure at the bottom of the tank.

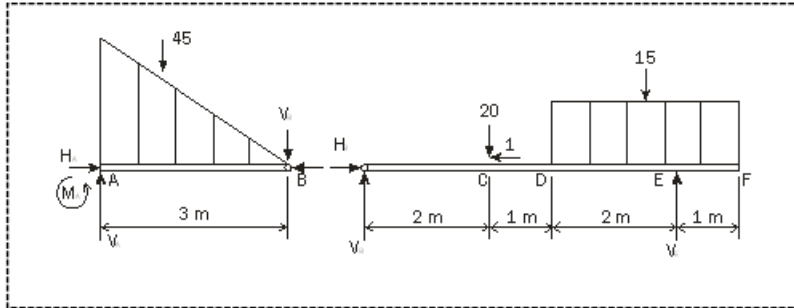
Solution:

$$H = h_1 + \frac{P}{\rho g} = 6.4 \text{ m} + \frac{\frac{180.6 \text{ kN}}{\text{m}^2}}{\left(\frac{825 \text{ kg}}{\text{m}^3} \right) \left(\frac{9.81 \text{ m}}{\text{s}^2} \right)} = 28.715 \text{ m}$$

38. Determine the maximum axial, shear and moment developed in the beam loaded as shown. The beam has an internal hinge at B.

Solution:





Considering free body diagram BF:

$$\sum F_h = 0: H_B - 15 = 0; H_B = 15\text{kN}$$

$$\sum M_B = 0: 20(2) + 15(4.5) - V_E(5) = 0; V_E = 21.5\text{kN}$$

$$\sum F_v = 0: V_B + 21.5 - 20 - 15 = 0; V_B = 13.5\text{kN}$$

39. What is the power dissipated when a $\frac{3}{4}$ A current flows through a $5\ \Omega$ resistor?

Solution:

$$P = I^2 R = \left(\frac{3}{4}\right)^2 (5\text{ohms}) = 2.81\text{W}$$

40. An inductor and a resistor are connected in series to a 120 V, 50Hz source. The inductive reactance and resistance was found to be 2.4 ohms and 5 ohms, respectively. The total impedance is

Solution:

$$Z = \sqrt{R^2 + (X_C - X_L)^2} = \sqrt{(5\text{ohms})^2 + (2.4)^2} = 5.54\text{ohms}$$

For number 41 – 44:

Three inductors are connected in parallel to a 230V 60 Hz supply. If $L_1 = L_2 = L_3 = 300\text{ H}$,

41. What is the total inductance?

Solution:

$$L_{\text{Total}} = \frac{1}{\frac{1}{300} + \frac{1}{300} + \frac{1}{300}} = 100\text{H}$$

42. What is the total inductive reactance (Ω)?

Solution:

$$X_L = 2\pi fL = 2\pi(60\text{Hz})(100\text{H}) = 37,699$$

43. What is the total reactive power?

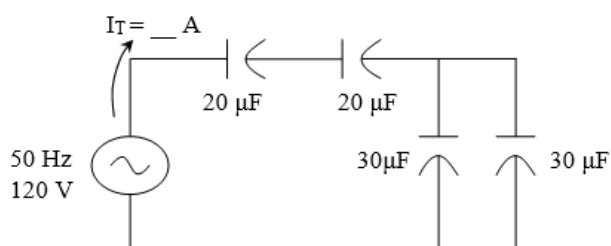
Solution:

$$P = \frac{E^2}{R} = \frac{(230\text{V})^2}{37,699\text{ ohms}} = 1.40\text{VA}$$

44. What is the total real power (W)?

Answer: The circuit is composed only by inductors, thus the real power is zero

For number 45 – 47:



45. What is the total capacitance?

Solution:

Since $30\mu\text{F}$ and $30\mu\text{F}$ are in parallel, the total is $60\mu\text{F}$ and the circuit becomes a series then:

$$C_{\text{Total}} = \frac{1}{\frac{1}{20\mu\text{F}} + \frac{1}{20\mu\text{F}} + \frac{1}{60\mu\text{F}}} = 8.60\mu\text{F}$$

46. What is the value of I_T ?

Solution:

$$Z = \sqrt{R^2 + (X_C - X_L)^2} = \sqrt{X_C^2} = X_C = \frac{1}{2\pi fC} = \frac{1}{2\pi(50)(8.6\mu\text{F})} = 370.12\text{ohms}$$

$$I_T = \frac{E_T}{Z} = \frac{120\text{V}}{370.12\text{ohms}} = 0.32\text{A}$$

47. What is the value of true power in watts?

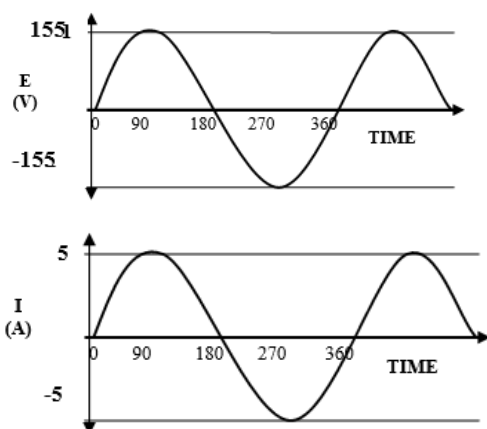
Answer: No true power, only reactive power is present since purely capacitive circuit

48. A 120 V generator delivers 30 kilowatts to an electric furnace. The current supplied by the generator is about

Solution:

$$I = \frac{P}{V} = \frac{30,000\text{Watts}}{120\text{V}} = 250\text{amps}$$

For numbers 49 – 51. Given a true sine voltage and current wave forms:



49. At 135° , the instantaneous current is

Solution:

$I(\theta) = A \sin \theta$ where A is the amplitude and θ is the angle in degrees

$$I(\theta) = A \sin \theta = 5 \sin(135^\circ) = 3.5\text{amps}$$

50. The phase shift angle is _____.

Solution:

There is no phase shift. The current and the voltage go through zero and through their peak values at the same time

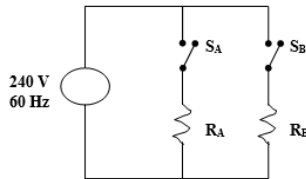
51. The effective or rms value of voltage (V_{rms}) equals _____ V

Solution:

$$E_{rms} = \frac{E_{peak}}{\sqrt{2}} = \frac{155}{\sqrt{2}} = 110V$$

52. What must be the value of the resistances for a 240 V, three-level heater if the desired output wattages are 500 W, 1000 W and 1500 W?

Solution:



The values of R_A and R_B can be 57.6Ω and 115.2Ω , arbitrarily.

Let's say $R_A = 115.2\Omega$.

When switch A is closed, the output wattage of the heater is 500W using the equation.

$$P = \frac{E^2}{R} = \frac{(240V)^2}{(115.2\Omega)} = 500Watts$$

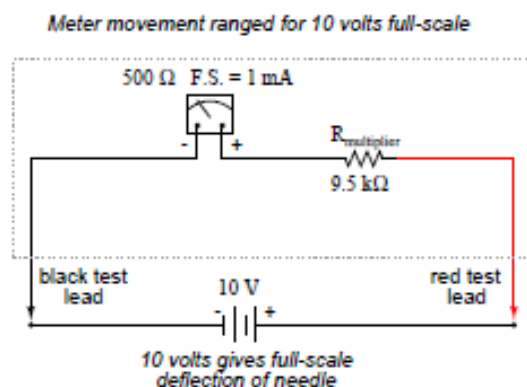
Then $R_B = 57.6\Omega$

$$P = \frac{E^2}{R} = \frac{(240V)^2}{(57.6\Omega)} = 1,000Watts$$

And when both switches are closed, the output wattage is 1500Watts.

53. When building a voltmeter, a resistance is placed in series with the movement coil. If the movement coil of a DC voltmeter has a resistance of 500 ohms and gives a full scale reading with a current of 1mA in the coil, what size resistor is needed in order to have the meter read full scale when measuring 10 V?

Solution:



54. How many 40W cool white fluorescent lamps would be required to have an initial lamp output of 10,000 lm? {A 40W cool white FL has the following specifications: Nominal length = 1200mm; Rated life = 20,000 hrs; Lamp Lumen Depreciation = 82% (@ 70% of rated life); Initial Lumens = 3150 lm}

Solution:

A single 40W CW FL has an initial lamp output of 3,150 lm. Thus, for us to have an initial lamp output of 10,000 lm: $10000/3150 = 3.17$ round up to 4.0. **We need 4 – 40W CW FL.**

55. From item #54, what is the output of five 40W cool white fluorescent lamp at 70 % of rated life?
Solution:

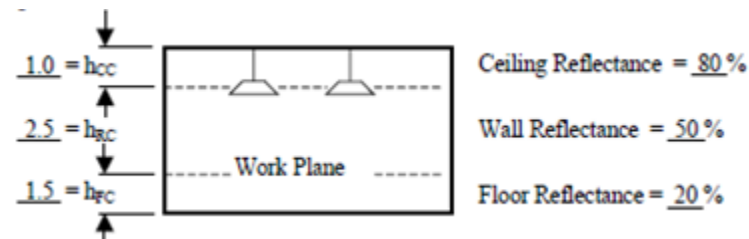
The initial lumen or lamp output of five 40W CW FL is 15,750lm (5x3150lm). At 70% of its rated life, apply LLD (Lamp Lumen Depreciation) factor of 0.82 (from item number 54). Thus, the output at 70% rated life of the 5-40W CW FL is **12,915lm.**

For numbers 56 to 57, consider the problem below:

A farm shop, 6.0 m wide, 14.0 m long and with a ceiling height of 5.0 m, is to be illuminated using fluorescent lamps which will be suspended 1.0 m below the ceiling. The work plane is to be 1.5 m above the floor. Ceiling wall and floor reflectances of 85%, 55 % and 20% can be assumed.

56. The room cavity ratio is _____.

Solution:



$$\text{Cavity Ratio} = \frac{5h(\text{Room Length} + \text{Room Width})}{\text{Room Length} \times \text{Room Width}}$$

where $h = h_{rc}$ for the room cavity ratio
 $= h_{cc}$ for the ceiling cavity ratio
 $= h_{fc}$ for the floor cavity ratio

$$\text{Room Cavity Ratio} = \frac{5(2.5\text{m})(14\text{m} + 6\text{m})}{14\text{m} \times 6\text{m}} = 3$$

57. The floor cavity ratio is _____.

Solution:

$$\text{Floor Cavity Ratio} = \frac{5(1.5\text{m})(14\text{m} + 6\text{m})}{14\text{m} \times 6\text{m}} = 1.8$$

58. Ten lux is equivalent to _____ footcandles?

Solution:

$$1\text{lux} = 10.76\text{footcandle} = (10.76\text{fc})(10) = 107.6\text{fc}$$

For numbers 59 to 61

A pony-break test of a motor gave the following data:

Force = 2lb

N = 1720 rpm

Length of arm = 1ft

Electrical meter readings:

Current = 11.4 A

Voltage=110 V Power = 960 W

59. What is the actual power delivered by the motor as measured by the pony break test?

Solution:

$$P = 2\pi F n = \frac{2\pi(2\text{lb})(1720\text{rpm})}{\left(33,000\text{ft.}\frac{\text{lb}}{\text{min}}\right)} 0.65\text{hp}$$

60. What is the efficiency of the motor in converting electric energy to mechanical energy?

Solution:

$$\text{Eff}_{\text{motor}} = \frac{(0.65\text{hp})(1000)}{(1.34\text{hp})(960\text{W})} = 51\%$$

61. What is the power factor?

Solution:

$$\text{Pf} = \frac{P}{VI} = \frac{960\text{W}}{(110\text{V})(11.4\text{A})} = 0.77$$

62. What would be the speed (rpm) of a 120V, 2-pole motor on a 60Hz source?

Solution:

$$f = \frac{PN}{120}; N = \frac{120f}{P} = \frac{120(60\text{Hz})}{2} = 3600\text{rpm}$$

63. A 120V generator delivers 3 kilowatts to an electric furnace. The current supplied by the generator is about:

Solution:

$$I = \frac{P}{V} = \frac{3,000\text{Watts}}{120\text{V}} = 25\text{A}$$

64. Oven method is primary method of moisture determination. If after three days at 105°C the initial 20g palay sample yields a final weight of 15g, what is the moisture content in percent wet basis?

Solution:

$$\text{MC}_{\text{WB}} = \frac{20 - 15}{20} = 25\%$$

65. Determine the quantity of corn in kilogram at 30% (db) required to produce 1 metric ton at 14% (wb).

Solution:

$$\text{MC}_{\text{wb}} = \frac{\text{MC}_{\text{db}}}{1 + \text{MC}_{\text{db}}} = \frac{0.3}{1 + 0.3} = 23.076\%$$

$$W_i = W_f \left(\frac{1 - \text{MC}_i}{1 - \text{MC}_f} \right) = (1000\text{kg}) \left(\frac{1 - 0.14}{1 - 0.23076} \right) = 1120\text{kg}$$

66. The total volume of biogas produced during anaerobic digestion of swine manure assuming the total volume of influent is 5m³/day, HRT is 30 days, and biogas production per unit digester is 2m³/m³.

Solution:

$$V = \left(\frac{5\text{m}^3}{\text{day}} \right) \left(\frac{2\text{m}^3}{\text{m}^3} \right) (30\text{days}) = 300\text{m}^3$$

67. Amount of water needed to bring the moisture content of a 5kg poultry-sawdust compost mixture from 45% wb to 65% wb

Solution:

$$W_w = \frac{5\text{kg}(1 - 0.45)}{1 - 0.65} - 5 = 3\text{kg}$$

68. A swine wastewater sample has a total solid concentration of 200mg/L. Determine the sample size used in the analysis if 0.05g of residue was recorded after drying the sample at 105°C.

Solution:

$$\text{Sample size} = \frac{0.05\text{g}}{\frac{0.2\text{g}}{\text{L}}} = 0.25\text{L} = 250\text{ml}$$

69. Daily manure inflow in the digester if loading rate is 200kg total solids per day and total solid concentration of influent is 5%. Liquid manure density is assumed to be 1000kg/m³

Solution:

$$\text{inflow} = \frac{\left(\frac{200\text{kg}}{\text{day}}\right)}{\left(\frac{1000\text{kg}}{\text{m}^3}\right)(0.05)} = \frac{4\text{m}^3}{\text{day}}$$

70. If 20kg of water is to be removed from 110kg of paddy, what is the percentage moisture content of paddy in wet basis?

Solution:

$$\text{MC}_{\text{wb}} = \frac{20\text{kg}}{110\text{kg}} = 18\%$$

71. What is the percentage moisture content of paddy (Problem #70) when expressed in dry basis?

Solution:

$$\text{MC}_{\text{db}} = \frac{110 - 90}{90} = 22.2\%$$

72. What is the amount of moisture to be removed from a material whose moisture content is 14.5% with dry weight of 76kg?

$$m_w = \frac{76\text{kg}}{1 - 0.145} = 88.88\text{kg} - 76\text{kg} = 12.88\text{kg}$$

73. IF a product has a percentage moisture content of 12% wet basis, the equivalent percentage moisture content dry basis is?

$$\text{MC}_{\text{db}} = \frac{0.12}{1 - 0.12} = 13.6\%$$

74. A product expressed in dry basis has a percentage moisture content of 25%, what is the percentage moisture content of the product in wet basis?

Solution:

$$\text{MC}_{\text{wb}} = \frac{0.25}{1 + 0.25} = 33.33\%$$

75. If 9 ton of a product is to be dried from 24% to 14%, the final weight of the product would be.

Solution:

$$W_f = W_i \left(\frac{1 - \text{MC}_f}{1 - \text{MC}_i} \right) = 9000\text{kg} \left(\frac{1 - 0.24}{1 - 0.14} \right) = 7,953.48\text{kg}$$

76. What is the moisture loss of the product in the item #75?

Solution:

$$W_w = 9,000\text{kg} - 7,953.48\text{kg} = 1046.5\text{kg}$$

77. If a pile of rice is 8m long, 6m wide and 3.6m high, how many bags of rice are safe to be piled?

Solution:

$$\# \text{ of bags} = (8\text{m} * 6\text{m} * 3.6\text{m}) \left(\frac{15\text{bags}}{\text{m}^3} \right) = 2,592\text{bags}$$

78. If 5,000 bags of paddy is to be piled 20 bags high, how many square meters of floor space is required?

Solution:

$$V = 5,000\text{bags} \left(\frac{\text{m}^3}{10\text{bags of paddy}} \right) = 500\text{m}^3$$

$$A = \frac{500\text{m}^3}{(0.23\text{m})(20\text{bags})} = 108.70\text{m}^2$$

79. What is the theoretical rpm of a bucket elevator whose head pulley diameter is 0.2m and the projection is 0.1m?

Solution:

$$N_{\text{theoretical}} = \frac{54.19}{\sqrt{R}} = \frac{54.19}{\sqrt{\left(\frac{0.2\text{m} \times 3.28}{2}\right) + \left(\frac{0.1\text{m} \times 3.28}{2}\right)}} = 77.30\text{rpm}$$

80. If the operational efficiency of the bucket elevator of the problem #79 is 85% of the theoretical speed, what is the operation speed of the elevator?

Solution:

$$N = (0.85)(77.30\text{rpm}) = 65.7\text{rpm}$$

81. What is the theoretical horsepower requirement of a bucket elevator whose capacity is 545kg/min and lift of 10.7m? The elevator is loaded downside.

Solution:

$$\text{HP}_{\text{th}} = \text{QHF} = \left(\frac{545\text{kg}}{\text{min}}\right)(10.7\text{m})(1.5)\left(\frac{1\text{min}}{60\text{s}}\right) = \frac{\frac{97.20\text{kg} \cdot \text{m}}{\text{s}}}{\frac{76.2\text{kg} \cdot \text{m}}{\text{s} \cdot \text{hp}}} = 1.91\text{hp}$$

82. Five tons of paddy milled in 6 hours produces 3950kg brown rice and 3250kg of milled rice. What is the milling recovery?

Solution:

$$\text{Milling Recovery} = \frac{W_{\text{MR}}}{W_{\text{Paddy}}} = \frac{3,250\text{kg}}{5000\text{kg}} = 65\%$$

83. What is the hulling coefficient of the huller of the item #82?

Solution:

$$\text{Coefficient of Hulling} = \frac{W_{\text{BR}}}{W_{\text{paddy}}} = \frac{3,950\text{kg}}{5000\text{kg}} = 79\%$$

84. If the head rice recovery (HRR) in the item #82 is equal to 85%, what is the amount of broken grains?

Solution:

Note: Head rice is 85% and 15% is broken grains.

$$\text{Broken grains} = 3,250(0.15) = 487.5\text{kg}$$

85. What is the amount of rice hull produced during milling in the item #82?

Solution:

$$\text{Rice Hull} = 5,000\text{kg} - 3950\text{kg} = 1050\text{kg}$$

86. A two-ton mechanical dryer dries a paddy from 24% to 14% in 8 hours. What is the final weight of the product after drying?

Solution:

$$W_f = W_i \left(\frac{1 - \text{MC}_f}{1 - \text{MC}_i} \right) = (2,000\text{kg}) \left(\frac{1 - 0.24}{1 - 0.14} \right) = 1,767.44\text{kg}$$

87. In item #86, what is the drying capacity of the dryer?

Solution:

$$\text{Drying Capacity} = \frac{2,000\text{kg}}{8\text{hrs}} = \frac{250\text{kg}}{\text{hr}}$$

88. What is the moisture reduction per hours in item #86?

Solution:

$$\text{Moisture Reduction} = \frac{(2000\text{kg} - 1767.44\text{kg})}{8\text{hrs}} = \frac{29.07\text{kg}}{\text{hr}}$$

89. The coefficient of friction of rice to a steel material is 0.41. What is the minimum angle of hopper required for the system?

Solution:

$$\theta = \arctan(0.41)$$

$$\theta = 22 \text{ degrees}$$

90. What is the peak storage capacity of a 5m diameter grain storage bin with an average height and minimum angle of fill of 6m and 28 degrees, respectively?

Solution:

$$V = \frac{\pi D^2 H}{4} + \frac{\pi D^2}{4} \left(\frac{r \tan \theta}{3} \right)$$

$$V = \frac{\pi (5\text{m})^2 (6\text{m})}{4} + \frac{\pi (5\text{m})^2}{4} \left(\frac{(2.5\text{m}) \tan(28^\circ)}{3} \right)$$

$$V = 126.50\text{m}^3$$

91. A 94lb-bag of Portland cement has a bulk volume of

Solution:

$$\text{Bulk Volume} = (94\text{lb}) \left(\frac{1\text{kg}}{2.2\text{lb}} \right) \left(\frac{1\text{bag}}{40\text{kg}} \right) \left(\frac{1\text{ft}^3}{\text{bag}} \right) = 1\text{ft}^3$$

92. The number of 10cm x 20cm glazed tiles need for 9m² floor area.

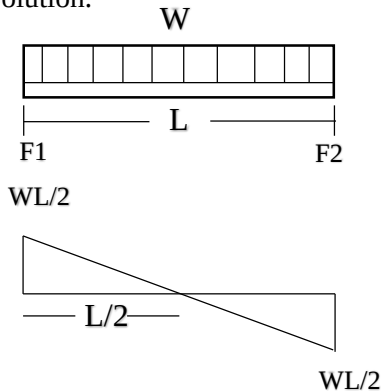
Solution:

$$A = (0.1\text{m})(0.2\text{m}) = 0.02\text{m}^2$$

$$\# \text{ of tiles} = \frac{9\text{m}^2}{0.02\text{m}^2} = 450 \text{ pieces}$$

93. The maximum moment M for a simple beam with a uniformly distributed load for W and a clear span of L? Assume weight of beam to be negligible.

Solution:



$$F_1 = F_2 = \frac{WL}{2}$$

$$A_T = \frac{1}{2}bh$$

$$M_{\max} = \frac{1}{2} \left(\frac{L}{2} \right) \left(\frac{WL}{2} \right) = \frac{WL^2}{8}$$

94. The tensile stress that will be developed if a force of 10N is applied to one end of a 10 inches diameter circular rod?

Solution:

$$\sigma_{\text{Tensile}} = \frac{F}{A} = \frac{10\text{N}}{\frac{\pi}{4} (10 \times 0.0254\text{m})^2} = 197.35\text{Pa or } 0.2\text{kPa}$$

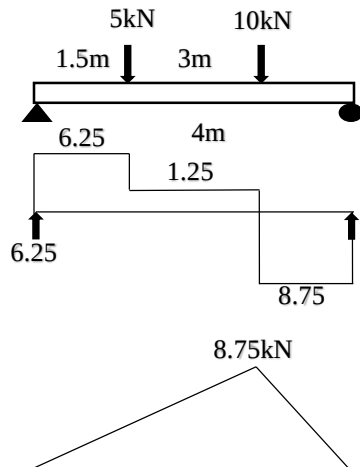
95. A 4" x 3" wood column has an unsupported length of 2.5m. Allowable compressive stress parallel to the grain is 13.2MPa and Modulus of elasticity of 8.47MPa. The actual slenderness ratio of the column is

Solution:

$$\text{Slenderness ratio} = \frac{\text{Length (Unsupported)}}{\text{dimension}} = \frac{2.5\text{m}}{3 \times 0.0254} = 32.81$$

96. A simply supported beam horizontal beam, with a span of 4m, is carrying a loads of 5kN and 10kN at 1m and 3m from the left. What is the maximum bending moment experienced by the beam?

Solution:



Base on the diagram, the maximum moment is:

$$M_{\max} = 8.75\text{kN}$$

97. The maximum cross section of the beam to resist bending stress in the item #96. $\sigma_{\text{Allowable}} = 16.5\text{MPa}$

Solution:

$$\sigma_{\max} = \frac{MC}{I} = \frac{M_{\max} \left(\frac{S}{2} \right)}{\frac{S(S^3)}{12}}$$

$$16.5\text{MPa} = \frac{M_{\max} \left(\frac{S}{2} \right)}{\frac{S(S^3)}{12}}$$

$$S^3 = \left[\frac{(8.75 \times 10^3 \text{ kN})(12)}{2(16500 \text{ kPa})} \right]$$

$$S = 0.147\text{m or } 5.8\text{in}$$

Cross Section or Dimension = 6" x 6"

98. The maximum deflection of a 5" x 10" wood beam with a span of 5m and supports a total load of 18kN/m (Modulus of Elasticity = 4GPa).

Solution:

$$\sigma_{\max} = \frac{5}{384} \left[\frac{WL^4}{EI} \right] = \frac{5}{384} \left[\frac{\left(\frac{1800\text{N}}{\text{m}} \right) (5\text{m})^4}{\left(\frac{4 \times 10^9 \text{ N}}{\text{m}^2} \right) \left(\frac{(0.127\text{m})(0.254)^3}{12} \right)} \right]$$

$$\delta_{\max} = 0.21\text{m}$$

99. A hollow tube well with inside diameter of 10mm must a tensile load of 400kN. Determine the outside diameter of the tube if the stress is limited to 120MPa.

Solution:

$$\sigma = \frac{F}{A}$$

$$\frac{120 \times 10^6 \text{ N}}{\text{m}^2} = \frac{400000 \text{ N}}{\frac{\pi}{4}(D_o^2 - (0.1 \text{ m})^2)}$$

$$D_o = 0.12 \text{ m or } 120 \text{ mm}$$

100. What force is required to punch a 20mm diameter hole in a plate that is 24mm thick? the shear strength is 350MPa

Solution:

$$A = \pi Dt$$

$$\sigma = \frac{F}{A}$$

$$F = \sigma A = \left(\frac{350 \times 10^6 \text{ N}}{\text{m}^2} \right) [\pi(0.02 \text{ m})(0.025 \text{ m})] = 550 \text{ kN}$$

101. A simply supported rectangular beam 50mm wide by 100mm deep carries a uniformly distributed load of 1200N/m over its entire length. What is the maximum length of the beam if the flexural stress is limited to 20MPa?

Solution:

$$\left(\frac{20 \times 10^6 \text{ N}}{\text{m}} \right) = \sigma = \frac{MC}{E} = \frac{M \left(\frac{0.1}{2} \right)}{\frac{(0.05 \text{ m})(0.1)^3}{12}}$$

$$M = 1666.67$$

$$A = L \times W = \frac{1}{2}(600L) \left(\frac{L}{2} \right) = 150L^2$$

$$150L^2 = 1666.67$$

$$L = 3.33 \text{ m}$$

102. Calculate the section modulus of a solid circle having a diameter of 5".

Solution:

$$\text{Section Modulus} = \frac{I}{C} = \frac{\frac{\pi}{64}(0.127)^4}{\frac{0.127}{2}} = 2.01 \times 10^{-4}$$

103. A simply supported beam having a span of 6m carries a concentrated loads of 80kN at its middle thirds. Compute the deflection at mid-span due to this load. Assume flexural stress rigidity of $30,670 \times 10^9 \text{ N-mm}^2$

Solution:

$$\delta_{\max} = \frac{23PL^3}{648EI} = \frac{23(80,000 \text{ N})(6 \text{ m})^3(1000)^3}{648(30,670 \times 10^9 \text{ N-mm}^2)} = 20 \text{ mm}$$

104. A wooden beam having a width of 50mm is subjected to a moment of 2,400N-m. Determine the depth if it has an allowable bending of 28.8MPa.

Solution:

$$\sigma = \frac{MC}{I}$$

$$\left(\frac{28.8 \times 10^6 \text{ N}}{\text{m}^2} \right) = \frac{(2,400 \text{ N-m}) \left(\frac{d}{2} \right)}{\frac{(0.05 \text{ m})(d^3)}{12}}$$

$$d = 0.1 \text{ m or } 100 \text{ mm}$$

105. A simple beam 6m long carries two equal concentrated load at the third points of the beam. If the concentrated load are 40kN each, what is the maximum deflection of the beam? $E = 200,000\text{N/m}^2$, $I = 50 \times 10^6 \text{mm}^4$

Solution:

$$EI = 1 \times 10^{13} \text{N} - \text{mm}^2$$

$$\delta = \frac{23}{648} \left[\frac{PL^3}{EI} \right] = \frac{23}{648} \left[\frac{(40 \times 10^3 \text{N})(6\text{m})^3 (1000)^3}{1 \times 10^{13} \text{N} - \text{mm}^2} \right] = 30.67 \text{mm}$$

106. Compute the safe uniform load that a simply supported beam having a span of 4m could carry if it has a section modulus of $1.04 \times 10^6 \text{mm}^4$. Allowable bending stress is 12MPa.

Solution:

$$\sigma = \frac{MC}{I} \text{ or } \frac{M}{I}$$

$$\frac{12 \times 10^6 \text{N}}{\text{m}^2} = \frac{M}{0.0104 \text{m}^4}; M = 124,800 \text{N} - \text{m}^2$$

$$\frac{W(4)^2}{8} = 124,800 \text{N} - \text{m}^2; W = \frac{6.24 \text{KN}}{\text{m}}$$

107. What is the power dissipated when a 1/2 ampere current flows through a 5Ω resistor?

Solution:

$$P = EI = I^2 R = (0.5)^2 (5\Omega) = 1.25 \text{Watts}$$

108. The amount of substance can be given on a molar basis in terms of kilomole. Given 1kg of air, what is its moles in kilomole?

Solution:

$$X = \left(\frac{1 \text{mole}}{28 \text{g}} \right) \left(\frac{1000 \text{g}}{1 \text{kg}} \right) \left(\frac{1 \text{kmol}}{1000 \text{mole}} \right) = \frac{0.036 \text{kmol}}{\text{g}}$$

109. An air-conditioned room is maintained at 24°C while outside air temperature is 32°C . What is the temperature difference in degrees Fahrenheit?

Solution:

$$^\circ\text{F} = (32 - 24) \left(\frac{9}{5} \right) = 14.4^\circ\text{F}$$

110. Air flow through a 10mm diameter pipe at an average velocity of 0.1m/s. What is the maximum flow in kg/s?

Solution:

$$X = \left(\frac{1.25 \text{kg}}{\text{m}^3} \right) \left(\frac{0.1 \text{m}}{\text{s}} \right) \left(\frac{\pi}{4} (0.01 \text{m})^2 \right) = \frac{9.82 \times 10^{-6} \text{kg}}{\text{s}}$$

111. The temperatures on the faces of a plane wall 15cm thick are 370°C and 93°C . The wall is constructed of a special glass with the following properties: $k = \frac{0.78 \text{W}}{\text{m}^\circ\text{C}}$, $\rho = \frac{2,700 \text{kg}}{\text{m}^3}$, $C_p = \frac{0.84 \text{kJ}}{\text{kg}^\circ\text{C}}$.

What is the heat flow through the wall at steady state?

Solution:

$$Q = \frac{k \Delta T A}{x} = \frac{\left(\frac{0.78 \text{W}}{\text{m}^\circ\text{C}} \right) (370^\circ\text{C} - 93^\circ\text{C}) (1 \text{m}^2)}{0.15 \text{m}} = 1,440.4 \text{Watts}$$

112. Water at a temperature of 150°F flows through a steel pipe having an inside diameter of 1" and an outside diameter of 1-1/4". The outer surface of the pipe is exposed to air at 70°F . The heat transfer coefficients on the water and air sides are $\frac{100 \text{BTU}}{\text{ft}^2 \cdot \text{hr}^\circ\text{F}}$ and $\frac{1 \text{BTU}}{\text{ft}^2 \cdot \text{hr}^\circ\text{F}}$ and K for the steel may be taken as $\frac{25 \text{BTU}}{\text{ft} \cdot \text{hr}^\circ\text{F}}$. The heat transfer rate per foot of pipe is:

Solution:

$$Q = \frac{\Delta T}{U} = \frac{\Delta T}{\frac{1}{2\pi H_o r_i} + \frac{\ln\left(\frac{r_o}{r_i}\right)}{2\pi k} + \frac{1}{2\pi H_i r_o}}$$

$$Q = \frac{150^\circ\text{F} - 70^\circ\text{F}}{\frac{1}{2\pi\left(\frac{100\text{BTU}}{\text{ft}^2 \cdot \text{hr} \cdot ^\circ\text{F}}\right)(0.042\text{ft})} + \frac{\ln\left(\frac{0.052\text{ft}}{0.042\text{ft}}\right)}{2\pi\left(\frac{25\text{BTU}}{\text{ft} \cdot \text{hr} \cdot ^\circ\text{F}}\right)} + \frac{1}{2\pi\left(\frac{1\text{BTU}}{\text{ft}^2 \cdot \text{hr} \cdot ^\circ\text{F}}\right)(0.052\text{ft})}}$$

$$Q = \frac{25.81\text{BTU}}{\text{ft} \cdot \text{hr}}$$

113. Water at a mean temperature of 120°F , flow through 6 inch diameter pipe at a rate of 80lbm/sec. If the mean temperature of the wall pipe is 180°F and if the length of the pipe is 20ft, Find the Reynolds Number at 120°F , $\rho = \frac{61.73\text{lbm}}{\text{ft}^3}$, $\mu = \frac{1.36\text{lbm}}{\text{ft} \cdot \text{hr}}$, $k = \frac{0.372\text{BTU}}{\text{hr} \cdot \text{ft} \cdot ^\circ\text{F}}$, $C_p = \frac{0.997\text{BTU}}{\text{lbm} \cdot ^\circ\text{F}}$.

Solution:

$$\text{Re} = \frac{\rho v D}{\mu} = \frac{\left(\frac{61.73\text{lbm}}{\text{ft}^3}\right)\left(\frac{80\text{lbm}}{\text{s}}\right)\left(\frac{6\text{in}}{12}\right)\left(\frac{1\text{ft}^3}{61.73\text{lbm}}\right)\left(\frac{3600\text{s}}{1\text{hr}}\right)}{\left(\frac{1.36\text{lbm}}{\text{ft} \cdot \text{hr}}\right)\left(\frac{\pi}{4}\left(\frac{6}{12}\right)^2\right)}$$

$$\text{Re} = 539,254$$

114. Heat is generated uniformly at a rate of $\frac{1,000\text{BTU}}{\text{hr} \cdot \text{ft}^3}$ in a 4" thick plane wall with a thermal conductivity of $\frac{12\text{BTU}}{\text{ft} \cdot \text{hr} \cdot ^\circ\text{F}}$. One side of the wall is maintained at a temperature of 140°F and the other at 70°F , at what rate is heat being transferred from the cooler side of the wall.

Solution:

$$\frac{Q}{A} = \frac{k\Delta T}{x} = \frac{\left(\frac{12\text{BTU}}{\text{ft} \cdot \text{hr} \cdot ^\circ\text{F}}\right)(140^\circ\text{F} - 70^\circ\text{F})}{(4\text{in})\left(\frac{1\text{ft}}{12\text{in}}\right)} = \frac{2520\text{BTU}}{\text{ft}^2 \cdot \text{hr}}$$

115. What is the ideal work for a refrigerator if the cold room temperature is -15°C , ambient temperature is 30°C and heat rejection is 10kW?

Solution:

$$W = Q_H - Q_L$$

$$Q_L = \frac{T_L Q_H}{T_H} = \frac{(-15 + 273.15)(10\text{kW})}{(30 + 273.15)} = 8.51\text{kW}$$

$$W = Q_H - Q_L = 10\text{kW} - 8.51\text{kW}$$

$$W = 1.48\text{kW}$$

116. Consider 100m^3 of moist air at 101.325kPa, 35°C , 70% RH. If the saturation pressure of water is 5.628kPa at 35°C , the partial pressure of air is?

Solution:

$$\text{RH} = \frac{\text{Pressure(Actual)}}{\text{Pressure(Saturation)}}$$

$$P_{\text{actual}} = \text{RH} * P_{\text{saturation}} = (0.7)(5.628\text{kPa}) = 3.94\text{kPa}$$

117. Two cubic meters per hour of water are cooled from 30°C to 4°C . What is the rate of heat transfer in kilowatts?

Solution:

$$Q = \dot{m}C_p\Delta T \text{ or } v\rho_w\dot{C}_p\Delta T$$

$$Q = \left(\frac{2\text{m}^3}{\text{hr}}\right) \left(\frac{1000\text{kg}}{\text{m}^3}\right) \left(\frac{4.19\text{kJ}}{\text{kg} \cdot ^\circ\text{K}}\right) (30^\circ\text{C} - 4^\circ\text{C}) \left(\frac{1\text{hr}}{3600\text{s}}\right) = 60.52\text{kW}$$

118. A refrigeration system operates on the reversed Carnot cycle. The higher temperature of the refrigerant in the system is 120°F and the lower is 10°F. The capacity is 20 tons. Determine COP and

Solution:

$$\text{COP} = \frac{Q_A}{W} = \frac{T_1}{T_2 - T_1} = \frac{10 + 460}{(120 + 460) - (10 + 460)} = 4.27$$

119. An electric pump has total resistance of 9.5 ohms. If its power source is from a 220-volt outlet, what should be the amperage of its safety fuse? Assume 27 % surge current.

Solution:

$$I = \frac{1.27V}{R} = \frac{1.27(220V)}{9.5\Omega} = 29.41\text{Amp (Use 30Amp)}$$

120. What is the power generated by a micro-hydropower plant having a gross head of 80 meters and reliable discharge of 50 liters/second. Assume typical efficiency.

Solution:

$$50\text{lite/s} = 0.05\text{m}^3/\text{s}$$

$$P = 9.8QHE = 9.8 \left(\frac{0.05\text{m}^3}{\text{s}}\right) (80\text{m})(0.6) = 23.52.\text{kW}$$

121. Give the estimated mechanical power output of a windmill having a rotor diameter of 8 meters if wind velocity is 3 m/s.

Solution:

$$P = 0.1AV^3 = 0.1 \left(\frac{\pi}{4}(8\text{m})^2\right) \left(\frac{3\text{m}}{\text{s}}\right)^3 = 135\text{kW}$$

122. What is the electrical energy consumption of a 10-kw coconut oil expeller plugged in a 220-volt source if it requires 3 hours to expel oil from 1,000 kg of grated copra?

Solution:

$$E_c = \frac{P_c T_o}{W_i} = \frac{(10\text{kW})(3\text{hrs})}{1000\text{kg}} = \frac{0.03\text{kWh}}{\text{kg}}$$

123. What is the peak solar power generation per charging channel of a solar battery charging station equipped with 12 units of 75 watt-peak 12-volt solar module? The system is divided into 3 charging channels, each channel has 4 modules connected in parallel. Assume 80% overall efficiency.

Solution:

$$P = \left(\frac{\# \text{ of Module}}{\text{Channel}}\right) (\text{Module power rating})(\text{Eff})$$

$$P = (4)(75\text{Wp})(0.8) = 240\text{Watt} - \text{peak}$$

124. An industrial heater has 8 pieces heating plates of 6 ohms each. Plates are arranged in series. If the heater power source is from a 220-volt outlet, how much current will flow?

Solution:

$$R_T = 8R = 8(6\text{ohms}) = 48\text{ohms}$$

$$I = \frac{V}{R} = \frac{220V}{48\text{ohms}} = 4.58\text{Amps}$$

125. How much should be budgeted annually for the electrical consumption of a 20-kw coconut oil expeller drawing 92 amperes current and having an electrical consumption of 0.04 kwh/kg

copra? It expels 250 kg of copra per hour and operates 16 h/day at 264 days/year. Electricity costs PHP5/kwh. Use 10% cost safety factor.

Solution:

$$\text{Budget} = \left(\frac{0.04 \text{ kWh}}{\text{kg}} \right) \left(\frac{250 \text{ kg}}{\text{hr}} \right) \left(\frac{16 \text{ hrs}}{\text{day}} \right) \left(\frac{264 \text{ days}}{\text{year}} \right) (1 + 0.1) \left(\frac{\text{P5}}{\text{kWh}} \right)$$

$$\text{Budget} = \frac{\text{P232,320}}{\text{year}}$$

126. What is the peak amperage of a solar battery charging station equipped with 16 units of 75 watt-peak 4.16-ampere solar module? The system is divided into 4 charging channels, each channel has 4 modules connected in parallel.

Solution:

$$I = \left(\frac{\# \text{ of modules}}{\text{Channel}} \right) (\text{module amperage rate}) = (4)(4.16 \text{ Amps})$$

$$I = 16.64 \text{ Amps}$$

127. What is the system voltage of a solar pumping system equipped with 2 units of 75 watt-peak solar modules? Each solar module has 4.16 amperes maximum, open circuit voltage of 21 volts maximum and nominal voltage of 12 volts. Modules are connected in series.

Solution:

$$\text{Voltage} = (\# \text{ of modules})(\text{Module Nomila Voltage}) = (2)(12 \text{ V})$$

$$V = 24 \text{ Volts}$$

128. Estimate the hydraulic power output of a multi-bladed windmill when wind-speed is 3 m/s. It has 0.3 rotor power coefficient, 90% transmission efficiency, 90% pump volumetric efficiency and 6 meters rotor diameter. Air density is 1.2 kg/m³.

Solution:

$$P_h = \frac{1}{2} \rho_{\text{air}} A V^3 = \frac{1}{2} \left(\frac{1.2 \text{ kg}}{\text{m}^3} \right) \left(\frac{\pi}{4} (6 \text{ m})^2 \right) \left(\frac{3 \text{ m}}{\text{s}} \right)^3 (0.9)(0.9)(0.3)$$

$$P_h = 111.30 \text{ Watts}$$

129. If a 10-meter high multi-bladed windmill for vegetable farm has a minimum running wind-speed of 1.5 m/s, what is the minimum wind-speed for the rotor to start turning?

Solution:

$$V = \frac{\text{Running windspeed}}{0.564} = \frac{\frac{1.5 \text{ m}}{\text{s}}}{0.564} = \frac{2.7 \text{ m}}{\text{s}}$$

130. A horizontal-axis windmill for poultry farm water pumping has a rotor diameter of 8 meters and registers a running torque of 920 Newton-meters. What is the torque needed to start its rotor turning?

Solution:

$$\text{Starting Torque} = \pi T = \pi(920 \text{ N} - \text{m}) = 2,890 \text{ N} - \text{m}$$

131. How many 75 watt-peak solar modules are needed for a solar powered incubator having a power rating of 144 watts, system voltage of 48 volts and amperage of 6 amperes? All solar modules used have 12 volts nominal voltage and 4.6 amperes peak amperage.

Solution:

$$N_{\text{total}} = (\# \text{ of module in series})(\# \text{ of modules in paralle})(\# \text{ of modules rounded up})$$

$$N_{\text{total}} = \left(\frac{\text{System voltage}}{\text{Nominal voltage}} \right) \left(\frac{\text{System amperage}}{\text{Module amperage}} \right) = \left(\frac{48 \text{ V}}{12 \text{ V}} \right) \left(\frac{6 \text{ amps}}{4.6 \text{ amps}} \right)$$

$$N_{\text{total}} = (4)(2 \text{ round it up}) = 8 \text{ modules}$$

132. What should be the tilt angle from the horizontal of the glass solar collector of a hybrid biomass-solar dryer located somewhere in Surigao, Philippines, having a longitude of 125 degrees and latitude of 9 degrees? Since Philippines is in northern hemisphere, the solar collector faces south or opposite hemisphere for maximum energy collection.

Solution:

Angle = Latitude of the site; but minimum of 10 degrees for self cleaning
 = 9 degrees; but constrained to 10 degrees minimum for self cleaning
 = 10 degrees minimum

133. Geothermal, biodiesel and wind-farm renewable energy projects are evaluated for financial feasibility but only a maximum of 2 projects can be implemented. Each of them requires PHP900,000,000 investment. The financial internal rate of return (IRR) are 28.2%, 9.8% and 9.6% for geothermal, biodiesel and wind project, respectively. Annual money discount and inflation rate is 10.2%. Which project shall not be implemented?

Solution:

All of the projects shall not be implemented since IRR of biofuel & ocean projects < monetary discount & inflation rate)

134. Determine the biogas consumption per day of a household using a 4-inch gas burner at 4 hours duration per day. The gas consumption of the burner is constant at 0.283m³/hour.

Solution:

C = NBT:

where C = Consumption, N = No. of units, B = biogas consumption of device, T = Time

$$C = (1\text{unit}) \left(\frac{0.283\text{m}^3}{\text{hr}} \right) \left(\frac{4\text{hrs}}{\text{day}} \right) = \frac{1.13\text{m}^3}{\text{day}}$$

135. Determine the biogas production per day of an animal farm having 10,000 heads of porkers of mixed ages. Use a retention period of 35 days. The mean daily manure production of porkers is 2.2 kg/head while the specific gas production for 35-day hydraulic retention time in the digester is 0.065m³/kg.

Solution:

P = NMG

where P = biogas production potential, N = No. of heads, M = manure production,

G = Specific gas production of the manure for a specific retention period

$$P = NMG = (10,000\text{heads}) \left(\frac{2.2\text{kg}}{\text{head}} \right) \left(\frac{0.065\text{m}^3}{\text{kg}} \right) = 1,430\text{m}^3$$

136. How many heads of breeding buffalo are needed as source of manure to generate 1,000m³ of biogas a day if the retention period is 25 days? The mean daily manure production of breeding buffalo is 13.5 kg/head while its specific gas production for 25-day manure retention period is half of that of chicken dung.

Solution:

G = 0.5 x G for chicken dung at 25-day retention period
 = 0.5 x 0.06 m³/kg
 = 0.03 m³/kg

$$N = \frac{P}{MG} = \frac{(1000\text{m}^3)}{\left(\frac{13.5\text{kg}}{\text{head}} \right) \left(\frac{0.03\text{m}^3}{\text{kg}} \right)} = 2469\text{heads}$$

137. Determine the required digester volume of a biogas plant that will accommodate manure from a broiler farm having a capacity of 20,000 heads of mixed ages. Use 30-day retention period and 1:1 manure-water ratio. The mean daily manure production of broiler is 0.025 kg/head while its specific gas production for 30-day manure retention period is 0.062m³/kg. Chicken dung density is 950 kg/m³.

Solution:

Use the following formulas:

$$V = IR, \text{ but, } I = 2I_m, I_m = 0.8P/GD, P = NMG$$

Where

V = digester volume, I = daily slurry input, R = retention period, I_m = Manure input, D = manure density, P = biogas production

$$\text{Volume} = \left(\frac{\text{Daily Slurry input}}{\text{day}} \right) (\text{Retention Period (days)}) = IR$$

$$P = NMG = (20,000\text{heads}) \left(\frac{0.025\text{kg}}{\text{head}} \right) \left(\frac{0.062\text{m}^3}{\text{kg}} \right) = 31\text{m}^3$$

$$I_{\text{Manure}} = \frac{0.8P}{GD} = \frac{0.8(31\text{m}^3)}{\left(\frac{0.062\text{m}^3}{\text{kg}} \right) \left(\frac{950\text{kg}}{\text{m}^3} \right)} = 0.42\text{m}^3$$

$$I = 2I_{\text{Manure}} = 2(0.42\text{m}^3) = \frac{0.842\text{m}^3}{\text{day}}$$

$$\text{Volume} = IR = \left(\frac{0.842\text{m}^3}{\text{day}} \right) (30\text{days}) = 25.26\text{m}^3$$

138. A biogas plant has the following specifications and utilization:

Number of swine (36-55 kg): 10

Manure production per head: 5.22 kg/day

Gas production per 1 kg of manure: 0.058m³

Retention time: 25 days

For one day utilization,

Light: 2 light bulbs at 3hrs operation

Stove: 1 5-cm stove at 3hrs operation

Refrigerator: 1 unit at 24hrs operation

If the biogas requirement for light is 0.071m³/hr. utilization, 0.226m³/hr. for the stove, and 0.053m³/hr. for the refrigerator, estimate the gasholder volume of the plant. Allow 30% allowance for biogas fluctuation.

Solution:

$$\text{Manure Production} = NM = \left(\frac{5.22\text{kg}}{\text{day} - \text{heads}} \right) (10\text{heads}) = \frac{52.2\text{kg}}{\text{day}}$$

$$\text{Biogas Production} = NMG = \left(\frac{52.2\text{kg}}{\text{day}} \right) \left(\frac{0.058\text{m}^3}{\text{kg}} \right) = \frac{3.03\text{m}^3}{\text{day}}$$

Daily Utilization:

$$\text{Light} = (2\text{lights}) \left(\frac{3\text{hrs}}{\text{light}} \right) \left(\frac{0.071\text{m}^3}{\text{hr}} \right) = \frac{0.426\text{m}^3}{\text{day}}$$

$$\text{Stove} = (1 \text{ stove}) \left(\frac{0.226 \text{m}^3}{\text{hr}} \right) (3 \text{hrs}) = \frac{0.687 \text{m}^3}{\text{day}}$$

$$\text{Refrigerator} = (1) \left(\frac{0.053 \text{m}^3}{\text{hr}} \right) (24 \text{hrs}) = \frac{1.272 \text{m}^3}{\text{day}}$$

$$\text{Total} = \frac{2.385 \text{m}^3}{\text{day}}$$

$$\text{Gas to be stored} = \left(\frac{3.03 \text{m}^3}{\text{day}} - \frac{2.385 \text{m}^3}{\text{day}} \right) (1 + 0.3 \text{ (for biogas fluctuation)}) = \frac{0.8385 \text{m}^3}{\text{day}}$$

139. If a multi-bladed wind-pump has a tip-speed ratio of 1:1, what winds-peed is needed to have its rotor tip move at 3 m/s?

Solution:

Wind-speed = tip speed, since tip-speed ratio is 1:1
= 3m/s

140. What is the electrical energy consumption of a 15-kw vermi cast production shredder plugged in a 220-volt source if it requires 2 hours to shred 1.5 tons of farm waste?

Solution:

$$E_{\text{consumption}} = \frac{P_c T_o}{W_i} = \frac{(15 \text{kW})(2 \text{hrs})}{1500 \text{kg}} = \frac{0.02 \text{kWh}}{\text{kg}}$$

141. A 2.4-tonne capacity Jatropha oil expeller was loaded full with dried Jatropha seeds in 4 minutes. It produced 600 kg of crude oil in 2 hours. What is its crude oil production rate?

Solution:

$$P_{\text{oil}} = \frac{\text{Weight of crude oil collected}}{\text{Total time}} = \frac{600 \text{kg}}{2 \text{hrs}} = \frac{300 \text{kg}}{\text{hr}}$$

142. A Jatropha oil expeller was loaded with 2.1 ton dried Jatropha seeds. It produced 600 kg of crude oil in 3 hours. What is its Jatropha cake production rate?

Solution:

$$P_{\text{cake}} = \frac{\text{Weight of resulting Jatropha cake produced}}{\text{Total time}} = \frac{2,100 \text{kg} - 600 \text{kg}}{3 \text{hours}}$$

$$P_{\text{cake}} = \frac{500 \text{kg}}{\text{hr}}$$

143. A 20-tonne capacity coconut oil expeller was loaded full with grated copra. It produced 8,000 kg of crude oil in 20 hours. What is the copra cake production rate of the machine at 98% cake collection efficiency?

Solution:

$$P_{\text{cake}} = \left(\frac{20,000 \text{kg} - 8000 \text{kg}}{20 \text{hrs}} \right) (0.98) = \frac{588 \text{kg}}{\text{hr}}$$

144. How many ton of agricultural waste is needed annually to optimally operate a 26-ampere 6kW vermi cast production shredder having a capacity of 0.25 ton of waste per hour? It optimally operates 16 hours daily during the 3-month peak sunny months and 8 h/day at 26 days/month for the rest of the year.

Solution

$$W = \left(\frac{0.25 \text{ton}}{\text{hr}} \right) \left[\left(\frac{16 \text{hrs}}{\text{day}} \right) \left(\frac{30 \text{days}}{\text{month}} \right) \left(\frac{3 \text{month}}{\text{year}} \right) + \left(\frac{8 \text{hrs}}{\text{day}} \right) \left(\frac{26 \text{days}}{\text{month}} \right) \left(\frac{9 \text{months}}{\text{year}} \right) \right]$$

$$W = \frac{828 \text{ ton}}{\text{year}}$$

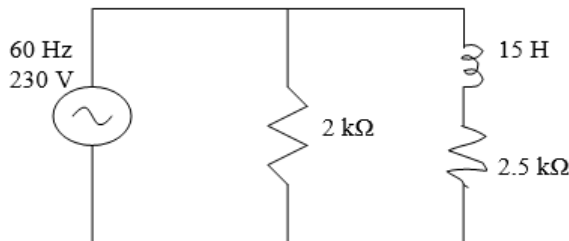
145. In open channel construction, how many 40-kg bags of cement are needed for a 10-km lateral channel if it requires 0.6 cubic meter concrete per linear meter? Use 5% safety factor.

Solution:

$$\text{Bags} = (\text{Safety factor}) \left(\frac{\text{Bags cement}}{\text{m}^3} \right) (\text{concrete volume})$$

$$\text{Bags} = (1.05) \left(\frac{9 \text{ bags}}{\text{m}^3} \right) \left(\frac{0.6 \text{ m}^3}{\text{m}} \right) (10,000 \text{ m}) = 56,700 \text{ bags}$$

From numbers 146 to 148, consider the circuit below:



146. What is the total impedance of the circuit?

Solution:

$$X_L = 2\pi fL = 2\pi(60 \text{ Hz})(15 \text{ H}) = (5.7 \text{ k}\Omega)j$$

$$Z_L = \sqrt{(2.5 \text{ k}\Omega)^2 + (5.7 \text{ k}\Omega)^2} = 6.22$$

$$\theta_1 = \arctan \left(\frac{5.7 \text{ k}\Omega}{2.5 \text{ k}\Omega} \right) = 66.32^\circ$$

$$Z_T = \frac{R_1 Z_L}{(R_1 + R_2) + X_L} = \frac{(2 \text{ k}\Omega)(6.22 \text{ k}\Omega)}{4.5 \text{ k}\Omega + (5.7 \text{ k}\Omega)j} = \frac{12.44 \text{ k}\Omega}{\sqrt{(4.5 \text{ k}\Omega)^2 + (5.7 \text{ k}\Omega)^2}} = 1.71 \text{ k}\Omega$$

$$\theta_2 = \arctan \left(\frac{5.7 \text{ k}\Omega}{4.5 \text{ k}\Omega} \right) = 51.70^\circ$$

147. What is the total current flowing through the circuit?

Solution:

$$I_T = \frac{E_T}{Z_T} = \frac{230 \text{ V}}{1.71 \text{ k}\Omega} = 134.5 \text{ mA}$$

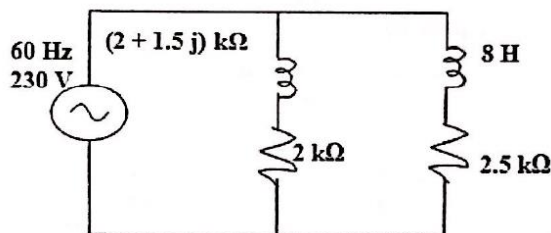
148. The phase shift angle of the electrical circuit is

Solution:

$$\theta = \theta_1 - \theta_2 = 66.32^\circ - 51.70^\circ$$

$$\theta = 14.60^\circ \approx 15^\circ$$

From numbers 149 to 151, consider the circuit below:



149. What is the total impedance of the circuit?

Solution:

$$Z_T = \frac{1}{\frac{1}{\sqrt{(2k\Omega)^2 + (\sqrt{(2k\Omega)^2 + (1.5k\Omega)^2})^2}} + \frac{1}{\sqrt{(2.5k\Omega)^2 + (3.015k\Omega)^2}}} = 1.76k\Omega$$

150. What is the total current flowing through the circuit?

Solution:

$$I_T = \frac{E_T}{Z_T} = \frac{230V}{1.76k\Omega} = 130.60mA$$

Alternative:

$$I_1 = \frac{E}{Z_1} = \frac{230\angle 0^\circ}{3.20k\Omega\angle 51.34^\circ} = 71.875mA\angle -51.34^\circ = 44.9 - 56.12j$$

$$I_2 = \frac{E}{Z_2} = \frac{230\angle 0^\circ}{3.92k\Omega\angle 50.38^\circ} = 58.70mA\angle -50.38^\circ = 37.43 - 45.22j$$

$$I_T = I_1 + I_2 = 82.33 - 101.34j = 130.60\angle -50.91^\circ$$

151. What is the power factor?

Solution:

$$Z = \frac{230\angle 0^\circ}{130.6k\Omega\angle 50.91^\circ} = 1.76\angle -50.91^\circ$$

$$\text{Power Factor} = \cos\theta = \cos(-50.91^\circ) = 0.63$$

152. How many truckloads of sand are needed for 25-km main irrigation canal requiring 3.5 cubic meters concrete per linear meter? 1 truck load is 2.5 cubic meters. Use 5% safety factor.

Solution:

$$\text{Truck loads} = (\text{Safety factor}) \left(\frac{\text{m}^3 \text{ sand}}{\text{m}^3 \text{ concrete}} \right) \left(\frac{\text{trucks}}{\text{m}^3} \right) (\text{concrete volume})$$

$$\text{Truck loads} = (1.05) \left(\frac{0.5\text{m}^3 \text{ sand}}{\text{m}^3 \text{ concrete}} \right) \left(\frac{1\text{truck}}{2.5\text{m}^3} \right) \left(\frac{3.5\text{m}^3 \text{ concrete}}{\text{m}} \right) (25,000\text{m})$$

$$\text{Truck loads} = 18,375 \text{ truck loads}$$

153. Estimate the truck loads of gravel needed for an 8-km main irrigation canal requiring 4 cubic meters concrete per linear meter. 1 truck load is 2.5 cubic meters. Use 6% safety factor.

Solution:

$$\text{Truck loads} = (\text{Safety factor}) \left(\frac{\text{m}^3 \text{ gravel}}{\text{m}^3 \text{ concrete}} \right) \left(\frac{\text{trucks}}{\text{m}^3} \right) (\text{concrete volume})$$

$$\text{Truck loads} = (1.06) \left(\frac{1\text{m}^3 \text{ gravel}}{\text{m}^3 \text{ concrete}} \right) \left(\frac{1\text{truck}}{2.5\text{m}^3} \right) \left(\frac{4\text{m}^3 \text{ concrete}}{\text{m}} \right) (8,000\text{m})$$

$$\text{Truck loads} = 13,568 \text{ truck loads}$$

154. How much should be budgeted for cement for 20-km irrigation canal if it requires half cubic meter concrete per linear meter and cement costs Php200/bag? Use 5% safety factor.

Solution:

$$\text{Budget} = (\text{Safety factor}) \left(\frac{\text{Bag scement}}{\text{m}^3 \text{ concrete}} \right) (\text{Concrete volume})(\text{Unit cost})$$

$$\text{Budget} = (1.05) \left(\frac{9\text{bags}}{\text{m}^3 \text{ concrete}} \right) \left(\frac{0.5\text{m}^3 \text{ concrete}}{\text{m}} \right) (20,000 \text{ m}) \left(\frac{\text{Php}200}{\text{bag}} \right)$$

$$\text{Budget} = \text{P}18,900,000$$

155. Give cement budget estimate for 20-km canal requiring 2/3 cubic meter concrete per linear meter. Cement costs Php200/bag while transport costs Php 10/bag. Use 5% safety factor.

Solution:

$$\text{Budget} = (\text{Safety factor}) \left(\frac{\text{Bag scement}}{\text{m}^3 \text{concrete}} \right) (\text{Concrete volume})(\text{Unit cost})$$

$$\text{Budget} = (1.05) \left(\frac{9\text{bags}}{\text{m}^3 \text{concrete}} \right) \left(\frac{0.667\text{m}^3 \text{concrete}}{\text{m}} \right) (20,000 \text{ m}) \left(\frac{\text{Php}200}{\text{bag}} + \frac{\text{Php}10}{\text{bag}} \right)$$

$$\text{Budget} = \text{Php } 26,473,230$$

156. A gravity dam weighing 65,000 tons is to be constructed with length along dam axis of 1 km and base of 100 meters. What should be the minimum soil bearing stress? Use factor of safety against sinking of 2.

Solution:

$$(\text{Sinking Factor of safety}) = \frac{\text{Min. defensive bearing stress}}{\text{Offensive bearing stress}}$$

$$\text{Offensive bearing stress} = \frac{\text{Weight}}{\text{Area}}$$

$$\text{Min. defensive bearing stress} = (2) \left(\frac{65,000 \times 1000}{1,000 \text{m} \times 100 \text{m}} \right)$$

$$\text{Min. defensive bearing stress} = \frac{1.3 \text{tons}}{\text{m}^2}$$

157. A 1-meter thick trapezoidal section of a concrete dam has resisting moment about the dam toe of 300,000 ton-meters. The water exerts 4,050 tons horizontally at 30 meters above the dam toe. What is the factor of safety against overturning

Solution:

$$\text{Overturning Factor of Safety} = \frac{\text{Resisting moment}}{\text{Overturning moment}}$$

$$\text{Overturning Factor of Safety} = \frac{300,000 \text{ton} - \text{m}}{4,050 \text{ton} \times 30 \text{m}}$$

$$\text{Overturning Factor of Safety} = 2.47$$

158. A concrete dam with 50,000m² submerged vertical upstream wall is to be constructed against a 546 tons/m² average water pressure. Coefficient of friction at dam base is 0.8. What should be the minimum dam weight? Use factor of safety against sliding of 2.

Solution:

$$\text{Sliding Factor of Safety} = \frac{\text{Min. defensive horizontal force}}{\text{Offensive horizontal force}}$$

$$\text{Min. defensive horizontal force} = \frac{\left(\frac{546 \text{tons}}{\text{m}^2} \right) (50,000 \text{m}^2) (2)}{0.8}$$

$$\text{Min. defensive horizontal force} = 68,250,000 \text{tons}$$

159. Determine the daily biogas consumption in cubic meters of an industrial boiler using 10 units of 50-cm gas burner 8 hours a day. The consumption of each gas burner is 1.2m³/hour.

Solution:

$$C = \text{NBT}$$

where C = Consumption, N = No. of units, B = biogas consumption of device, T = Time

$$C = (10 \text{units}) \left(\frac{1.2 \text{m}^3}{\text{hours}} \right) \left(\frac{8 \text{hrs}}{\text{day}} \right) = 96 \text{m}^3$$

160. Estimate the biogas production in cubic meters per day from a large balloon type digester using 1:1 water-manure ratio. The source farm has 1,000 heads of porkers of mixed ages. Use a retention period of 30 days. The mean daily manure production of porkers is 2.2 kg/head while the specific gas production for 30-day manure retention period is 0.063 m³/kg.

Solution:

$$P = NMG$$

Where:

P = biogas production potential, N = No. of heads, M = manure production,

G = Specific gas production of the manure for a specific retention period

$$P = NMG = (1,000\text{heads}) \left(\frac{2.2\text{kg}}{\text{head}} \right) \left(\frac{0.063\text{m}^3}{\text{kg}} \right) = 138.6\text{m}^3$$

161. How many heads of breeding cattle are needed as source of manure to generate 75m³ of biogas a day if the retention period is 25 days and water-manure ratio is 1:1? The mean daily manure production of breeding cattle is 13 kg/head while its specific gas production for 25-day manure retention period is half of that of chicken dung.

Solution:

$$\begin{aligned} G &= 0.5 \times G \text{ for chicken dung at 25-day retention period} \\ &= 0.5 \times 0.06\text{m}^3/\text{kg} \\ &= 0.03\text{m}^3/\text{kg} \end{aligned}$$

$$N = \frac{P}{MG} = \frac{(75\text{m}^3)}{\left(\frac{13\text{kg}}{\text{head}} \right) \left(\frac{0.03\text{m}^3}{\text{kg}} \right)} = 1923 \text{ heads}$$

162. In the Clean Development Mechanism (CDM), how much will the project owner of a new covered lagoon bio-digester receive annually from CDM fund for the first 10 years if his bio-digester produces 7,000 tons of methane and 3,000 ton of carbon dioxide per year? Assume that the current carbon credit costs US\$10 per ton of carbon equivalent.

Solution:

Considering that methane is 21 times more destructive than carbon,

$$\text{Amount} = (\text{Carbon cost per tonne}) \left[\left(\frac{\text{tonnes carbon}}{\text{year}} \right) + 21 \left(\frac{\text{tonnes of methane}}{\text{year}} \right) \right]$$

$$\text{Amount} = \left(\frac{\$10}{\text{ton}} \right) \left[\left(\frac{3,000\text{tons}}{\text{year}} \right) + 21 \left(\frac{7,000\text{tons}}{\text{year}} \right) \right]$$

$$\text{Amount} = \frac{\$1,500,000}{\text{year}}$$

163. A farm residence has 37 general purpose outlets. How many 20-ampere branch circuits are needed?

Solution:

$$N = \frac{37}{8} = 4.6 \approx 5$$

164. A farm house has a total electrical load of 15,000 watts. What is the total demand load?

Solution:

$$\text{Total Demand} = (3,000) + (15,000 - 3,000)(0.35) = 7,200\text{Watts}$$

165. A trapezoidal concrete gravity dam has inclined upstream wall. Resisting moment relative to the dam toe is 263,953.2 tonne-meters while overturning moment is 96,233.3 tonne-meters per meter of dam with. What is the safety factor against overturning? It is safe?

Solution:

$$\text{Overturing Factor of Safety} = \frac{\text{Resisting moment}}{\text{Overturning moment}}$$

$$\text{Overturing Factor of Safety} = \left(\frac{263,953.2 \text{ ton} - \text{m}}{96,233.3 \text{ ton} - \text{m}} \right) = 2.74$$

166. A trapezoidal concrete gravity dam has vertical upstream wall. Water depth in the upstream side is 50 meters above the dam base. Base of dam is 78 meters along water flow. What is the uplift force of seepage water below the dam per meter of dam width?

Solution:

$$\text{Uplift Force} = 0.5\rho_w HB = 0.5 \left(\frac{1000\text{kg}}{\text{m}^3} \right) (50\text{m})(78\text{m}) = \frac{1950\text{tons}}{\text{m}}$$

167. A trapezoidal concrete gravity dam has vertical upstream wall. Water depth in the upstream side is 50 meters above the dam base. Freeboard of 15% based on design depth. What is the force of water exerted against dam per meter of dam width?

Solution:

$$F = 0.5\rho_w H^2 = 0.5 \left(\frac{1000\text{kg}}{\text{m}^3} \right) (50\text{m})^2 = \frac{1,250\text{tons}}{\text{m}}$$

168. The farm's class AA slaughterhouse has installed a 15m³ water tank and a 5m³ scalding tank. Every 5:00 to 5:30 A.M., two-thirds of the capacity of the water tank and one-fifth of the capacity of the scalding tank need to be filled. Pumping depth is 50 m. A single phase 240-volt water pump with electric motor needs to be purchased. What should be the horsepower rating of the motor? Assume 66% pump efficiency and 15% friction head.

Solution:

$$\text{Total water volume needed} = \frac{2}{3}(\text{water tank capacity}) + \frac{1}{5}(\text{scalding tank capacity})$$

$$\text{Total water volume needed} = \frac{2}{3}(15\text{m}^3) + \frac{1}{5}(5\text{m}^3) = 11\text{m}^3$$

$$\text{HP} = \frac{\left(\frac{11\text{m}^3}{30\text{min}} \right) (50\text{m} + 0.15(50\text{m})) \left(\frac{1000\text{kg}}{\text{m}^3} \right) \left(\frac{1\text{min}}{60\text{s}} \right)}{\frac{76.2\text{kg} - \text{m}}{\text{s}}} = 7\text{hp use } 7.5\text{hp}$$

169. A 480-watt farm-to-market road street light is 48 meters away from 240-volt AC electricity source. A 5.26 square mm copper conductor is used. What is the estimated voltage drop? Assume 100% power factor.

Solution:

$$I = \frac{P}{V} = \frac{480\text{W}}{240\text{V}} = 2\text{A}$$

$$\text{For } 5.26\text{mm}^2 \text{ copper wire, single phase \& 100\% power factor, voltage drop is } \frac{0.00761\text{V}}{\text{m-A}}$$

$$\text{Voltage drop} = (\text{Amperage})(\text{Total Length of Wire in meters})(\text{Unit Voltage Drop})$$

$$\text{Voltage drop} = (2\text{A})(48\text{m} \times 2) \left(\frac{0.00761\text{V}}{\text{m-A}} \right) = 1.46\text{Volts}$$

170. A 4.6 KVA 240-volt farmshop bar cutter is to be provided with a copper ground wire. What should be the AWG number of the ground wire?

Solution:

$$I = \frac{4600\text{VA}}{240\text{V}} = 19.17\text{A}$$

For 16 – 20A rating, the required copper ground wire size is AWG 12 (PEC)

171. What is the true power in kW of an AC farm circuit having a 5-hp pump motor? Supply voltage is between 230-240 volts and the phase shift between voltage and current is 20 degrees.

Solution:

$$I = \frac{\text{power considering 30\% standard motor power surge}}{\text{safe voltage level}} = \frac{((5\text{hp})(746\text{W})(1.3))}{230\text{V}}$$

$I = 21.08\text{A}$, the lower voltage of 230 V is used to avoid overcurrent

$$\text{True P} = (\text{Amperage})(\text{Safe Voltage})(\text{Power factor}) = (21.08\text{A})(230\text{V})(\cos 20^\circ)$$

$$\text{True P} = 4556\text{W or } 4.556\text{kW}$$

172. A broiler house has twenty 20-watt LED AC florescent lamps, a 5-hp pump motor and a 700-watt chick heaters made from incandescent lamps. Estimated wire temperature will not exceed 60 degrees Celsius. Supply voltage fluctuates between 210 and 240. What should be the size of copper drop wire in square mm?

Solution:

$$P = \text{LED power} + \text{motor power considering 30\% standard motor power surge} + \text{incubator power}$$

$$P = (20)(20\text{W}) + (1.3)(5\text{hp})(746\text{W}) + (700\text{W})$$

$$P = 5949\text{W or } 5.949\text{kW}$$

$$I = \frac{P}{V} = \frac{5949\text{W}}{210\text{V}} = 28.3\text{A, minimum voltage of 210 is used to avoid overcurrent}$$

Next higher amperage of circuit breaker available in the market is 30A (max. current)

For 21 – 30A max. current & 60°C wire temp. the required size is 5.26mm²(PEC)

173. A layer house has 220-volt supply, twenty five 20-watt LED AC florescent lamps, and a single phase 6-hp pump motor and scheduled to purchase a 700-watt incubator. Average ambient temperature is 29 °C. What should be the circuit breaker amperage?

Solution:

$$P = \text{LED power} + \text{motor power considering 30\% standard motor power surge} + \text{incubator power}$$

$$P = 25(20\text{W}) + 6 \text{ hp } (746\text{W})(1.3) + 700\text{W}$$

$$P = 7,018.8\text{W or } 7.0188\text{kW}$$

$$I = \frac{P}{V} = \frac{7,018.8\text{W}}{220\text{V}} = 31.90\text{A, next higher amperage available in the market is 60A}$$

174. A 500-watt farm-to-market road street light is to be placed 99.5 meters away from 230-volt source. A 5.26 square mm copper conductor is planned to be used. Voltage drop is to be determined if it is within the recommended maximum of 2% otherwise a bigger conductor will be used. What is the percent voltage drop? Assume 100% power factor.

Solution:

$$I = \frac{P}{V} = \frac{5000\text{W}}{230\text{V}} = 2.2\text{A}$$

$$\text{For } 5.26\text{mm}^2 \text{ copper wire, single phase \& 100\% power factor, voltage drop is } \frac{0.00761\text{V}}{\text{m} - \text{A}}$$

$$\text{Voltage drop} = (\text{Amperage})(\text{Total Length of Wire in meters})(\text{Unit Voltage Drop})$$

$$\text{Voltage drop} = (2.2\text{A})(99.5\text{m} \times 2) \left(\frac{0.00761\text{V}}{\text{m} - \text{A}} \right) = 3.33\text{V}$$

$$\% \text{Voltage Drop} = \left(\frac{\text{Voltage Drop}}{\text{Source Voltage}} \right) 100\%$$

$$\text{Voltage drop} = \left(\frac{3.33\text{V}}{230\text{V}} \right) \times 100 = 1.45\%$$

175. A proposed layer house has 220-volt supply, thirty two 20-watt LED AC florescent lamps, and a 6-hp pump motor and scheduled to purchase a 0.5-kw incubator. Wire temperature cannot exceed 60 °C. What should be the wire gauge (AWG) number of copper drop wire?

Solution:

$P = \text{LED power} + \text{motor power considering 30\% standard motor power surge} + \text{incubator power}$

$$P = 32(20W) + 6\text{hp}(746W)(1.3) + 500W$$

$$P = 6,958.8W$$

$$I = \frac{P}{V} = \frac{6,958.8W}{220V} = 31.63A$$

Exceeds 30A, 30Amp market – available circuit breaker cannot be used

For 31.63A, the next higher amp. of market available circuit breaker is 60A (Allowable)

For 56 – 70A allowable current & 60°C copper wire temp. wire gauge is No. 4 (PEC)

176. Compute the heat to be removed from 110kg of lean beef if it were to be cooled from 20°C to 4°C, after which it is frozen and cooled to -18°C. Specific heat of beef above freezing is 3.23kJ/(kg-°C) and below freezing is 1.68kJ/(kg-°C). Freezing point is -2.2°C and latent heat of fusion is 233kJ/kg.

Solution:

$$Q_T = Q_1 + Q_2 + Q_3 + Q_n$$

$$Q_1 = (110\text{kg}) \left(\frac{3.23\text{kJ}}{\text{kg} - ^\circ\text{C}} \right) (20^\circ\text{C} - (-2.2^\circ\text{C})) = 7885.66\text{kJ}$$

$$Q_2 = (110\text{kg}) \left(\frac{233\text{kJ}}{\text{kg}} \right) = 25,630\text{kJ}$$

$$Q_3 = (110\text{kg}) \left(\frac{1.68\text{kJ}}{\text{kg} - ^\circ\text{C}} \right) (-2.2^\circ\text{C} - (-18^\circ\text{C})) = 2,919.84\text{kJ}$$

$$Q_T = 7885.66\text{kJ} + 25,630\text{kJ} + 2,919.84\text{kJ}$$

$$Q_T = 36,435.5\text{kJ}$$

177. What is the heat rate input to a water heater if 0.4kg/s of water enters 82°C and leaves at 93°C? The water has approximate specific heat of 4.19kJ/kg.°K

Solution:

$$Q = \dot{m}C_p\Delta T = \left(\frac{0.4\text{kg}}{\text{s}} \right) \left(\frac{4.19\text{kJ}}{\text{kg}^\circ\text{K}} \right) (93^\circ\text{C} - 82^\circ\text{C})$$

$$Q = \frac{18.436\text{kJ}}{\text{s}} = 18.436\text{kW}$$

178. A flow rate of 0.06kg/s of water enters a boiler at 90°C, at which temperature the enthalpy is 379.9kJ/kg. The water leaves as a steam and at 100°C (2,676kJ/kg). What is the rate of heat transfer added to this boiler?

Solution:

$$Q = \dot{m}(h_2 - h_1) = \left(\frac{0.06\text{kg}}{\text{s}} \right) \left(\frac{2,676\text{kJ}}{\text{kg}} - \frac{379.9\text{kJ}}{\text{kg}} \right) = \frac{137.80\text{kJ}}{\text{s}} \text{ or } 137.80\text{kW}$$

179. Water is pumped from a driller in the basement, where $z = 0\text{m}$ to a cooling coil located on the top floor of a building, where $z = 80\text{m}$. What is the minimum pressure rise the pump must be capable of providing if the temperature of the water is 4°C? The density of water is 1000kg/m³.

Solution:

$$\frac{\text{kg} - \text{m}}{\text{s}^2} = \text{N}; \frac{\text{N}}{\text{m}^2} = \text{Pascal}$$

$$P = \rho_w g h = \left(\frac{1000\text{kg}}{\text{m}^3} \right) \left(\frac{9.81\text{m}}{\text{s}^2} \right) (80\text{m}) = 784.8\text{kPa}$$

180. Air flowing at the rate of 2.5kg/s is heated in a heat exchanger from -10°C to 30°C. What is the rate of heat transfer?

Solution:

$$Q = \dot{m}C_p\Delta T = \left(\frac{2.5\text{kg}}{\text{s}}\right)\left(\frac{1.0035\text{kJ}}{\text{kg} - ^\circ\text{C}}\right)(30^\circ\text{C} - (-10)^\circ\text{C})$$

$$Q = \frac{100.35\text{kJ}}{\text{s}} \text{ or } 100.35\text{kW}$$

For problems 181 and 182:

The masonry wall of a building consist of an outer layer of facing brick ($k_b = \frac{1.32\text{W}}{\text{m-k}}$), 10cm thick followed by a 15cm thick of common brick ($k_{cb} = \frac{0.69\text{W}}{\text{m-k}}$), followed by 1.25cm thick layer of gypsum plaster ($k_b = \frac{0.48\text{W}}{\text{m-k}}$). An outside coefficient of ($h_o = \frac{30\text{W}}{\text{m}^2\text{-k}}$), may be expected and a coefficient of ($h_i = \frac{8\text{W}}{\text{m}^2\text{-k}}$), is reasonable value to use for the inner surface of ventilated room.

181. What will be the rate of heat gain per unit area when the outside air is 35°C and the inside air is conditioned to 22°C ?

Solution:

$$\frac{Q}{A} = \frac{\Delta T}{\frac{1}{h_o} + \frac{x_1}{k_1} + \frac{x_2}{k_2} + \frac{x_3}{k_3} + \frac{1}{h_i}} = \frac{35^\circ\text{C} - 22^\circ\text{C}}{\frac{1}{30} + \frac{0.1}{1.32} + \frac{0.15}{0.69} + \frac{0.0125}{0.48} + \frac{1}{8}}$$

$$\frac{Q}{A} = \frac{27.22\text{W}}{\text{m}^2}$$

182. What will be the temperature of the exposed surface of the plate?

$$\frac{Q}{A} = h_1(T_2 - T_1)$$

$$T_2 = \frac{\left(\frac{27.22\text{W}}{\text{m}^2}\right)}{\frac{8\text{W}}{\text{m}^2\text{-k}}} + 22 = 25.4^\circ\text{C}$$

183. Two infinite black plates at 800°C and 300°C exchange heat by radiation. The heat transfer rate per unit area is?

Solution:

$$\frac{q}{A} = \varepsilon\sigma T^4 = (1)\left(\frac{5.669 \times 10^{-8}\text{W}}{\text{m}^2\text{-K}}\right)((800 + 273)^4 - (300 + 273)^4)$$

$$\frac{q}{A} = \frac{69\text{kW}}{\text{m}^2}$$

184. Consider a 10cm wall made of polyurethane board ($k = \frac{0.0260\text{W}}{\text{m-}^\circ\text{C}}$) which is exposed to still air ($h_i = \frac{9.37\text{W}}{\text{m}^2\text{-}^\circ\text{C}}$) on the inside and outer surface exposed to 24km/hr. wind ($h_o = \frac{34\text{W}}{\text{m}^2\text{-}^\circ\text{C}}$).

Determine the overall coefficient of heat transfer in $\frac{\text{W}}{\text{m}^2\text{-}^\circ\text{C}}$

Solution:

$$U = \frac{1}{\frac{1}{h_i} + \frac{x}{k} + \frac{1}{h_o}} = \frac{1}{\frac{1}{\frac{9.37\text{W}}{\text{m}^2\text{-}^\circ\text{C}}} + \frac{0.1\text{m}}{\frac{0.0260\text{W}}{\text{m-}^\circ\text{C}}} + \frac{1}{\frac{34\text{W}}{\text{m}^2\text{-}^\circ\text{C}}}}$$

$$U = \frac{0.25\text{W}}{\text{m}^2\text{-}^\circ\text{C}}$$

185. A shrimp harvested in a pond situated in Bohol are packed in insulated sheet metal containers and mixed with crushed ice while in transit to the shrimp processing plant in Manila. Initial temperature of the product is 25°C and it is desired to cool the shrimp at 0°C during

shipment. Estimate the product load in kJ/kg, for shrimp highest freezing temperature is -2.2°C . C_p Above freezing is $\frac{3.38\text{kJ}}{\text{kg}^{\circ}\text{K}}$, C_p below freezing is $\frac{1.79\text{kJ}}{\text{kg}^{\circ}\text{K}}$ and latent heat of fusion is $\frac{254\text{kJ}}{\text{kg}}$.

Solution:

$$\frac{Q}{m} = C_p \Delta T + \Delta h + C_p \Delta T = \left(\frac{3.38\text{kJ}}{\text{kg}^{\circ}\text{K}} \right) (25^{\circ}\text{C} - 0^{\circ}\text{C}) + \left(\frac{254\text{kJ}}{\text{kg}} \right) + \left(\frac{1.79\text{kJ}}{\text{kg}^{\circ}\text{K}} \right) (0^{\circ}\text{C} - (-2.2^{\circ}\text{C}))$$

$$\frac{Q}{m} = \frac{342.438\text{kJ}}{\text{kg}}$$

186. Oven method is primarily method of moisture determination. If after 3 days at 105°C the initial 25g yields a final weight of 15g, what is the moisture content in percent wet basis?

Solution:

$$\text{MC}_{\text{wb}} = \frac{5\text{g}}{25\text{g}} = 25\%$$

187. Paddy grain with 14% MC wet basis has approximately _____ MC dry basis?

Solution:

$$\text{MC}_{\text{db}} = \frac{\text{MC}_{\text{wb}}}{100 - \text{MC}_{\text{wb}}} = \frac{0.14}{1 - 0.14} = 16.30\%$$

or

$$\text{MC}_{\text{db}} = \frac{14}{86} = 16.30\%$$

188. Determine the quantity of heat evolved in transit by 1000 tons of given banana ($\text{HV} = \frac{7,000\text{BTU}}{\text{ton-day}}$) when it is shipped at 54 degrees Fahrenheit to a destination requiring 5 days of travel time. If the other heat loads constitutes 50% of the product load, compute for the tons of refrigeration?

Solution:

$$(1000\text{tons}) \left(\frac{7,000\text{BTU}}{\text{ton-day}} \right) \left(\frac{1\text{day}}{24\text{hrs}} \right) \left(\frac{1\text{hr}}{3600\text{s}} \right) \left(\frac{1.055\text{kJ}}{\text{BTU}} \right) = 85.50\text{kJ or } 85.50\text{kW}$$

$$(85.50\text{kW})(1.50) \left(\frac{1\text{TR}}{3.52\text{kW}} \right) = 36.42\text{TR}$$

189. A Carnot refrigerator absorbing heat at -10°C requires 1kW of work for each 6kW of heat absorbed. The temperature (K) of heat rejection is?

Solution:

$$W = 6\text{kW}; Q = 1\text{kW}; e = \frac{W}{Q} = \frac{6}{1} = 6$$

$$e = \frac{T_1}{T_2 - T_1}$$

$$T_2 = \frac{T_1}{e} + T_1 = \frac{-10^{\circ}\text{C} + 273}{6} + (-10 + 273) = 307.88^{\circ}\text{K}$$

190. One face of a copper plate 3cm thick is maintained at 400°C and the other face is maintained at 100°C . How much heat is transferred through the plate? Assume $k = \frac{370\text{W}}{\text{m}^{\circ}\text{C}}$ and $\rho = \frac{8.954\text{kg}}{\text{m}^3}$.

Solution:

$$\frac{Q}{A} = \frac{k \Delta T}{x} = \frac{\left(\frac{370\text{W}}{\text{m}^{\circ}\text{C}} \right) (400^{\circ}\text{C} - 100^{\circ}\text{C})}{0.03\text{m}} = 3700\text{kW}$$

191. Air at 20°C blows over a hot plate $50 \times 75\text{cm}$ maintained at 250°C . The convection heat transfer coefficient is $\frac{25\text{W}}{\text{m}^2^{\circ}\text{K}}$. The heat transfer is?

Solution:

$$Q = kA\Delta T = \left(\frac{25W}{m^2 - ^\circ K}\right)(0.5m \times 0.75m)(250^\circ C - 20^\circ C) = 2,156.25W$$

192. How much weight (24% wet basis) palay does a trader need to buy if he wants to get 2 tons of dried (14% wet basis) grain?

Solution:

$$W_i = \frac{(2000kg)(1 - 0.14)}{1 - 0.24} = 2,263.16kg$$

193. How many kilograms of rice hulls are needed to dry a 1 ton of Palay from 24% to 14% wet basis. Assume latent heat of evaporation is $\frac{2500kJ}{kg}$ and efficiency is 100%.

Solution:

Note: Heating Value of rice hull is $\frac{14,000kJ}{kg}$

$$W_{H_2O} = 1000kg - \frac{(1000kg)(1 - 0.24)}{1 - 0.14} = 116.28kg$$

$$W_{RH} = \frac{W_w LHE_w}{HV_{RH}} = \frac{(116.28kg) \left(\frac{2,500kJ}{kg}\right)}{\frac{14,000kJ}{kg}} = 20.76kg$$

194. A 10 ohms resistor is connected in parallel with a capacitive reactance of 24 ohms. What is the total impedance?

Solution:

$$I = \sqrt{I^2 + I_C^2}$$

$$\frac{V}{R} = \sqrt{\left(\frac{V}{R}\right)^2 + \left(\frac{V}{R}\right)^2} = \sqrt{\left(\frac{V}{10\Omega}\right)^2 + \left(\frac{V}{24\Omega}\right)^2} = \sqrt{\frac{V^2}{100\Omega} + \frac{V^2}{576\Omega}} = \sqrt{\frac{676V^2}{57600}} = 0.108V$$

$$\frac{1}{0.108} = R; R = 9.26\Omega$$

195. An automobile tire is inflated to 32 psig at 50°F. After being driven, the temperature rises to 75°F. Assuming that the volume remains constant, the final gauge pressure is?

Solution:

$$\frac{T_1}{P_1} = \frac{T_2}{P_2}$$

$$P_2 = \frac{P_1 T_2}{T_1} = \frac{(32psig + 14.70)(75^\circ F + 460)}{(50^\circ F + 460)} = 48.99psig - 14.7 = 34.30psig$$

196. What is the temperature of 2 liters of water at 30°C after 500 Cal of heat have been added to it?

Solution:

$$2 \text{ liters} = 0.002m^3$$

$$500Cal \left(\frac{1J}{0.2389Cal}\right) = 2,091.92kJ = Q$$

$$Q = mC_p\Delta T = \dot{v}_w C_p\Delta T = (0.002m^3) \left(\frac{1,000kg}{m^3}\right) \left(\frac{4.187kJ}{kg - ^\circ C}\right) (T_2 - 30^\circ C)$$

$$T_2 = \frac{2.092kJ}{(0.002m^3) \left(\frac{1,000kg}{m^3}\right) \left(\frac{4.187kJ}{kg - ^\circ C}\right)} + 30^\circ C$$

$$T_2 = 30.25^\circ\text{C}$$

197. The maximum thermal efficiency possible for a power cycle operating between 1200°F and 225°F is?

Solution:

$$\eta_{\text{th}} = \frac{T_H - T_L}{T_H} = \frac{(1200^\circ\text{F} + 460) - (225^\circ\text{F} + 460)}{1200^\circ\text{F} + 460} = 58.73\%$$

198. A rigid tank contains 5kg of an ideal gas at 4atm and 40°C . Now a valve is opened and half of mass of the gas is allowed to escape. If the final pressure in the tank is 1.5atm, the final temperature in the tank is?

Solution:

$$\frac{m_1 T_1}{P_1} = \frac{m_2 T_2}{P_2}; T_2 = \frac{m_1 T_1 P_2}{m_2 P_1}$$

$$T_2 = \frac{(5\text{kg})(40^\circ\text{C} + 273)(1.5\text{atm} \times 101325\text{Pa})}{(2.5\text{kg})(4\text{atm} \times 101325\text{Pa})} = 234.75^\circ\text{F}$$

199. The atmospheric pressure at the top and bottom of a building are read by a barometer 96kPa and 98kPa. If the density of air is $1.2\text{kg}/\text{m}^3$, the height of the building is?

Solution:

$$\Delta P = \rho_w g h$$

$$h = \frac{98\text{kPa} - 96\text{kPa}}{\left(\frac{1.2\text{kg}}{\text{m}^3}\right)\left(\frac{9.81\text{m}}{\text{s}^2}\right)} = 0.170\text{km or } 170\text{m}$$

200. A 5m^3 rigid steel contains steam at 20MPa and 400°C . The mass of the steam is?

Solution:

Gas constant for water vapor or steam is $462\text{J}/\text{kg}\cdot\text{K}$

$$Pv = mRT; m = \frac{Pv}{RT}$$

$$m = \frac{\left(\frac{20 \times 10^6 \text{N}}{\text{m}^2}\right)(5\text{m}^3)}{\left(\frac{461.5\text{J}}{\text{kg}\cdot^\circ\text{K}}\right)(400^\circ\text{C} + 273)} = 322\text{kg}$$

201. A piston cylinder initially contains 0.4m^3 air at 100kPa and 80°C . The air is compressed to 0.1m^3 in such a way that the temperature inside the cylinder remains constant. Determine the work done during the process?

Solution:

Work done Formula for Constant Temperature:

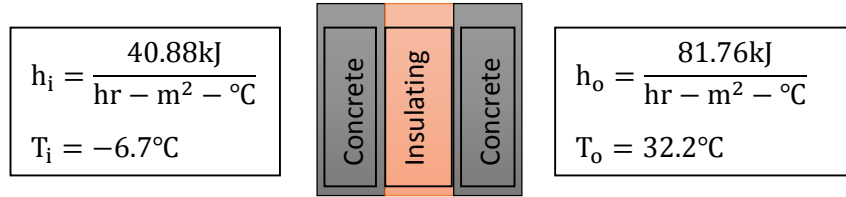
$$W = P_1 V_1 \ln\left(\frac{V_2}{V_1}\right) = (100\text{kPa})(0.4\text{m}^3) \ln\left(\frac{0.1\text{m}^3}{0.4\text{m}^3}\right) = 55.45\text{kJ}$$

202. The wall of a cold storage plant are composed of an insulating material ($k_{\text{in}} = \frac{0.2336\text{kJ}}{\text{hr}\cdot\text{m}\cdot^\circ\text{C}}$)

10.16cm thick held between two layers of concrete of ($k_{\text{C}} = \frac{3.7382\text{kJ}}{\text{hr}\cdot\text{m}\cdot^\circ\text{C}}$) each 10.16cm thick. The film coefficients are ($h_o = \frac{81.76\text{kJ}}{\text{hr}\cdot\text{m}^2\cdot^\circ\text{C}}$) on the outside and ($h_i = \frac{40.88\text{kJ}}{\text{hr}\cdot\text{m}^2\cdot^\circ\text{C}}$) on inside. Cold storage temperature is -67°C and ambient temperature is 32.2°C . Determine the heat transmitted in kW through the area of 55.74m^2 .

Solution:

$$Q = \frac{A\Delta T}{U} = \frac{A(T_2 - T_1)}{\frac{1}{h_o} + \frac{x_1}{k_1} + \frac{x_2}{k_2} + \frac{1}{h_i}}$$



$$Q = \frac{(55.74 \text{ m}^2)(32.2^\circ\text{C} + 6.7^\circ\text{C})}{\frac{1}{\frac{81.76 \text{ kJ}}{\text{hr} - \text{m}^2 - ^\circ\text{C}}} + \frac{0.1016 \text{ m}}{\frac{0.2336 \text{ kJ}}{\text{hr} - \text{m} - ^\circ\text{C}}} + \frac{2(0.1016 \text{ m})}{\frac{3.7382 \text{ kJ}}{\text{hr} - \text{m} - ^\circ\text{C}}} + \frac{1}{\frac{40.88 \text{ kJ}}{\text{hr} - \text{m}^2 - ^\circ\text{C}}}}$$

$$Q = \frac{4122.36 \text{ kJ}}{\text{hr}} \left(\frac{1 \text{ hr}}{3600 \text{ s}} \right)$$

$$Q = \frac{1.145 \text{ kJ}}{\text{s}} \text{ or } 1.145 \text{ kW}$$

203. A 12 inch thick furnace wall with a dimension of 5m by 2m has a temperature difference of 60°C . The wall has thermal conductivity of $\frac{0.75 \text{ BTU}}{\text{hr} - \text{ft} - ^\circ\text{F}}$. Calculate the heat transmitted across the wall.

Solution:

$$Q = \frac{kA\Delta T}{x} = \frac{\left(\frac{0.75 \text{ BTU}}{\text{hr} - \text{ft} - ^\circ\text{F}} \right) \left(\frac{1055 \text{ J}}{1 \text{ BTU}} \right) \left(\frac{1 \text{ hr}}{3660 \text{ s}} \right) (10 \text{ m}^2) \left(\frac{3.28 \text{ ft}}{1 \text{ m}} \right)^2 \left(60 \left(\frac{9}{5} \right) \right)}{\frac{12 \text{ in}}{12 \text{ in}} (1 \text{ ft})}$$

$$Q = 2,553.78 \text{ Watts}$$

204. How much heat in kJ must be transferred to 20kg of air to increase the temperature from 20°C to 280°C if the pressure is to remain constant?

Solution:

$$Q = mC_p\Delta T = (20 \text{ kg}) \left(\frac{1 \text{ kJ}}{\text{kg} - ^\circ\text{K}} \right) (280^\circ\text{C} - 20^\circ\text{C})$$

$$Q = 5200 \text{ kJ}$$

205. At an average temperature of 100°C , hot air flows through a 2.5m long tube with an inside diameter of 50mm. The temperature of the tube is 20°C along its length. Convective heat transfer is $\frac{20.1 \text{ W}}{\text{m}^2 - \text{K}}$. Determine the heat transfer from air to the tube?

Solution:

$$Q = hA\Delta T = \left(\frac{20.1 \text{ W}}{\text{m}^2 - \text{K}} \right) (2\pi(0.025 \text{ m})(2.5 \text{ m}))(100^\circ\text{C} - 20^\circ\text{C})$$

$$Q = 631.46 \text{ Watts}$$

206. Calculate the energy transfer rate across 6 inches wall with a temperature difference of 50°C . The thermal conductivity of the fire brick is $\frac{0.65 \text{ BTU}}{\text{hr} - \text{ft} - ^\circ\text{F}}$

Solution:

$$\frac{Q}{A} = \frac{k\Delta T}{x} = \frac{\left(\frac{0.65 \text{ BTU}}{\text{hr} - \text{ft} - ^\circ\text{F}} \right) [50^\circ\text{C} \left(\frac{9}{5} \right)] \left(\frac{1.055 \text{ kJ}}{1 \text{ BTU}} \right) \left(\frac{1 \text{ hr}}{3600 \text{ s}} \right) (3.28 \text{ ft})^2}{\frac{6}{12}}$$

$$\frac{Q}{A} = \frac{368.88 \text{ W}}{\text{m}^2}$$

207. A 50cm long, 0.2cm diameter electric resistance wire submerged in water is used to determine the boiling heat transfer coefficient in water at 1atm experimentally. The surface temperature of the wire is measured to be 130°C when a wattmeter indicates the electric power consumption to be 4.1kW, then the heat transfer coefficient is?

Solution:

Note: @ 1atm, boiling temperature of water is 100°C

$$Q = hA\Delta T; h = \frac{Q}{A\Delta T}$$

$$h = \frac{4.1\text{kW}}{(2\pi(0.001\text{m})(0.5\text{m}))(130^\circ\text{C} - 100^\circ\text{C})}$$

$$h = \frac{43.50\text{kW}}{\text{m}^2 - ^\circ\text{C}}$$

208. Heat is transferred steadily through a 0.2m thick 8m by 4m wall at a rate of 1.6kW. The inner and outer surface temperatures of the wall are measured to be 15°C and 5°C. The average heat conductivity of the wall is?

Solution:

$$Q = \frac{kA\Delta T}{x}; k = \frac{Qx}{A\Delta T}$$

$$k = \frac{(1.6\text{kW})(0.2\text{m})}{(8\text{m} \times 4\text{m})(15^\circ\text{C} - 5^\circ\text{C})}$$

$$k = \frac{1\text{W}}{\text{m}^2 - ^\circ\text{C}}$$

209. A brand new air conditioner has an EER of $\frac{11\text{kJ}}{\text{kg-hr}}$. What is the equivalent COP?

Solution:

$$\text{COP} = \frac{\text{EER}}{3.6} = \frac{11}{3.6}$$

$$\text{COP} = 3.06$$

210. A refrigerator removes heat from a refrigerated space at -5°C at a rate of $\frac{0.35\text{kJ}}{\text{s}}$ and rejects it to an environment of 20°C. The minimum required power input is?

Solution:

$$e = \frac{Q_R}{W} = \frac{T_L}{T_H - T_L} = \frac{-5 + 273}{(20 + 273) - (-5 + 273)} = 10.72$$

$$e = \frac{Q_R}{W}; W = \frac{Q_R}{e} = \frac{\left(\frac{0.35\text{kJ}}{\text{s}}\right)}{10.72}$$

$$W = 32.65\text{Watts}$$

211. The drinking water needs of an office are met by cooling tap water in a refrigerated water fountain from 22°C to 8°C at an average rate of 8kg/hr. If the COP of this refrigerator is 3.1, the required power input to this refrigerator is?

Solution:

$$\text{COP} = \frac{Q_R}{W}; W = \frac{Q_R}{\text{COP}}$$

$$Q_R = \dot{m}C_p\Delta T = \left(\frac{8\text{kg}}{\text{hr}}\right)\left(\frac{1\text{hr}}{3600\text{s}}\right)\left(\frac{4.187\text{kJ}}{\text{kg} - ^\circ\text{K}}\right)(22^\circ\text{C} - 8^\circ\text{C}) = 130.26\text{Watts}$$

$$W = \frac{Q_R}{\text{COP}} = \frac{130.26\text{Watts}}{3.1}$$

$$W = 42\text{Watts}$$

212. A refrigerator is removing heat from a cold medium at 30°C at a rate of $\frac{7,200\text{kJ}}{\text{hr}}$ and rejecting waste heat to a medium at 30°C. If COP of the refrigerator is 2, the power consumed by refrigerator is?

Solution:

$$\text{COP} = \frac{Q}{R};$$

$$W = \frac{Q}{\text{COP}} = \frac{\left(\frac{7,200\text{kJ}}{\text{hr}}\right)\left(\frac{1\text{hr}}{3,600\text{s}}\right)}{2} = 1\text{kW}$$

213. A window air-conditioner that consumes 2kW of electricity when running and has a COP of 4 is placed in the middle of a room and is plugged in the rate of cooling or heating, this air condition will provide to the air in the room when running is

Solution:

$$\text{COP} = \frac{Q_L}{W};$$

$$Q_L = \text{COP} * W = (4)(2\text{kW}) = 8\text{kW}$$

$$W = Q_H - Q_L = 10\text{kW} - 8\text{kW} = 2\text{kW}$$

214. A reverse Carnot cycle requires a 3hp and extracts energy from a lake to heat a house. If the house is kept at 70°F and requires 2,000BTU per minute, what is the temperature of the lake?

Solution:

$$W = 3\text{hp} \left(\frac{0.746\text{kJ}}{\text{s}} \right) \left(\frac{1\text{hp}}{\text{hp}} \right) = \frac{2.238\text{kJ}}{\text{s}} \left(\frac{1\text{BTU}}{1.055\text{kJ}} \right) \left(\frac{60\text{s}}{\text{min}} \right) = \frac{127.30\text{BTU}}{\text{min}}$$

$$T_2 = 70^\circ\text{F} + 460 = 530^\circ\text{R}$$

$$Q_R = T_2(S_1 - S_4)$$

$$(S_1 - S_4) = \frac{2,000}{530}$$

$$Q_A = T_1(S_1 - S_4) = T_1 \left(\frac{2,000}{530} \right)$$

$$W = Q_R - Q_A = 2,000 - T_1 \left(\frac{2,000}{530} \right)$$

$$T_1 = 496^\circ\text{R} - 460 = 36^\circ\text{F}$$

215. Ton of refrigeration equivalent in kN-m/hr.

Solution:

$$1\text{TR} = \left(\frac{3.516\text{kJ}}{\text{s}} \right) \left(\frac{3600\text{s}}{\text{hr}} \right) = \frac{12,657\text{kJ} - \text{m}}{\text{hr}}$$

216. What is the coefficient of vapor compression refrigeration system having the following data? Enthalpy entering the compressor is 181.79kJ/kg; enthalpy after compression work is done 207.3kJ/kg; after condensation the enthalpy is 58.2kJ/kg and throttled from 0.19MPa to 0.18MPa.

Solution:

$$\text{COP} = \frac{Q_A}{W} = \frac{h_1 - h_4}{h_2 - h_1} = \frac{\frac{181.79\text{kJ}}{\text{kg}} - \frac{58.2\text{kJ}}{\text{kg}}}{\frac{207.3\text{kJ}}{\text{kg}} - \frac{181.79\text{kJ}}{\text{kg}}} = 4.84$$

217. The refrigerating capacity of R-12 system is 22kW. Compressor power is 7.8hp. Determine the COP of the refrigeration system.

Solution:

$$\text{COP} = \frac{Q_A}{W} = \frac{22\text{kW}}{7.8 \times 0.746\text{kW}} = 3.78$$

218. A refrigeration cycle has a coefficient of performance of 5. The power input to the compressor is 5kW. Compute the rate of heat of rejection from the refrigerant to the environment.

Solution:

$$\text{COP} = \frac{Q_A}{W}$$

$$Q_A = W * \text{COP} = (5\text{kW})(5) = 25\text{kW}$$

$$Q_R = W + Q_A = 25\text{kW} + 5\text{kW} = 30\text{kW}$$

219. In the refrigeration cycle, the heat rejected in the condenser is 150kW and the coefficient of performance is 3. Find the refrigerating effect.

Solution:

$$\text{COP} = \frac{Q_A}{W} = \frac{Q_A}{Q_R - Q_A} = \frac{Q_A}{150\text{kW} - Q_A} = 3$$

$$Q_A = 112.5\text{kW}$$

220. A refrigerating machine has the capability of producing a cooling effect of 4,500kcal per hour. What is the refrigerating load in Tons of Refrigeration?

Solution:

$$Q_A = \left(\frac{4,500\text{kCal}}{\text{hr}}\right) \left(\frac{1\text{hr}}{3600\text{s}}\right) \left(\frac{4.187\text{kJ}}{\text{kCal}}\right) \left(\frac{1\text{TR}}{3.516\text{kW}}\right) = 1.48\text{TR}$$

221. A reverse Carnot cycle is operating under the temperature of limits -10°C and 20°C . Find the COP.

Solution:

$$\text{COP} = \frac{T_L}{T_H - T_L} = \frac{-10^\circ\text{C} + 273}{(20^\circ\text{C} + 273) - (-10^\circ\text{C} + 273)} = 8.77$$

222. There are 800kg of fish to be cooled from 18°C to -8°C . The specific heat above freezing is $\frac{0.7\text{kCal}}{\text{kg}-^\circ\text{K}}$ and below freezing is $\frac{0.3\text{kCal}}{\text{kg}-^\circ\text{K}}$ with latent heat of fusion of $\frac{61.2\text{kCal}}{\text{kg}}$. The freezing temperature is -3°C . Find the heat removed from the product.

Solution:

$$Q = (800\text{kg}) \left(\frac{0.7\text{kCal}}{\text{kg} - ^\circ\text{K}}\right) (18^\circ\text{C} + 3^\circ\text{C}) + (800\text{kg}) \left(\frac{61.2\text{kCal}}{\text{kg}}\right) + (800\text{kg}) \left(\frac{0.3\text{kCal}}{\text{kg} - ^\circ\text{K}}\right) (-3 + 8)$$

$$Q = 61,920\text{kCal}$$

223. In a refrigeration cycle, the heat rejected in the condenser is 100kW and the coefficient of performance is 2.56. Find the refrigerating effect.

Solution:

$$\text{COP} = \frac{Q_A}{W} = \frac{Q_A}{Q_R - Q_A} = \frac{Q_A}{100\text{kW} - Q_A} = 2.56$$

$$Q_A = 71.91\text{kW}$$

224. One metric ton of fish is to be refrigerated from a temperature of 30°C to -5°C in 10 hours. The freezing temperature is -1°C , the specific heats below and above freezing temperature are $\frac{0.4\text{kJ}}{\text{kg}-^\circ\text{C}}$ and $\frac{0.8\text{kJ}}{\text{kg}-^\circ\text{C}}$, respectively, and the latent heat of fusion is 435kJ/kg. Allow 20% additional heat load to compensate for heat losses. Determine the refrigerating load in ton of refrigeration.

Solution:

$$Q = \frac{1,000\text{kg}}{10\text{hrs} * 3600\text{s}} \left[\left(\frac{0.8\text{kJ}}{\text{kg} - ^\circ\text{C}}\right) (30^\circ\text{C} + 1^\circ\text{C}) + \frac{435\text{kJ}}{\text{kg}} + \left(\frac{0.4\text{kJ}}{\text{kg} - ^\circ\text{C}}\right) (-1^\circ\text{C} + 5^\circ\text{C}) \right]$$

$$Q = (12.82\text{kW})(1.2) \left(\frac{1\text{TR}}{3.516\text{kW}} \right)$$

$$Q = 4.37\text{TR}$$

225. A reversed Carnot Cycle is used for cooling. The input work is 12kW, while the COP is 3.8. Calculate the refrigerating effect in Tons of Refrigeration.

Solution:

$$\text{COP} = \frac{Q_A}{W}$$

$$Q_A = (12\text{kW})(3.8)$$

$$Q_A = 45.6\text{kW} \left(\frac{1\text{TR}}{3.516\text{kW}} \right)$$

$$Q_A = 12.97\text{TR}$$

226. Express 1TR in kcal/min

Solution:

$$1\text{TR} \left(\frac{3.516\text{kW}}{\text{TR}} \right) = \frac{3.516\text{kJ}}{\text{s}} \left(\frac{1\text{kCal}}{4.182\text{kJ}} \right) \left(\frac{60\text{s}}{\text{min}} \right)$$

$$1\text{TR} = 50.44\text{kCal/min}$$

227. What is the coefficient of performance of a Carnot refrigeration cycle operating between -10°F and -190°F?

Solution:

$$\text{COP} = \frac{T_L}{H_L - T_L} = \frac{-190^\circ\text{F} + 460}{(-10^\circ\text{F} + 460) - (-190^\circ\text{F} + 460)}$$

$$\text{COP} = 1.5$$

228. A refrigeration plant is rated at 20 ton capacity. How many pounds of air per hour will it cool from 90 to 70°F at constant pressure? $C_p = 0.24$

Solution:

$$\text{TR} = \frac{mC_p(t_1 - t_2)}{12,000}$$

$$m = \frac{20\text{ton} \times 12,000}{(0.24)(90^\circ\text{F} - 70^\circ\text{F})} = \frac{50,000\text{lb}}{\text{hr}}$$

229. A refrigeration system produces 150BTU/lbm of cooling. In order to have a rating of 1TR, what must be the flow rate required to freeze 1 ton of ice in a day?

Solution:

$$1\text{TR} = \frac{200\text{BTU}}{\text{min}}$$

$$m = \frac{\frac{200\text{BTU}}{\text{min}}}{\frac{150\text{BTU}}{\text{lbm}}} = \frac{1.33\text{lbm}}{\text{min}} \left(\frac{60\text{min}}{1\text{hr}} \right) = \frac{80\text{lbm}}{\text{hr}}$$

230. Fish weighing 5,000kg with a temperature of 6°C is brought to a cold storage and which shall be cooled to -14°C in 24 hours. Find the cooling load required if the specific heat of fish is $\frac{0.7\text{kcal}}{\text{kg}^\circ\text{C}}$ above freezing and $\frac{0.3\text{kcal}}{\text{kg}^\circ\text{C}}$ below freezing which is -2.2°C. The latent heat of freezing is 55.5 kcal/kg.

Solution:

$$Q = \frac{5,000\text{kg}}{86,400\text{s}} \left[\left(\frac{0.7\text{kcal}}{\text{kg}^\circ\text{C}} \right) (6^\circ\text{C} + 2.2^\circ\text{C}) + \frac{55.5\text{kcal}}{\text{kg}} + \left(\frac{0.3\text{kcal}}{\text{kg}^\circ\text{C}} \right) (-2.2^\circ\text{C} + 14^\circ\text{C}) \right]$$

$$Q = \frac{3.75 \text{ kCal}}{s} \left(\frac{4.185 \text{ kJ}}{1 \text{ kCal}} \right) = 15.70 \text{ kW} \left(\frac{1 \text{ TR}}{3.516 \text{ kW}} \right)$$

$$Q = 4.46 \text{ TR}$$

231. A reversed Carnot refrigeration used to ice at 0°C, water is available at 301°K, brine mixture is -16°C, find the coefficient of performance.

Solution:

$$\text{COP} = \frac{T_L}{T_H - T_L} = \frac{-16^\circ\text{C} + 273}{(301) - (-16^\circ\text{C} + 273)}$$

$$\text{COP} = 5.84$$

232. Determine the power per kW of refrigeration if the COP is 4.2786.

Solution:

$$\text{Power per kW ref} = \frac{W}{Q_A} = \frac{1}{\text{COP}} = \frac{1}{4.2786}$$

$$\text{Power per kW ref} = 0.2337$$

233. A vapor compression refrigeration system is designed to have a capacity of 100tons. The actual COP is 5.86 and 35% of the power supplied to the compressor is lost in the form of friction and cylinder cooling losses. Determine the motor power in kW.

Solution:

$$Q_A = 100 \text{ tons} \left(\frac{3.516 \text{ kW}}{\text{TR}} \right) = 351.6 \text{ kW}$$

$$\text{COP} = \frac{Q_A}{W};$$

$$W = \frac{Q_A}{\text{COP}} = \frac{351.6 \text{ kW}}{5.86} = 60 \text{ kW}$$

$$\text{Motor power} = \frac{60 \text{ kW}}{0.65} = 92.31 \text{ kW}$$

234. The refrigeration system has a refrigeration cycle per kg of 220kJ. The heat required to remove is 630kJ/hr. Calculate the mass circulated per hour.

Solution:

$$\text{Cooling load} = m\Delta h$$

$$\frac{630 \text{ kJ}}{\text{hr}} = m \left(\frac{220 \text{ kJ}}{\text{kg}} \right)$$

$$m = 2.86 \text{ kg}$$

235. A 3.5kg mass of refrigerant is circulated per hour in a refrigeration system, whose refrigerating effect is 250 kJ per kg. Compute the heat that has to be removed from the system per hour.

Solution:

Heat removed from the sytem = heat absorbed by refriegeerants.

$$\text{Heat removed from the sytem} = m\Delta h = (3.5 \text{ kg}) \left(\frac{250 \text{ kJ}}{\text{kg}} \right) = 875 \text{ kJ}$$

236. 1,000 kilograms of fish at a temperature was placed in a chiller box storage. The product temperature is four degrees. Determine the quantity of heat required to bring the temperature of the fish to its freeing temperature. (Note: specific heat above freezing = $\frac{0.76 \text{ kcal}}{\text{kg}^\circ\text{C}}$; specific heat below

$$\text{freezing} = \frac{0.41 \text{ kcal}}{\text{kg}^\circ\text{C}} \text{ and freezing temperature is } = -2.2^\circ\text{C}.$$

Solution:

$$\text{Cooling load} = mC_p\Delta T = (1,000\text{kg})\left(\frac{0.76\text{kcal}}{\text{kg} - ^\circ\text{C}}\right)(4^\circ\text{C} + 2.2^\circ\text{C})$$

$$\text{Cooling load} = 4,712\text{kcal}$$

237. A vapor compression cycle has a coefficient of performance of 4.2. The power input to the compressor is 1.85kW. Compute the rate of heat rejection from the refrigerant to the environment.

Solution:

$$\text{COP} = \frac{Q_A}{W};$$

$$Q_A = W * \text{COP} = (1.85\text{kW})(4.2) = 7.77\text{kW}$$

$$W = Q_R - Q_A$$

$$Q_R = W + Q_A = 1.85\text{kW} + 7.77\text{kW}$$

$$Q_R = 9.62\text{kW}$$

238. What is the tons of refrigeration required to cool 15,000lb of fresh pork from a temperature of 89°F to 32°F in 24 hours? Specific heat above freezing of fresh pork is $\frac{0.68\text{BTU}}{\text{lb} - ^\circ\text{F}}$

Solution:

$$Q = mC_p\Delta T = \frac{(15,000\text{lb})\left(\frac{0.68\text{BTU}}{\text{lb} - ^\circ\text{F}}\right)(89^\circ\text{F} - 32^\circ\text{F})}{86,400\text{s}}$$

$$Q = \frac{6.73\text{BTU}}{\text{s}}\left(\frac{1.055\text{kJ}}{1\text{BTU}}\right) = 7.2\text{kW}\left(\frac{1\text{TR}}{3.516\text{kW}}\right)$$

$$Q = 2.02\text{TR}$$

239. A refrigerator is maintained at 5°C. Heat is removed from the stored food at a rate of 330kJ/min. What is the refrigerator's coefficient of performance if the necessary power input to the refrigerator is 3.5kW?

Solution:

$$\text{COP} = \frac{Q_A}{W} = \frac{\left(\frac{330\text{kJ}}{\text{min}}\right)\left(\frac{1\text{min}}{60\text{s}}\right)}{3.5\text{kW}} = 1.57$$

240. A building to be maintained at 18°C at all times. A heat pump is required for this. When the temperature outside the building drops to -6°C, the building loses heat at a rate of 120,000kJ/hr. Compute the least power necessary to drive the heat pumps.

Solution:

$$Q_R = \left(\frac{120,000\text{kJ}}{\text{hr}}\right)\left(\frac{1\text{hr}}{3600\text{s}}\right) = \frac{33.33\text{kJ}}{\text{s}}$$

$$\text{Carnot cycle Efficiency} = \frac{T_H - T_L}{T_H} = \frac{Q_A - Q_R}{Q_A}$$

$$\frac{(18^\circ\text{C} + 273) - (-6^\circ\text{C} + 273)}{18^\circ\text{C} + 273} = \frac{Q_A - \frac{33.33\text{kJ}}{\text{s}}}{Q_A}$$

$$Q_A = 36.31\text{kW}$$

$$W = Q_A - Q_R = 36.31\text{kW} - 33.33\text{kW}$$

$$W = 3\text{kW}$$

241. A refrigerating machine that is classified as one-ton machine has the capacity to produce a cooling effect of

Solution:

$$1\text{ton}\left(\frac{3.516\text{kW}}{1\text{TR}}\right) = \frac{3.516\text{kJ}}{\text{s}}\left(\frac{3600\text{s}}{\text{hr}}\right)\left(\frac{1\text{kcal}}{4.185\text{kJ}}\right) = \frac{3024\text{kcal}}{\text{hr}}$$

242. 300kJ of heat flow by conduction from the outside to the inside of a cold storage in one hour. If the temperature and all other conditions, what is the heat flowing through the cold storage in two hours?

Solution:

$$Q = \left(\frac{300\text{kJ}}{\text{hr}} \right) (2\text{hrs}) = 600\text{kJ}$$

243. An air-conditioning system has a capacity of 300kW refrigeration and uses R-12 with evaporating temperature of 0°C ($h_f = \frac{200\text{kJ}}{\text{kg}}$, $h_g = \frac{351.48\text{kJ}}{\text{kg}}$). The mass of flash gas per kilogram of refrigerant circulated is 0.2212. Determine the COP if the work of compression 42kW.

Solution:

$$\text{COP} = \frac{Q_A}{W} = \frac{300\text{kW}}{42\text{kW}} = 7.14$$

244. A 2.1kg of refrigerant is circulated per hour in a refrigeration system, whose refrigerating effect is 290kJ/kg. Compute the heat that has to be removed from the system per hour.

Solution:

$$Q_A = m\Delta h = \left(\frac{2.1\text{kg}}{\text{hr}} \right) \left(\frac{290\text{kJ}}{\text{kg}} \right)$$

$$Q_A = \frac{609\text{kJ}}{\text{hr}}$$

245. A TR in BTU/24hrs is

Solution:

$$1\text{TR} \left(\frac{200\text{BTU}}{\text{min}} \right) \left(\frac{60\text{min}}{\text{hr}} \right) (24\text{hrs}) = \frac{288,000\text{BTU}}{\text{hr}}$$

246. Magnolia Dairy Products plan must cool 4000 gallons of fresh milk receive from the farm per day from an initial temperature of 80°F to a temperature of 38°F in 5 hours. If the density of milk is 8.6lbs/gallon, specific gravity is 1.03, and specific heat is 0.935BTU per lb per °F. What must be the capacity of the refrigerating machine in tons?

Solution:

$$\text{Note: Milk has } C_p = \frac{0.935\text{BTU}}{\text{lb-}^\circ\text{F}} \text{ or } \frac{3.93\text{kJ}}{\text{kg-}^\circ\text{K}}$$

$$m = \left(\frac{4,000\text{gallons}}{5\text{hrs}} \right) \left(\frac{8.6\text{lb}}{\text{gal}} \right) = \frac{6,880\text{lb}}{\text{hr}}$$

$$Q_A = mC_p\Delta T = \left(\frac{6,880\text{lb}}{\text{hr}} \right) \left(\frac{0.935\text{BTU}}{\text{lb-}^\circ\text{F}} \right) (80^\circ\text{F} - 38^\circ\text{F})$$

$$Q_A = \frac{\frac{270,177.6\text{BTU}}{\text{hr}} \left(\frac{1\text{hr}}{60\text{min}} \right)}{\frac{200\text{BTU}}{\text{min}}}$$

$$Q_A = 22.50\text{TR}$$

247. A refrigeration system needs 9hp to compress a liquid, 15kW was extracted from the cooled space. What is the coefficient of performance?

Solution:

$$\text{COP} = \frac{Q_A}{W} = \frac{15\text{kW}}{9\text{hp} \times 0.746\text{kW}} = 2.23$$

248. A San Miguel Corporation processing plant that requires cooling of 120 gallons per minute of water from 15.6°C to 10°C, assume that the cooler heat transfer surface area will enable a 5.5°C

differential between the chilled water leaving the cooler and the R-12 evaporating temperature, also assume that the condenser heat transfer surface area will enable 5.5°C differential between the condenser water out and the R-12 condensing temperature. Water be available for the condensing medium at 29.4°C inlet and 35°C outlet. Assume no liquid sub-cooling, or gas superheating. Find the tons of refrigeration.

Solution:

$$\text{Refrigerating load} = mC_p\Delta T$$

$$m = \left(\frac{120\text{gallons}}{\text{min}}\right)\left(\frac{3.7854\text{kg}}{1\text{gal}}\right)\left(\frac{1\text{min}}{60\text{s}}\right) = \frac{7.57\text{kg}}{\text{s}}$$

$$\text{Refrigerating load} = mC_p\Delta T = \left(\frac{7.57\text{kg}}{\text{s}}\right)\left(\frac{4.187\text{kJ}}{\text{kg} \cdot ^\circ\text{C}}\right)(15.6^\circ\text{C} - 10^\circ\text{C})$$

$$\text{Refrigerating load} = 177.50\text{kW}\left(\frac{1\text{TR}}{3.516\text{kW}}\right) = 50.5\text{TR}$$

249. An ideal gas at 45 psig and 80°F is heated in a closed container to 130°F. What is the final pressure?

Solution:

$$\frac{P_1}{T_1} = \frac{P_2}{T_2}$$

$$P_2 = \frac{P_1}{T_1}(T_2) = \frac{45\text{psig} + 14.7}{80^\circ\text{F} + 460}(130^\circ\text{F} + 460)$$

$$P_2 = 65.23\text{psia}$$

250. A 1kg steel ($C_p = \frac{0.75\text{kJ}}{\text{kg} \cdot ^\circ\text{K}}$) rivet cools from 700°K and 400°K upon being installed in a riveted structure. The entropy change in kJ/°K of this rivet is:

Solution:

$$\Delta S = mC_p \ln\left(\frac{T_2}{T_1}\right) = (1\text{kg})\left(\frac{0.75\text{kJ}}{\text{kg} \cdot ^\circ\text{K}}\right)\ln\left(\frac{400^\circ\text{K}}{700^\circ\text{K}}\right)$$

$$\Delta S = 0.419$$

251. A volume of 400cc of air is measured at a pressure of 740mmHg abs and a temperature of 18°C. What will be the volume at 760mmHg abs and 0°C?

Solution:

$$\frac{P_1 V_1}{T_1} = \frac{P_2 V_2}{T_2}; V_2 = \frac{P_1 V_1 T_2}{T_1 P_2}$$

$$V_2 = \frac{(740\text{mmHg})(400\text{cm}^3)(0^\circ\text{C} + 273)}{(18 + 273)(760\text{mmHg})} = 365\text{cm}^3$$

252. A gaseous mixture has dew point temperature of 15°C. The total pressure is 143.27kPa. Determine the amount of water vapor present in 100 moles of the mixture. Note: saturation pressure at 15°C is 1.7051kPa

Solution:

$$P_v = \left(\frac{V_v}{V}\right)P$$

$$1.7051\text{kPa} = \left(\frac{V_v}{100}\right)(143.27\text{kPa})$$

$$V_v = 1.19$$

253. A certain gas at 101.325kPa and 16°C whose volume is 2.83m³ are compressed into a storage vessel of 0.31m³ capacity. Before admission, the storage vessel contained the gas at a

pressure and temperature of 137.9kPa and 24°C; after admission the pressure has increased to 1171.8kPa. What should be the final temperature of the gas in the vessel in Kelvin?

Solution:

Compressed gas

$$PV = mRT$$

$$(101.325\text{kPa})(2.83\text{m}^3) = mR(16 + 273); m = \frac{(101.325\text{kPa})(2.83\text{m}^3)}{R(16 + 273)} = \frac{0.992}{R}$$

The gas initially:

$$PV = mRT$$

$$(1171.8\text{kPa})(0.31\text{m}^3) = mR(24 + 273); m = \frac{(1171.8\text{kPa})(0.31\text{m}^3)}{R(24 + 273)} = \frac{0.144}{R}$$

Solving for the final temperature:

$$PV = mRT$$

$$(1171.8\text{kPa})(0.31\text{m}^3) = \left(\frac{0.144}{R} + \frac{0.992}{R} \right) RT$$

$$T = 320^\circ\text{K}$$

254. A large furnace can supply energy in the form of heat at 2,000°K at a steady rate of 3,143kW. Assuming an environment temperature of 25°C, what is the available energy for the furnace in kW?

Solution:

$$E_a = Q - T_o\Delta S = 3,143\text{kW} - (25 + 273) \left(\frac{3,143}{2000} \right)$$

$$E_a = 2,674.70\text{kW}$$

255. What is the total required heating energy in raising the temperature of a given amount of water when the energy applied is 1000kWh with heat losses of 25%

Solution:

$$Q_T - 0.25Q = 1000\text{kWh}$$

$$0.75Q_T = 1000\text{kWh}$$

$$Q_T = 1,333.33\text{kWh}$$

256. Determine the average C_p value in kJ/kg-°K of a gas if 522kJ of heat is necessary to raise the temperature from 300°K to 800°K making the pressure constant:

Solution:

$$Q = mC_p\Delta T; C_p = \frac{Q}{m\Delta T}$$

$$C_p = \frac{522\text{kJ}}{(1\text{kg})(800^\circ\text{K} - 300^\circ\text{K})}$$

$$C_p = \frac{1.044\text{kJ}}{\text{kg} - ^\circ\text{K}}$$

257. A large mining company was provided with a 3m³ of compressed air tank. Air pressure in the tank drops from 700kPa to 180kPa while the temperature remains unchanged at 28°C. What percentage has the mass in the tank been reduced?

Solution:

The mass of the air at 700kPa

$$PV = mRT;$$

$$(700\text{kPa})(3\text{m}^3) = m \left(\frac{0.287\text{kJ}}{\text{kg} - ^\circ\text{K}} \right) (28 + 273)$$

$$m = 24.31\text{kg}$$

Mass of air at 180kPa

$$(180\text{kPa})(3\text{m}^3) = m \left(\frac{0.287\text{kJ}}{\text{kg} \cdot ^\circ\text{K}} \right) (28 + 273)$$

$$m = 6.25\text{kg}$$

Mass percentage reduced:

$$\% = \frac{24.31 - 6.25}{24.31} = 74.30\%$$

258. A heat engine is operated between temperature limits of 1,370°C and 260°C. Engine supplied with 14,142kJ per kWh. Find the Carnot cycle efficiency.

Solution:

$$\text{CCE} = 1 - \frac{(260^\circ\text{C} + 273)}{(1,370^\circ\text{C} + 273)} = 67.56\%$$

259. A close vessel contains air at a pressure of 160kN/m² gauge and temperature of 30°C. The air is heated at constant volume to 60°C with the atmospheric pressure as 750mmHg. What is the final gauge pressure?

Solution:

$$\frac{P_1}{T_1} = \frac{P_2}{T_2}; P_2 = \frac{P_1 T_2}{T_1}$$

$$P_2 = \frac{\left(\frac{160\text{kN}}{\text{m}^2} + 101.325\text{kPa} \right) (60 + 273)}{(30 + 273)} = 287.325\text{kPa} = 185.87\text{kPag}$$

260. What is the temperature in degree Celsius of 2 liters of water at 30 degree Celsius after 500 calories of heat have been added to?

Solution:

$$Q = mC_p \Delta T = mC_p (T_2 - T_1)$$

$$(0.5\text{kCal}) \left(\frac{4.187\text{kJ}}{\text{Cal}} \right) = (2\text{lit}) \left(\frac{1\text{kg}}{\text{lit}} \right) \left(\frac{4.187\text{kJ}}{\text{kg} \cdot ^\circ\text{C}} \right) (T_2 - 30^\circ\text{C})$$

$$T_2 = 30.25^\circ\text{C}$$

261. A volume of 450cm³ of air is measured at a pressure of 740mmHg absolute and a temperature of 20°C. What is the volume in cc at 760mmHg absolute and 0°C?

Solution:

$$\frac{P_1 V_1}{T_1} = \frac{P_2 V_2}{T_2}; \frac{(740\text{mmHg})(450\text{cm}^3)}{20^\circ\text{C} + 273} = \frac{(760\text{mmHg})V_2}{(0^\circ\text{C} + 273)}$$

$$V_2 = 408.25\text{cm}^3$$

262. How much heat in kJ must be transferred to 20kg of air to increase the temperature from 20°C to 280°C if the pressure is to remain constant?

Solution:

$$Q = mC_p \Delta T = (20\text{kg}) \left(\frac{1\text{kJ}}{\text{kg} \cdot ^\circ\text{C}} \right) (280^\circ\text{C} - 20^\circ\text{C})$$

$$Q = 5,200\text{kJ}$$

263. Assuming compression is according to the law pV = constant. Calculate the initial volume of gas at a pressure of 2bar which will occupy a volume of 6m³ when it is compressed to a pressure of 42 bar.

Solution:

$$P_1 V_1 = P_2 V_2; (2\text{bar})V_1 = (42\text{bar})(6\text{m}^3)$$

$$V_1 = 126\text{m}^3$$

264. A 4m x 4m x 3m office room contains air at 27°C. If the pressure of dry air is 95kPa, calculate the mass of air if the air fills completely the room.

Solution:

$$PV = mRT; (95\text{kPa})(4\text{m} \times 4\text{m} \times 3\text{m}) = m \left(\frac{0.287\text{kJ}}{\text{kg} \cdot ^\circ\text{K}} \right) (27^\circ\text{C} + 273)$$

$$m = 52.96\text{kg}$$

265. The drinking water needs of an office are met by cooling tap water in a refrigerated water fountain from 22°C to 8°C at an average rate of 8kg/h. If the COP of this refrigerator is 3.1, the required power input to this refrigerator is?

Solution:

$$Q_A = mC_p\Delta T = \left(\frac{8\text{kg}}{\text{hr}} \right) \left(\frac{1\text{hr}}{3600\text{s}} \right) \left(\frac{4.187\text{kJ}}{\text{kg} \cdot ^\circ\text{C}} \right) (22^\circ\text{C} - 8^\circ\text{C}) = 0.13\text{kW}$$

$$\text{COP} = \frac{Q_A}{W}; 3.1 = \frac{0.13\text{kW}}{W}$$

$$W = 0.042\text{kW} \text{ or } 42\text{Watts}$$

266. A closed rigid tank has a capacity of 2.5m³. It holds air at 400kPa and 18°C. Heat is added until the temperature is 350°C. Determine the heat added in kJ.

Solution:

$$PV = mRT; (400\text{kPa})(2.5\text{m}^3) = (m) \left(\frac{0.287\text{kJ}}{\text{kg} \cdot ^\circ\text{K}} \right) (18^\circ\text{C} + 273)$$

$$m = 11.97\text{kg}$$

$$Q = mC_p\Delta T = (11.97\text{kg}) \left(\frac{0.716\text{kJ}}{\text{kg} \cdot ^\circ\text{C}} \right) (350^\circ\text{C} - 18^\circ\text{C})$$

$$Q = 2,845.41\text{kJ}$$

267. An orange with an average mass of 0.15kg and average specific heat of 3.65kJ/kg-°C is cooled from 20°C to 5°C. Compute the amount of heat transferred from the orange.

Solution:

$$Q = mC_p\Delta T = (0.15\text{kg}) \left(\frac{3.65\text{kJ}}{\text{kg} \cdot ^\circ\text{C}} \right) (20^\circ\text{C} - 5^\circ\text{C})$$

$$Q = 8.21\text{kJ}$$

268. Given a vacuum pressure of 165mmHg. The barometric pressure is 745mmHg. Find the absolute pressure in kPa.

Solution:

$$P_{\text{abs}} = P_{\text{gage}} + P_{\text{atm}} = -165\text{mmHg} + 745\text{mmHg} = 580\text{mmHg} \left(\frac{0.133\text{kPa}}{\text{mmHg}} \right)$$

$$P_{\text{abs}} = 77.14\text{kPa}$$

269. There are 150kg/min of water enters at 5°C and leaves at 40°C. Find the heat added during the process.

Solution:

$$Q = mC_p\Delta T = \left(\frac{150\text{kg}}{\text{min}} \right) \left(\frac{1\text{min}}{60\text{s}} \right) \left(\frac{4.187\text{kJ}}{\text{kg} \cdot ^\circ\text{C}} \right) (40^\circ\text{C} - 5^\circ\text{C})$$

$$Q = 366.36\text{kW}$$

270. The pump suction has a pressure of -150mmHg. If atmospheric pressure is 700mmHg. Find the absolute pressure.

Solution:

$$P_{\text{abs}} = P_{\text{gage}} + P_{\text{atm}} = (-150\text{mmHg} + 700\text{mmHg})(0.133\text{kPa})$$

$$P_{\text{abs}} = 73.15\text{kPa}$$

271. A piston-cylinder set-up contains methane at 100kPa. The 1kg piston opens to the atmosphere and has a 0.10m diameter. What must be the pressure of the methane in order to lift the piston?

Solution:

$$F = PA$$

$$F = (1\text{kg}) \left(\frac{9.81\text{m}}{\text{s}^2} \right) = 9.81\text{N or } 0.00981\text{kN}$$

$$0.00981\text{kN} = P \left[\frac{\pi}{4} (0.1\text{m})^2 \right]$$

$$P = 1.25\text{kPa} + 101.325\text{kPa} = 103\text{kPa}$$

272. A closed piston cylinder set-up contains liquid and water vapor at 100°C. The 0.1m diameter piston, exposed to the atmosphere is externally loaded with force of 20kN. What pressure of water is needed to push the piston?

Solution:

$$F = PA$$

$$20\text{kN} = P \left[\frac{\pi}{4} (0.1\text{m})^2 \right]$$

$$P = 2546.48\text{kPa} + 101.325\text{kPa}$$

$$P = 2,647.80\text{kPa}$$

273. A 2kg piston with 0.5m diameter is acted by a pressure gas. What should be the pressure of the gas that would move the piston upward?

Solution:

$$F = PA$$

$$\frac{(2\text{kg}) \left(\frac{9.81\text{m}}{\text{s}^2} \right)}{1000} = P \left[\frac{\pi}{4} (0.5\text{m})^2 \right]$$

$$P = 0.1\text{kPa} + 101.325\text{kPa} = 101.4\text{kPa}$$

274. An ideal gas at 0.60 atmospheres and 87°C occupies 0.450liter. How many moles are in the sample? ($R = 0.0821\text{atm/mole-}^\circ\text{K}$)

Solution:

$$PV = nRT$$

$$(0.60\text{atm})(0.450\text{liters}) = (n) \left(\frac{0.0821\text{atm}}{\text{mole-}^\circ\text{K}} \right) (87^\circ\text{C} + 273)$$

$$n = 0.00913\text{moles}$$

275. What is the change in internal energy of air (assumed to be an ideal gas) cooled from 1000°F to 200°F.

Solution:

$$\Delta U = C_V(T_2 - T_1) = \left(\frac{0.171\text{kJ}}{\text{kg-}^\circ\text{F}} \right) (1000^\circ\text{F} - 200^\circ\text{F})$$

$$\Delta U = \frac{140\text{BTU}}{\text{min}}$$

276. How much does 30lbm object weight in the moon?

(Gravity of moon is 5.47ft/s² & earth is 32.2ft/s²)

Solution:

$$\frac{W_E}{G_E} = \frac{W_M}{G_M}$$

$$\frac{30\text{lbm}}{\frac{32.2\text{ft}}{\text{s}^2}} = \frac{W_M}{\frac{5.47\text{ft}}{\text{s}^2}}$$

$$W_M = 5.096\text{lbm}$$

277. 100g of water is mixed with 150g of alcohol ($d = 790\text{kg/m}^3$). What is the specific volume of the resulting mixtures, assuming that the fluids mixed completely?

Solution:

$$\text{Total mass} = 150\text{g} + 100\text{g} = 250\text{g}$$

$$\text{Total volume} = \frac{0.1\text{kg}}{\frac{1000\text{kg}}{\text{m}^3}} + \frac{0.15}{\frac{790\text{kg}}{\text{m}^3}} = 2.899 \times 10^{-4}\text{m}^3 \text{ or } 289.87\text{cm}^3$$

$$\text{Specific volume} = \frac{289.87\text{cm}^3}{250\text{g}} = \frac{1.16\text{cm}^3}{\text{g}}$$

278. Given a steam pressure of 900lb/ft^2 , the temperature of 300°F , specific volume of $5.8\text{ft}^3/\text{lb}$. If the specific enthalpy is $9,500\text{lb-ft/lb}$., what is the internal energy of the system?

Solution:

$$h = u + Pv$$

$$\frac{9,500\text{lb-ft}}{\text{lb}} = u + \left(\frac{900\text{lb}}{\text{ft}^2}\right)\left(\frac{5.8\text{ft}^3}{\text{lb}}\right)$$

$$u = \frac{4,280\text{lb-ft}}{\text{lb}}$$

279. The pressure reading of 35psi in kPa abs is:

Solution:

$$\text{Pressure} = (35\text{psi})\left(\frac{101.325\text{kPa}}{14.7\text{psi}}\right) + 101.325\text{kPa}$$

$$\text{Pressure} = 342.57\text{kPa}$$

280. A 1,500kg automobile accelerates from 15km/hr . to 140km/hr . How much work in kJ does this require?

Solution:

$$W = \text{KE} = \frac{1}{2}mV^2 = \frac{1}{2}(1,500\text{kg})\left[\left(\frac{140\text{km}}{\text{hr}}\right)^2 - \left(\frac{15\text{km}}{\text{hr}}\right)^2\right]$$

$$W = 1,121.24\text{kJ}$$

281. If air is at a pressure of $3,200\text{lb/ft}^2$ and at a temperature of 800°R , what is the specific volume?

Solution:

$$PV = mRT$$

$$v = \frac{V}{m} = \frac{RT}{P} = \frac{\left(\frac{53.34\text{lb-ft}}{\text{lbm-}^\circ\text{R}}\right)(800^\circ\text{R})}{\frac{3,200\text{lb}}{\text{ft}^2}} = \frac{13.335\text{ft}^3}{\text{lbm}}$$

282. The barometer reads 88kPa . What is the absolute pressure of a 127.5cm liquid with a specific gravity of 0.32?

Solution:

$$P_g = \gamma h = (0.32)\left(\frac{9.81\text{m}}{\text{s}^2}\right)(1.275\text{m}) = 4\text{kPa}$$

$$P_{\text{abs}} = P_{\text{atm}} + P_g = 88\text{kPa} + 4\text{kPa}$$

$$P_{\text{abs}} = 92\text{kPa}$$

283. If 120kg of air has a dry bulb temperature of 18°C. What is the standard air volume in cubic meters?

Solution:

$$PV = mRT$$

$$(101.325\text{kPa})V = (120\text{kg})\left(\frac{0.287\text{kJ}}{\text{kg} \cdot ^\circ\text{K}}\right)(18^\circ\text{C} + 273)$$

$$V = 98.9\text{m}^3$$

284. An engine is required to produce 4,500kg of dry steam per hour. The quality of steam in it is 90%. What must be the weight of the wet steam?

Solution:

$$\text{Qualit, } x = \frac{m_{\text{vapor}}}{m_{\text{fluid}}}$$

$$0.9 = \frac{4,500\text{kg}}{m_{\text{fluid}}}$$

$$m_{\text{fluid}} = 5,000\text{kg}$$

285. Air is flowing through a 20cm diameter pipe with a velocity of 5m/s. If the temperature of air is 25°C and 120kPa pressure, find the rate of air in the pipe.

Solution:

$$\text{flow rate (Q)} = AV = \frac{\pi}{4}(0.2\text{m})^2\left(\frac{5\text{m}}{\text{s}}\right) = \frac{0.157\text{m}^3}{\text{s}}$$

$$PV = mRT$$

$$(120\text{kPa})\left(\frac{0.157\text{m}^3}{\text{s}}\right) = (m)\left(\frac{0.287\text{kJ}}{\text{kg} \cdot ^\circ\text{K}}\right)(25^\circ\text{C} + 273)$$

$$m = \frac{0.22\text{kg}}{\text{s}}$$

286. Fluid stored in a tank, is stored by a paddle disk. If the heat transfer from the tank is 1,300kJ and the work input to the paddle disk is 4,800kJ, then what is the change in internal energy if the tank and paddle disk is considered inside a control surface?

Solution:

$$Q = \Delta U + W_p$$

$$1300\text{kJ} = \Delta U + 4,800\text{kJ}$$

$$\Delta U = -3,500\text{kJ}$$

287. Gas is enclosed in a cylinder with a weighted piston as the top boundary. The gas is heated and expands from a volume of 0.04 to 0.10m³ at a constant pressure of 200kPa. Calculate the work done by the system.

Solution:

$$W = P(V_2 - V_1) = (200\text{kPa})(0.10\text{m}^3 - 0.04\text{m}^3) = 12\text{kJ}$$

288. Determine the amount of heat energy rejected by 0.5m³ of water when it is cooled from 35°C to 6°C.

Solution:

$$Q = mC_p\Delta T = \rho_w VC_p\Delta T = \left(\frac{1,000\text{kg}}{\text{m}^3}\right)(0.5\text{m}^3)\left(\frac{4.187\text{kJ}}{\text{kg} \cdot ^\circ\text{C}}\right)(35^\circ\text{C} - 6^\circ\text{C})$$

$$Q = 60,711.5\text{kJ}$$

289. The thermal efficiency of a particular engine operating on an ideal cycle is 35%. Calculate the heat in kJ supplied to the engine if the engine develops 1,200W-hr.

Solution:

$$\eta_{th} = \frac{W}{Q_A}$$

$$0.35 = \frac{\left(\frac{1,2\text{kJ} - \text{hr}}{\text{s}}\right) \left(\frac{3600\text{s}}{\text{hr}}\right)}{Q_A}$$

$$Q_A = 12,343\text{kJ}$$

290. Ninety kilograms of ice at 0°C completely melted. Find the entropy change in kJ/°K. (Take latent heat of ice as 320kJ/kg).

Solution:

$$\Delta S = \frac{Q}{T} = \frac{\left(\frac{320\text{kJ}}{\text{kg}}\right) (90\text{kg})}{0^\circ\text{C} + 273}$$

$$\Delta S = \frac{105.5\text{kJ}}{^\circ\text{K}}$$

291. A piston cylinder contains 2kg of steam, which expands from state 1 with an internal energy of 2,700kJ/kg to state 2 with an internal energy of 2,650kJ/kg. During the expansion process, the system receive 30kJ of heat. Determine the system work.

Solution:

$$W = m(U_2 - U_1) = (2\text{kg}) \left(\frac{2,700\text{kJ}}{\text{kg}} - \frac{2,650\text{kJ}}{\text{kg}} \right) = 100\text{kJ}$$

$$W_{\text{system}} = \Delta U + Q_A = 30\text{kJ} + 100\text{kJ}$$

$$W_{\text{system}} = 130\text{kJ}$$

292. A spherical tank 10m in diameter contains methane at 400kPa and 32°C after 8kg had been used. Calculate the original mass of methane in the tank in kg.

Solution:

$$V = \frac{4}{3}\pi(5\text{m})^3 = 523.60\text{m}^3$$

$$PV = mRT$$

$$(400\text{kPa})(523.60\text{m}^3) = m \left(\frac{0.5182\text{kJ}}{\text{kg} - ^\circ\text{K}} \right) (32^\circ\text{C} + 273)$$

$$m = 1,325.14\text{kg}$$

293. A clock is powered by a 1.8kg mass hanging on a chain passing over and engaging a 5cm diameter pulley of the clock mechanism. How much energy in joules has been put into the mechanism when the weight drops to 1.2m?

Solution:

$$\text{Energy} = mgh = (1.8\text{kg}) \left(\frac{9.81\text{m}}{\text{s}^2} \right) (1.2\text{m}) = 21.20\text{J}$$

294. A thermodynamic system contains 15m³ of air at 320kPa and 144°C. The weight of this system in newton is.

Solution:

$$PV = mRT$$

$$(320\text{kPa})(15\text{m}^3) = m \left(\frac{0.287\text{kJ}}{\text{kg} - ^\circ\text{K}} \right) (144^\circ\text{C} + 273)$$

$$m = (40.1\text{kg}) \left(\frac{9.81\text{m}}{\text{s}^2} \right)$$

$$m = 393.45\text{N}$$

295. R-134A flows in a pipe at 28°C with a specific volume of 0.04525m³/kg. The internal energy and enthalpy of R-134A are 242.2kJ/kg and 280.4kJ/kg respectively. The pressure of the refrigerant in MPa is.

Solution:

$$h = u + Pv$$

$$\frac{280.4\text{kJ}}{\text{kg}} = \frac{242.2\text{kJ}}{\text{kg}} + P \left(\frac{0.04525\text{m}^3}{\text{kg}} \right)$$

$$P = 844.20\text{kPa or } 0.844\text{MPa}$$

296. Calculate the average constant pressure specific heat in kJ/kg-°K, if at constant pressure 100kJ/kg of heat is needed to raise the temperature from 32°C to 500°C.

Solution:

$$Q = mC_v\Delta T$$

$$\frac{1000\text{kJ}}{\text{kg}} = C_v(500^\circ\text{C} - 32^\circ\text{C})$$

$$C_v = \frac{2.14\text{kJ}}{\text{kg} - ^\circ\text{K}}$$

297. The sun generates 1kW/m² when used as a source for solar collectors. A collector with an area of 1m² heat water. the flow rate is 3liter/min. What is the temperature rise in the water? The specific heat of water is 4200J/kg-°C.

Solution:

$$Q = \left(\frac{1\text{kW}}{\text{m}^2} \right) (1\text{m}^2) = 1\text{kW or } 1000\text{Watts.}$$

$$m = \left(\frac{3\text{liters}}{\text{min}} \right) \left(\frac{1\text{kg}}{\text{lit}} \right) \left(\frac{\text{min}}{60\text{s}} \right) = \frac{0.05\text{kg}}{\text{s}}$$

$$Q = mC_p\Delta T$$

$$\Delta T = \frac{Q}{mC_p} = \frac{1000\text{Watts}}{\left(\frac{0.05\text{kg}}{\text{s}} \right) \left(\frac{4,200\text{J}}{\text{kg} - ^\circ\text{C}} \right)}$$

$$\Delta T = 4.76^\circ\text{C}$$

298. At an average temperature of 100°C, hot air flows through a 2.5m long tube with an inside diameter of 50mm. The temperature of the tube is 20°C along its entire length. Convective film coefficient is 20.1W/m²-°K. Determine the convective heat transfer from air to the tube.

Solution:

$$Q = Ah_o(T_2 - T_1) = (\pi(0.05\text{m})(2.5\text{m})) \left(\frac{20.1\text{W}}{\text{m}^2 - ^\circ\text{K}} \right) (100^\circ\text{C} - 20^\circ\text{C})$$

$$Q = 631.46\text{Watts}$$

299. Calculate the energy transfer rate across 6" wall of firebrick with a temperature difference across the wall of 50°C. The thermal conductivity of the firebrick is 0.65BTU/hr-ft-°F at the temperature interest.

Solution:

$$Q = \frac{kA\Delta T}{x} = \frac{\left(\frac{0.65\text{BTU}}{\text{hr} - \text{ft} - ^\circ\text{F}} \right) (1) \left(50^\circ\text{C} \left(\frac{9}{5} \right) \right)}{0.5\text{ft}} = \frac{117\text{BTU}}{\text{hr} - \text{ft}^2}$$

$$Q = \left(\frac{117\text{BTU}}{\text{hr} - \text{ft}^2} \right) \left(\frac{1055\text{J}}{\text{BTU}} \right) \left(\frac{1\text{hr}}{3600\text{s}} \right) \left(\frac{3.28\text{ft}}{\text{m}} \right)^2$$

$$Q = \frac{368.88\text{W}}{\text{m}^2}$$

300. What is the heat transfer in the glass surface area of 0.7m^2 having an inside temperature (room) of 25°C and 13°C outside temperature (surrounding). The thickness of glass surface is 0.007m . The glass thermal conductivity is 1.8W per meter per degree Kelvin.

Solution:

$$Q = \frac{kA\Delta T}{x} = \frac{\left(\frac{1.8\text{W}}{\text{m}^\circ\text{K}}\right)(0.7\text{m}^2)(25^\circ\text{C} - 13^\circ\text{C})}{0.007\text{m}}$$

$$Q = 2,160\text{W} \text{ or } 2.16\text{kW}$$

301. Calculate the logarithmic mean temperature difference for a condenser if the temperature difference between condensing steam and water inlet is 790°C and that steam inlet and water is 100°C .

Solution:

$$\text{LMTD} = \frac{790^\circ\text{C} - 100^\circ\text{C}}{\ln\left(\frac{790^\circ\text{C}}{100^\circ\text{C}}\right)} = 333.84^\circ\text{C}$$

302. One side of a 40cm stock wall ($k = 1.5\text{W}/\text{m}^\circ\text{K}$) is exposed to the air whose temperature is 18°C ($h = 15\text{W}/\text{m}^2\text{-}^\circ\text{K}$) and the other side is exposed to the furnace glasses whose temperature is 900°C ($h = 15\text{W}/\text{m}^2\text{-}^\circ\text{K}$). Estimate the heat transfer through the wall per m^2 area.

Solution:

$$\frac{Q}{A} = \frac{\Delta T}{\frac{1}{h_o} + \frac{x}{k} + \frac{1}{h_i}} = \frac{900^\circ\text{C} - 18^\circ\text{C}}{\frac{1}{15\text{W}} + \frac{0.4\text{m}}{1.5\text{W}} + \frac{1}{15\text{W}}}$$

$$\frac{Q}{A} = \frac{2,205\text{W}}{\text{m}^2}$$

303. At $\text{P}2.50/\text{kWh}$, how much will it cost to maintain a temperature of 96°F for 24 hours in a box 2ft^2 on each side if the outside temperature is 72°F and the overall heat transfer coefficient is $0.1\text{BTU}/\text{hr}\text{-ft}^2\text{-}^\circ\text{R}$?

Solution:

$$Q = UA\Delta T = \left(\frac{0.1\text{BTU}}{\text{hr}\text{-ft}^2\text{-}^\circ\text{F}}\right)(6\text{sides} * 2\text{ft} * \text{ft})(96^\circ\text{F} - 72^\circ\text{F})$$

$$Q = \left(\frac{57.6\text{BTU}}{\text{hr}}\right)\left(\frac{1.055\text{kJ}}{\text{BTU}}\right)\left(\frac{1\text{hr}}{3600\text{s}}\right)(2\text{hr}) = 0.40512\text{kWh}$$

$$\text{Cost} = (0.40512\text{kWh})\left(\frac{\text{P}2.50}{\text{kWh}}\right)$$

$$\text{Cost} = \text{P}1.01$$

304. Heat flow inside a storage by conduction. The amount of heat flowing is 250kJ in one hour. Compute the amount of heat flowing to the storage in 3hours, assuming all conditions are the same.

Solution:

$$Q = \left(\frac{250\text{kJ}}{\text{hr}}\right)(3\text{hr}) = 750\text{kJ}$$

305. The filament of a 150-watt bulb maybe considered a black body radiating into a black enclosure at 70°C . Considering only radiation, calculate the filament temperature in $^\circ\text{C}$, if the filament diameter is 0.20mm and the length is 8cm .

Solution:

$$Q = \frac{20,408.4 \times 10^{-8}(1)(\pi(0.0002\text{m})(0.08\text{m}))(T_2^4 - (70^\circ\text{C} + 273))}{3600}$$

$$T_2 = \frac{(3600)(150\text{Watt})}{20,408.4 \times 10^{-8}(1)(\pi(0.0002\text{m})(0.08\text{m}))} + 70^\circ\text{C} + 273$$

$$T_2 = 2693^\circ\text{K} - 273$$

$$T_2 = 2,420^\circ\text{C}$$

306. A cold soda initially at 3°C gains 30kJ of heat in a room at 22°C during a 20minute period. What is the average rate of heat transfer during the process?

Solution:

$$\text{Rate of heat transfer} = \frac{30,000\text{J}}{(20\text{min})(60\text{s})} = 25\text{Watts}$$

307. A plate type solar energy collector with an absorbing surface covered by a glass plate is to receive an incident radiation of $1,000\text{W}/\text{m}^2$. The glass plate has a reflectivity of 0.12 and transmissivity of 0.90. The absorbing surface has an absorptivity of 0.80. The area of the collector is 8m^2 . How many solar energy in watts is absorbed by the collector?

Solution:

$$\text{Energy} = \left(\frac{1,000\text{W}}{\text{m}^2}\right)(8\text{m}^2)(0.90)(0.8)$$

$$\text{Energy} = 5,760\text{Watts}$$

308. A tank containing liquid nitrogen at -120°C is suspended in a vacuum shell by three stainless steel rods 0.9cm in diameter and 3 meters long with a thermal conductivity of $18.3\text{W}/\text{m}\cdot^\circ\text{C}$. If the ambient air outside the vacuum shell is 28°C , calculate the magnitude of the conductive heat flow in watts along the support rods.

Solution:

$$Q = hA(T_2 - T_1) = \left(\frac{18.3\text{W}}{\text{m}\cdot^\circ\text{C}}\right)\left(\frac{\pi}{4}(0.009\text{m})^2\right)(28^\circ\text{C} + 120^\circ\text{C})$$

$$Q = 0.172\text{Watts}$$

309. Heat is transferred from hot water to an oil in a double pipe counter-flow heat exchanger. Water enter the outlet pipe at 100°C and exits at 50°C while the oil enters the inner pipe at 30°C and exits at 45°C . Calculate the Logarithmic mean temperature difference.

Solution:

$$\text{LMTD} = \frac{\Delta T_A - \Delta T_B}{\ln\left(\frac{\Delta T_A}{\Delta T_B}\right)} = \frac{(100^\circ\text{C} - 45^\circ\text{C}) - (50^\circ\text{C} - 30^\circ\text{C})}{\ln\left(\frac{(100^\circ\text{C} - 45^\circ\text{C})}{(50^\circ\text{C} - 30^\circ\text{C})}\right)}$$

$$\text{LMTD} = 34.60^\circ\text{C}$$

310. The surface of a furnace wall is at temperature of $1,400^\circ\text{C}$. The outside wall temperature is 50°C . The furnace wall is made of 30cm of refractory material having a thermal conductivity of $1.3\text{W}/\text{m}\cdot^\circ\text{K}$. The outside wall is steel, 2cm thick with thermal conductivity of $45\text{W}/\text{m}\cdot^\circ\text{K}$. Calculate the thickness in meters of brick to be installed in between the refractory material and steel if its thermal conductivity is $0.35\text{W}/\text{m}\cdot^\circ\text{K}$ and the heat loss is not to exceed $900\text{W}/\text{m}^2$

Solution:

$$\frac{Q}{A} = \frac{\Delta T}{\frac{x_1}{k_1} + \frac{x_2}{k_2} + \frac{x_3}{k_3}}$$

$$\frac{900\text{W}}{\text{m}^2} = \frac{1,400^\circ\text{C} - 50^\circ\text{C}}{\frac{0.3\text{m}}{\frac{1.3\text{W}}{\text{m}\cdot^\circ\text{K}}} + \frac{x_2}{\frac{0.35\text{W}}{\text{m}\cdot^\circ\text{K}}} + \frac{0.02\text{m}}{\frac{45\text{W}}{\text{m}\cdot^\circ\text{K}}}}$$

$$x_2 = 0.444\text{m}$$

311. An evaporator consisting of 34mm pipes is immersed in ordinary water flowing normal to the pipe at the velocity of 0.5m/s in the narrowest cross-section. The surface temperature is 2°C. Calculate the surface heat transfer coefficient in W/m²-°C. The pipes are staggered.

Solution:

$$h = \frac{370(1 + 0.0067T)V^{0.6}}{D^{0.4}} \left(\frac{\text{BTU}}{\text{hr} - \text{ft}^2 - ^\circ\text{F}} \right)$$

$$D = \text{inch}; T = ^\circ\text{F}; V = \frac{\text{ft}}{\text{s}}$$

$$h = \frac{370 \left(1 + 0.0067 \left(2 \left(\frac{9}{5} \right) + 32 \right) \right) \left(\frac{1.64\text{ft}}{\text{s}} \right)^{0.6}}{(1.34\text{in})^{0.4}} = \frac{548.49\text{BTU}}{\text{hr} - \text{ft}^2 - ^\circ\text{F}}$$

$$h = \left(\frac{548.49\text{BTU}}{\text{hr} - \text{ft}^2 - ^\circ\text{F}} \right) \left(\frac{1055\text{J}}{\text{BTU}} \right) \left(\frac{1\text{hr}}{3600\text{s}} \right) \left(\frac{3.23\text{ft}}{\text{m}} \right)^2 \left(\frac{1.8^\circ\text{F}}{^\circ\text{C}} \right)$$

$$h = \frac{3,113\text{W}}{\text{m}^2 - ^\circ\text{C}}$$

312. A 6in x 20ft (15cm x 6m) uninsulated B.I pipe conveys steam ($h = 1,000\text{BTU/hr-ft}^2\text{-}^\circ\text{F}$) at 385°F with an average ambient air ($h = 2\text{BTU/hr-ft}^2\text{-}^\circ\text{F}$) temperature of 85°F. Calculate of the heat transfer

Pipe properties of 6 inch for steam use

Outside diameter = 6.625in

Inside diameter = 5.761in

Conductivity = 52W/m-°K

Solution:

$$\frac{52\text{W}}{\text{m} - ^\circ\text{K}} = \frac{30\text{BTU}}{\text{hr} - \text{ft} - ^\circ\text{F}}$$

$$A_i = \pi \left(\frac{5.761\text{in}}{12} \right) (20\text{ft}) = 30.16\text{ft}^2$$

$$A_o = \pi \left(\frac{6.625\text{in}}{12} \right) (20\text{ft}) = 34.70\text{ft}^2$$

$$Q = \frac{T_1 - T_2}{\frac{1}{A_o h_o} + \frac{\ln \left(\frac{D_o}{D_i} \right)}{2\pi k L} + \frac{1}{A_i h_i}}$$

$$Q = \frac{385^\circ\text{F} - 85^\circ\text{F}}{\frac{1}{(34.70\text{ft}^2) \left(\frac{2\text{BTU}}{\text{hr} - \text{ft}^2 - ^\circ\text{F}} \right)} + \frac{\ln \left(\frac{6.625\text{in}}{5.761\text{in}} \right)}{2\pi \left(\frac{30\text{BTU}}{\text{hr} - \text{ft} - ^\circ\text{F}} \right) (20\text{ft})} + \frac{1}{(30.16\text{ft}^2) \left(\frac{1,000\text{BTU}}{\text{hr} - \text{ft}^2 - ^\circ\text{F}} \right)}}$$

$$Q = \frac{20,719\text{BTU}}{\text{hr}}$$

313. What is the rate of heat flow through a brick wall that is constructed 30cm thick and 5m x 4m in area? The temperature on one side is 180°C and 60°C on the other side. The average coefficient of thermal conductivity is equal to 0.80W/m-°K.

Solution:

$$Q = \frac{kA\Delta T}{x} = \frac{\left(\frac{0.80\text{W}}{\text{m} - ^\circ\text{K}} \right) (20\text{m}^2) (180^\circ\text{C} - 60^\circ\text{C})}{0.3\text{m}}$$

$$Q = 6400\text{Watts or } 6.4\text{kW}$$

314. A series of composite walls of a furnace consists of 20cm of fire brick, 15cm of a high temperature insulating material, 12cm of ordinary brick, and, lastly, 80mm of asbestos cement board. The inside of the furnace wall is 900°C and the temperature of the outside wall is 100°C. Determine the heat transfer through the wall per m².

Fire brick	$k = 1.4\text{W/m}^\circ\text{K}$
High temperature insulation	$k = 0.22\text{W/m}^\circ\text{K}$
Ordinary brick	$k = 0.90\text{W/m}^\circ\text{K}$
Asbestos cement board	$k = 0.39\text{W/m}^\circ\text{K}$

Solution:

$$\frac{Q}{A} = \frac{\Delta T}{\frac{x_1}{k_1} + \frac{x_2}{k_2} + \frac{x_3}{k_3} + \frac{x_4}{k_4}}$$

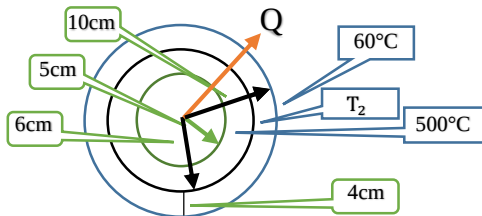
$$\frac{Q}{A} = \frac{900^\circ\text{C} - 100^\circ\text{C}}{\frac{0.2\text{m}}{1.4\text{W/m}^\circ\text{K}} + \frac{0.15}{0.22\text{W/m}^\circ\text{K}} + \frac{0.12}{0.90\text{W/m}^\circ\text{K}} + \frac{0.08\text{m}}{0.39\text{W/m}^\circ\text{K}}}$$

$$\frac{Q}{A} = \frac{688\text{W}}{\text{m}^2}$$

For Number 315 to 316:

A 12cm outside diameter and 10cm inside diameter cast iron pipe with conductivity of 80W/m-°K has 4cm polystyrene insulation with a conductivity of 0.027W/m-°K. Saturated steam at 500°C is flowing through the pipe and the outside surface temperature is 60°C.

315. Determine the heat transfer per 150m of pipe.



Solution:

$$Q = \frac{\Delta T}{\frac{\ln(r_2/r_1)}{2\pi k_1 L_1} + \frac{\ln(r_3/r_2)}{2\pi k_2 L_2}} = \frac{500^\circ\text{C} - 60^\circ\text{C}}{\frac{\ln(12\text{cm}/10\text{cm})}{2\pi(80\text{W/m}^\circ\text{K})(150\text{m})} + \frac{\ln(20\text{cm}/12\text{cm})}{2\pi(0.027\text{W/m}^\circ\text{K})(150\text{m})}}$$

$$Q = 21.92\text{kW}$$

316. Determine the temperature T_2

Solution:

$$Q = \frac{\Delta T}{\frac{\ln(r_3/r_2)}{2\pi k_2 L_2}} = \frac{T_2 - 60^\circ\text{C}}{\frac{\ln(20\text{cm}/12\text{cm})}{2\pi(0.027\text{W/m}^\circ\text{K})(150\text{m})}}$$

$$21,916 = \frac{T_2 - 60^\circ\text{C}}{\frac{\ln\left(\frac{20\text{cm}}{12\text{cm}}\right)}{2\pi\left(\frac{0.027\text{W}}{\text{m} \cdot ^\circ\text{K}}\right)(150\text{m})}}$$

$$T_2 = 500^\circ\text{C}$$

317. An iron pipe $d = 3\text{in}$ and $D = 6\text{in}$, contains steam such that its inner surface is at 340°F and its outer surface is at 220°F . Assuming that the thermal conductivity is $40\text{BTU/hr}\cdot\text{ft}\cdot^\circ\text{F}$, the heat transfer rate per foot of pipe length is:

Solution:

$$Q = \frac{2\pi kL(T_2 - T_1)}{\ln\left(\frac{D_o}{D_i}\right)}$$

$$Q = \frac{2\pi\left(\frac{40\text{BTU}}{\text{hr}\cdot\text{ft}\cdot^\circ\text{F}}\right)(1\text{ft})(340^\circ\text{F} - 220^\circ\text{F})}{\ln\left(\frac{0.5\text{ft}}{0.25\text{ft}}\right)}$$

$$Q = \frac{43,510\text{BTU}}{\text{hr} \cdot ^\circ\text{F}}$$

318. A certain body at 20°C is displayed on a top of a building during the night. The body sees nothing but the sky which has an effective temperature of 110°K . Determine the heat transfer rate from the body to the sky if the body temperature is maintained at 23°C , the surface emissivity of the body is equal to 0.92, and none of the radiation going out of the comes back.

Solution:

$$Q = \varepsilon \sigma A T^4 = \left(\frac{5.669 \times 10^{-8}\text{W}}{\text{m}^2 \cdot \text{K}^4}\right)(0.92)(1\text{m}^2)((23^\circ\text{C} + 273)^4 - (110\text{K})^4)$$

$$Q = 400\text{W}$$

319. A 2.5cm diameter electric power wire used in transmission line transmits 14,000 Volts and 750amps. The resistance of the wire is $0.0006\Omega/\text{m}$ and is wrapped with 0.01cm insulation of $k = 0.07\text{W/m}\cdot\text{K}$. The convection coefficient is an average of $27\text{W/m}^2\cdot\text{K}$ and the outside temperature is a -5°C . the power in kW dissipated in the wire per km is most nearly

Solution:

$$Q = I^2 R = (750)^2 \left(\frac{0.0006\Omega}{\text{m}}\right) = \frac{337.5\text{W}}{\text{m}}$$

$$\text{Total } Q = \frac{337.5(1,000)}{1000} = \frac{337.5\text{kW}}{\text{km}}$$

320. The heat transfer from a 3m diameter sphere to a 27°C air stream over a time interval of one hour is 4,000kJ. Estimate the surface temperature of the sphere if the heat transfer coefficient is $15\text{W/m}^2\cdot\text{K}$.

Solution:

$$\frac{4,000,000\text{J}}{\text{hr}} \left(\frac{1\text{hr}}{3600\text{s}}\right) = hA(T_o - T_i) = \left(\frac{15\text{W}}{\text{m}^2 \cdot ^\circ\text{K}}\right)(4\pi(1.5\text{m})^2)(T_o - 27)$$

$$T_o = 30^\circ\text{C}$$

321. Estimate the rate of heat transfer from a 240°C sphere which has an emissivity of 0.9. If it is suspended in a cold volume maintained at -10°C . The sphere has a diameter of 40cm.

Solution:

$$Q = \varepsilon \sigma A T^4 = \left(\frac{5.669 \times 10^{-8}\text{W}}{\text{m}^2 \cdot \text{K}^4}\right)(0.9)(4\pi(0.02\text{m})^2)(240^\circ\text{C} + 273)^4 - (-10^\circ\text{C} + 273)^4)$$

$$Q = 1653.50 \text{ Watts}$$

322. A plane wall is 3.5m high by 3.5m wide and 30cm thick. It is made of material that has a thermal conductivity of 0.5W/m-K. A temperature difference of 50°C is imposed on the two large faces. Find the heat flow in Watts.

$$Q = \frac{kA\Delta T}{x} = \frac{\left(\frac{0.5W}{m-\text{K}}\right)(3.5m * 3.5m)(50^{\circ}\text{C})}{0.3m}$$

$$Q = 1021 \text{ Watts}$$

323. In the problem #322, determine the heat flux in W/m².

Solution:

$$\frac{Q}{A} = \frac{1021 \text{ Watts}}{(3.5m)(3.5m)}$$

$$\frac{Q}{A} = \frac{83.33W}{m^2}$$

324. In problem #322, determine the conductive resistance in °K/W.

Solution:

$$R = \frac{x}{kA} = \frac{0.3m}{\left(\frac{0.5W}{m-\text{K}}\right)(3.5m * 3.5m)}$$

$$R = 0.05^{\circ}\frac{K}{\text{Watt}}$$

325. Water at an average temperature of 23°C flows through a 10cm diameter pipe that is 2.5m long. The pipe wall is heated by steam and is held at 100°C. The convective heat transfer coefficient is 2.25x10⁴W/m²-K. Find the heat flow in Watts.

Solution:

$$Q = hA(T_2 - T_1) = \left(\frac{2.25 \times 10^4 W}{m^2 - \text{K}}\right)(\pi(0.1m)(2.5m))(100^{\circ}\text{C} - 23^{\circ}\text{C})$$

$$Q = 13.6 \times 10^5 \text{ Watts}$$

326. In problem #425, determine the heat flux in Watt per square meters.

Solution:

$$\frac{Q}{A} = \frac{13.6 \times 10^5 \text{ Watts}}{(\pi)(0.1m)(2.5m)}$$

$$\frac{Q}{A} = \frac{1.73 \times 10^6 \text{ Watts}}{m^2}$$

327. Find the convective resistance in °K/Watt.

Solution:

$$R = \frac{1}{hA} = \frac{1}{\left(\frac{2.25 \times 10^4 W}{m^2 - \text{K}}\right)((\pi)(0.1m)(2.5m))}$$

$$R = \frac{5.66 \times 10^{-5} \text{ K}}{\text{Watts}}$$

328. A steam pipe with a surface area of 7.2m² and a surface temperature of 700°C radiates into a large room (which acts as a black body), the surfaces of which are at 27°C. The pipe-gray body surface emissivity is 0.70. Find the heat flow from the surface to the room in Watts.

Solution:

$$Q = \varepsilon \sigma A T^4 = \left(\frac{5.669 \times 10^{-8} W}{m^2 - K^4}\right)(0.70)(7.2m^2)((700^{\circ}\text{C} + 273)^4 - (27^{\circ}\text{C} + 273)^4)$$

$$Q = 25.38 \times 10^4 \text{ Watts}$$

329. In the problem #328, determine the heat flux from the surface to the room in W/m^2 .

Solution:

$$\frac{Q}{A} = \frac{25.38 \times 10^4 \text{ Watts}}{7.2 \text{ m}^2}$$

$$\frac{Q}{A} = \frac{35246 \text{ Watts}}{\text{m}^2}$$

330. A quantity of 70 cubic meters of water passes through a heat exchanger and absorbs 2,500 MJ. The exit temperature is 90°C . Find the entrance water temperature in $^\circ\text{C}$.

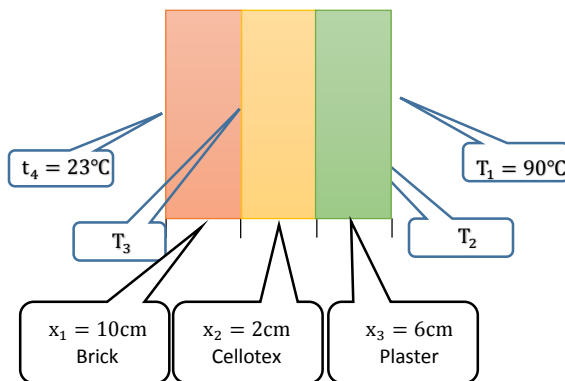
Solution:

$$Q = mC_p(T_2 - T_1)$$

$$2,500,000 \text{ kJ} = (70 \text{ m}^3) \left(\frac{1,000 \text{ kg}}{\text{m}^3} \right) \left(\frac{4.187 \text{ kJ}}{\text{kg} - ^\circ\text{K}} \right) (90 - T_1)$$

$$T_1 = 81.50^\circ\text{C}$$

331. The composite wall which is composed of 10cm brick ($k = 0.40 \text{ W/m-C}$), 2cm cellotex ($k = 0.03 \text{ W/m-C}$), and 6cm plaster ($k = 0.30 \text{ W/m-C}$), has an outer (brick) temperature $T = 23^\circ\text{C}$ and an inner (plaster) temperature $T = 90^\circ\text{C}$. Find the temperature between cellotex and plaster in $^\circ\text{C}$.



Solution:

$$R_{\text{brick}} = \frac{x}{k} = \frac{0.10 \text{ m}}{0.4 \frac{\text{W}}{\text{m} - ^\circ\text{C}}} = \frac{0.25 \text{ m}^2 - ^\circ\text{C}}{\text{W}}$$

$$R_{\text{cellotex}} = \frac{x}{k} = \frac{0.02 \text{ m}}{0.03 \frac{\text{W}}{\text{m} - ^\circ\text{C}}} = \frac{0.667 \text{ m}^2 - ^\circ\text{C}}{\text{W}}$$

$$R_{\text{plaster}} = \frac{x}{k} = \frac{0.06 \text{ m}}{0.3 \frac{\text{W}}{\text{m} - ^\circ\text{C}}} = \frac{0.2 \text{ m}^2 - ^\circ\text{C}}{\text{W}}$$

$$R_T = \frac{0.25 \text{ m}^2 - ^\circ\text{C}}{\text{W}} + \frac{0.667 \text{ m}^2 - ^\circ\text{C}}{\text{W}} + \frac{0.2 \text{ m}^2 - ^\circ\text{C}}{\text{W}}$$

$$R_T = \frac{1.117 \text{ m}^2 - ^\circ\text{C}}{\text{W}}$$

$$\frac{Q}{A} = \frac{\Delta T}{\Delta R_T} = \frac{90^\circ\text{C} - 23^\circ\text{C}}{\frac{1.117 \text{ m}^2 - ^\circ\text{C}}{\text{W}}} = \frac{60 \text{ W}}{\text{m}^2}$$

$$\frac{Q}{A} = \frac{T_1 - T_2}{R_{\text{plaster}}} = \frac{(90^\circ\text{C} + 273) - T_2}{\frac{0.2\text{m}^2 - ^\circ\text{C}}{\text{W}}} = \frac{60\text{W}}{\text{m}^2}$$

$$T_3 = 351^\circ\text{K} - 273 = 78^\circ\text{C}$$

332. In problem #331, determine the temperature between brick and cellox.

Solution:

$$\frac{Q}{A} = \frac{T_3 - T_4}{R_{\text{brick}}} = \frac{T_3 - (23 + 273)}{\frac{0.25\text{m}^2 - ^\circ\text{C}}{\text{W}}} = \frac{60\text{W}}{\text{m}^2}$$

$$T_3 = 311^\circ\text{K} - 273 = 38^\circ\text{C}$$

333. A plaster wall which is 20cm thick with a thermal conductivity of $0.30\text{W/m}^2\text{-}^\circ\text{C}$ is used in certain house. The inner wall temperature is at 40°C , and the outer wall is exposed to ambient wind and the surroundings at 12°C . The film coefficient, h , for convective heat transfer in a 10m/s wind is about $21\text{W/m}^2\text{-}^\circ\text{C}$. Ignoring any radiation losses, find the overall coefficient for the conduction and convection losses in $\text{W/m}^2\text{-}^\circ\text{C}$.

Solution:

$$U = \frac{1}{\frac{1}{h} + \frac{x}{k}} = \frac{1}{\frac{1}{\frac{21\text{W}}{\text{m}^2 - ^\circ\text{C}}} + \frac{0.2\text{m}}{\frac{0.30\text{W}}{\text{m}^2 - ^\circ\text{C}}}}$$

$$U = \frac{1.4\text{W}}{\text{m}^2 - ^\circ\text{C}}$$

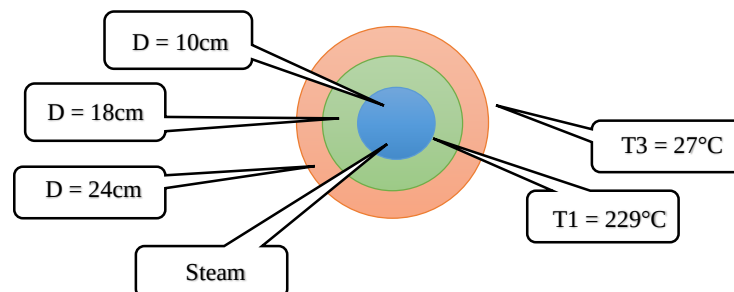
334. Heat is transferred by conduction through composite wall from the outer layer of concrete to the outer layer of wood. Assume the three materials are in good thermal contact and that has no significant thermal resistance exists at any of the interfaces. The material are composed of 30cm concrete ($k = 0.80\text{W/m}^2\text{-}^\circ\text{C}$), 15cm rock wool ($k = 0.04\text{W/m}^2\text{-}^\circ\text{C}$) and 15cm wood ($k = 0.07\text{W/m}^2\text{-}^\circ\text{C}$). Find the overall coefficient U in $\text{W/m}^2\text{-}^\circ\text{C}$.

Solution:

$$U = \frac{1}{\frac{x_1}{k_1} + \frac{x_2}{k_2} + \frac{x_3}{k_3}} = \frac{1}{\frac{0.3\text{m}}{\frac{0.80\text{W}}{\text{m}^2 - ^\circ\text{C}}} + \frac{0.15\text{m}}{\frac{0.04\text{W}}{\text{m}^2 - ^\circ\text{C}}} + \frac{0.15\text{m}}{\frac{0.07\text{W}}{\text{m}^2 - ^\circ\text{C}}}}$$

$$U = \frac{0.16\text{W}}{\text{m}^2 - ^\circ\text{C}}$$

335. A 10cm diameter steam pipe is covered by 2 layers of lagging. The inside layer is 4cm thick and has a coefficient of thermal conductivity of 0.08W/m-K . The outside layer is 3cm thick and has a coefficient of thermal conductivity of 0.15W/m-K . The steam main conveys steam at a pressure of 1.7MPa with 25°C superheat. The outside temperature of the lagging is 27°C . If the steam main is 30m long, find the heat loss in kW. (From the steam table: T_{sat} @ 1.7MPa is 204°C).



Solution:

$$D_1 = 10\text{cm}$$

$$D_2 = 10\text{cm} + 2(4\text{cm}) = 18\text{cm}$$

$$D_3 = 18\text{cm} + 2(3\text{cm}) = 24\text{cm}$$

$$Q = \frac{T_1 - T_3}{\frac{\ln\left(\frac{r_2}{r_1}\right)}{2\pi k_1 L} + \frac{\ln\left(\frac{r_3}{r_2}\right)}{2\pi k_2 L}} = \frac{229^\circ\text{C} - 27^\circ\text{C}}{\frac{\ln\left(\frac{18\text{cm}}{10\text{cm}}\right)}{2\pi\left(\frac{0.08\text{W}}{\text{m} - ^\circ\text{K}}\right)(30\text{m})} + \frac{\ln\left(\frac{24\text{cm}}{18\text{cm}}\right)}{2\pi\left(\frac{0.15\text{W}}{\text{m} - ^\circ\text{K}}\right)(30\text{m})}}$$

$$Q = 4,109.60\text{Watts or } 4.1096\text{kW}$$

336. In problem #335, determine the interface temperature of the lagging.

Solution:

$$Q = \frac{T_1 - T_2}{\frac{\ln\left(\frac{r_2}{r_1}\right)}{2\pi k_1 L}}$$

$$4109.60\text{Watts} = \frac{(229^\circ\text{C} - T_2)}{\frac{\ln\left(\frac{18}{10}\right)}{2\pi\left(\frac{0.08\text{W}}{\text{m} - ^\circ\text{K}}\right)(30\text{m})}}$$

$$T_2 = 68.81^\circ\text{C}$$

337. A composite furnace wall has an external wall of 120mm thick brickwork, middle wall of 80mm thick insulating brick and internal wall of 30mm thick high temperature firebrick. Surface coefficients are $5\text{W/m}^2\text{-}^\circ\text{K}$ for external wall surface and $10\text{W/m}^2\text{-}^\circ\text{K}$ for internal wall surface. Temperature of hot gas is 40°C and of cold gas is 18°C . Thermal conductivities are 1.7W/m-K for external wall, 0.05W/m-K for middle wall; and 0.08W/m-K for internal wall. Determine the overall heat transfer coefficient.

Solution:

$$U = \frac{1}{\frac{i}{\frac{5\text{W}}{\text{m}^2 - ^\circ\text{K}}} + \frac{0.12\text{m}}{1.7\text{W}} + \frac{0.08\text{m}}{0.05\text{W}} + \frac{0.03\text{m}}{0.08\text{W}} + \frac{1}{\frac{10\text{W}}{\text{m}^2 - ^\circ\text{K}}}}$$

$$U = \frac{0.426\text{W}}{\text{m}^2 - ^\circ\text{K}}$$

338. In the problem #337, determine the heat loss per square meter of surface area.

Solution:

$$\frac{Q}{A} = \left(\frac{0.426\text{W}}{\text{m}^2 - ^\circ\text{K}}\right)(40^\circ\text{C} - 18^\circ\text{C})$$

$$\frac{Q}{A} = \frac{9.38\text{W}}{\text{m}^2}$$

339. A 25cm x 25cm test panel, 3cm thick is placed between two plates and the whole is properly insulated. The surface of the plate is maintained at 90°C by an electrically heated energy of 500 Watts. The other surface of the plate has a temperature of 25°C . Find the thermal conductivity of the test panel in W/m-K .

Solution:

$$Q = \frac{kA\Delta T}{x}$$

$$500\text{W} = \frac{k(0.25\text{m} * 0.25\text{m})(90^\circ\text{C} - 25^\circ\text{C})}{0.03\text{m}}$$

$$k = \frac{3.70W}{m - ^\circ C}$$

340. Determine the thermal conductivity of a material used in a 1.5m x 1.5m test panel, 5cm thick, and if during a 4hr test period there conducted 150kcal through the panel with a temperature differential of 10°C between the surfaces.

Solution:

$$Q = \frac{kA\Delta T}{x}$$

$$\frac{(150kcal) \left(\frac{4.184kJ}{kCal} \right)}{(4hrs)(3600s)} = \frac{k(1.5m * 1.5m)(10^\circ C)}{0.05m}$$

$$k = \frac{9.70 \times 10^{-5} W}{m - ^\circ C}$$

341. What is the critical radius in cm for an insulated pipe ($k = 0.35W/m-K$) if the external heat transfer coefficient is $10W/m^2-K$.

Solution:

$$R_{critical} = \frac{k}{h} = \frac{\frac{0.35W}{m-K}}{\left(\frac{10W}{m^2-K} \right)} = 3.5cm$$

342. Carnot ideal refrigerator operates between 0 and 30°C. What fraction of the cooling effect in kW is required as work input?

Solution:

$$COP = \frac{T_L}{T_H - T_L} = \frac{0^\circ C + 273}{(30^\circ C + 273) - (0^\circ C + 273)} = 9.1$$

$$Work\ input = \frac{Q}{COP} = \frac{1kW}{9.1} = 0.109kW$$

343. A Carnot reversed engine has a coefficient of performance of 3.5, when discharging heat at 45°C. What is the heat –supply temperature in (°C)

Solution:

$$COP = \frac{T_L}{T_H - T_L}$$

$$3.5 = \frac{T_L}{(45^\circ C + 273) - T_L}$$

$$T_L = 247.44^\circ K - 273$$

$$T_L = -25.70^\circ C$$

344. A reversed Carnot ideal cycle has a refrigerating COP of 4.5. What is the ratio $\frac{T_H}{T_L}$?

Solution:

$$COP = \frac{T_L}{T_H - T_L}$$

$$4.5 = \frac{T_L}{T_H - T_L}; T_L = 4.5(T_H - T_L) = 4.5T_H - 4.5T_L$$

$$4.5T_L + T_L = 4.5T_H$$

$$5.5T_L = 4.5T_H$$

$$\frac{T_H}{T_L} = 1.22$$

345. In an “ideal” situation, what is the COP of a refrigerating cycle that is operating between minus -40°C and 40°C ?

Solution:

$$\text{COP} = \frac{T_L}{T_H - T_L} = \frac{-40^{\circ}\text{C} + 273}{(40^{\circ}\text{C} + 273) - (-40^{\circ}\text{C} + 273)}$$

$$\text{COP} = 2.9 \approx 3$$

346. What is the heat leakage, in kW, on a 1 in thick plywood wall which has an area 4 x 8ft given a 20°C difference between the surface temperatures. Thermal conductivity of plywood is $0.115\text{W/m}\cdot\text{K}$.

Solution:

$$Q = \frac{kA\Delta T}{x} = \frac{(4\text{ft} * 8\text{ft}) \left(\frac{0.115\text{W}}{\text{m}\cdot^{\circ}\text{K}}\right) (20^{\circ}\text{C})}{(1\text{in} * 0.0254\text{m})(3.28)^2}$$

$$Q = 267.33\text{Watt or } 0.269\text{kW}$$

347. A refrigerated room has 4in. thick hollow block wall ($k = 0.7\text{ W/m}\cdot^{\circ}\text{C}$). If the convection coefficient inside and outside the room is $9.3\text{ W/m}^2\cdot^{\circ}\text{C}$ and $30\text{ W/m}^2\cdot^{\circ}\text{C}$, respectively, calculate the over-all heat transfer coefficient ($\text{W/m}^2\cdot^{\circ}\text{C}$).

Solution:

$$U = \frac{1}{\frac{1}{\frac{9.3\text{W}}{\text{m}^2\cdot^{\circ}\text{C}}} + \frac{4 * 0.0254\text{m}}{0.7\text{W}} + \frac{1}{\frac{30\text{W}}{\text{m}^2\cdot^{\circ}\text{C}}}}$$

$$U = \frac{3.5\text{W}}{\text{m}^2\cdot^{\circ}\text{C}}$$

348. How much ice, in kg, is needed to cool a gallon of punch from 20°C to 4°C .? Assume specific heat of punch is $4.23\text{ kJ/kg}\cdot^{\circ}\text{C}$, heat of fusion of ice is 344 kJ/kg and ice is initially at freezing temperature

Solution:

$$1\text{gal} = 3.7854\text{liters} = 3.7854\text{kg}$$

$$1\text{liter or water or punch} = 1\text{kg}$$

$$Q_i = Q_p$$

$$m_i h_f + m_i C_{p/i} \Delta T_i = m_p C_{p/p} \Delta T_p$$

$$m_i = \frac{m_p C_{p/p} \Delta T_p}{h_f + C_{p/i} \Delta T} = \frac{(3.7854\text{kg}) \left(\frac{4.23\text{kJ}}{\text{kg}}\right) (16^{\circ}\text{C})}{\frac{344\text{kJ}}{\text{kg}} + \left(\frac{4.187\text{kJ}}{\text{kg}}\right) (4^{\circ}\text{C})}$$

$$m_i = 0.71\text{kg}$$

349. Calculate the heat removal (kJ) to freeze 1 kg water for 30°C to -4°C . Assume C_p above freezing of $4.19\text{ kJ/kg}\cdot\text{K}$, C_p below freezing of $1.94\text{ kJ/kg}\cdot\text{K}$ and latent heat of fusion 335 kJ/kg .

Solution:

$$Q = (1\text{kg}) \left[\left(\frac{4.19\text{kJ}}{\text{kg}}\right) (30^{\circ}\text{C} - 0^{\circ}\text{C}) + \frac{335\text{kJ}}{\text{kg}} + \left(\frac{1.94\text{kJ}}{\text{kg}\cdot^{\circ}\text{K}}\right) (0^{\circ}\text{C} + 4^{\circ}\text{C}) \right]$$

$$Q = 468.46\text{kJ}$$

350. In the problem # 349. Refrigeration system capacity for products brought into refrigerated space determined from the time allotted for heat removal and assumes that the product is properly exposed to remove the heat in that time, supposing it takes 3 hours to freeze the water in the previous problem, calculate the refrigerating capacity in W.

Solution:

$$\text{Refrigerating Capacity} = \frac{468.46\text{kJ}}{3\text{hr}} \left(\frac{1\text{hr}}{3,600\text{s}} \right)$$

$$\text{Refrigerating Capacity} = 43.38\text{Watts}$$

351. Most food products freeze in the range of -3 to -0.5°C . When the exact freezing temperature is not known, we can assume that it is

Solution:

$$T_{\text{freezing}} = \frac{-3^\circ\text{C} - 3^\circ\text{C} - 0.5^\circ\text{C}}{3}$$

$$T_{\text{freezing}} = -2.2^\circ\text{C}$$

352. 100 kg of lean beef is to be cooled from 18 to 4°C . Specific heat of beef before freezing is 3.23kJ/kg-K , after freezing is 1.68kJ/kg-K . The latent heat of fusion is 233kJ/kg . Determine the product load in kJ.

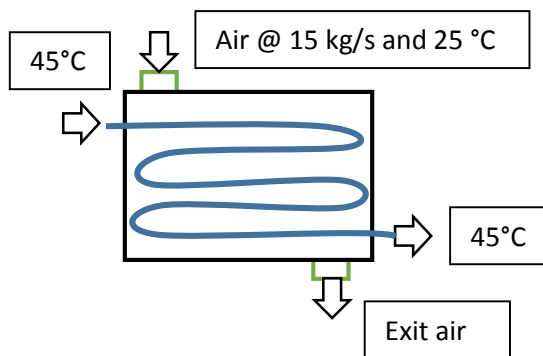
Solution:

$$Q = (100\text{kg}) \left(\frac{3.23\text{kJ}}{\text{kg} - ^\circ\text{K}} \right) (18^\circ\text{C} - 4^\circ\text{C})$$

$$Q = 4,522\text{kJ}$$

353. A condenser is to transfer 60 kW with an air flow rate of 15 kg/s entering at 25°C and a condensing temperature of 45°C . The exit temperature, in Celsius, is

Solution:



$$Q = mC_p(T_2 - T_1)$$

$$60\text{kW} = \left(\frac{15\text{kg}}{\text{s}} \right) \left(\frac{1\text{kJ}}{\text{kg} - \text{K}} \right) (T_2 - (25 + 273))$$

$$T_2 = 302^\circ\text{K} - 273 = 29^\circ\text{C}$$

354. In problem #354, determine the Mean Logarithmic Temperature Difference.

Solution:

$$\text{MLTD} = \frac{(45^\circ\text{C} - 25^\circ\text{C}) - (45^\circ\text{C} - 29^\circ\text{C})}{\ln \left(\frac{45^\circ\text{C} - 25^\circ\text{C}}{45^\circ\text{C} - 29^\circ\text{C}} \right)} = 17.92^\circ\text{C}$$

355. The following properties of water are at 10°C : $\mu = 0.00131\text{ Pa-s}$; $\rho = 1000\text{ kg/m}^3$; $k = 0.573\text{ W/m-K}$; $C_p = 4190\text{ J/kg-K}$. Consider an evaporator with water at 10°C flowing inside the tubes (8 mm ID) with velocity of 2 m/s . The Reynolds number is

Solution:

$$Re = \frac{\rho DV}{\mu} = \frac{\left(1000 \frac{\text{kg}}{\text{m}^3} \right) (0.008\text{m}) \left(\frac{2\text{m}}{\text{s}} \right)}{\frac{0.00131\text{kg} - \text{m} - \text{s}}{\text{m}^2 - \text{s}^2}}$$

$$Re = 12,213.74$$

356. In problem #355, determine the Prandtl number.

Solution:

$$Pr = \frac{C_p \mu}{k} = \frac{\left(\frac{4,190 \text{ J}}{\text{kg} \cdot \text{K}}\right) \left(\frac{0.00131 \text{ kg} \cdot \text{m} \cdot \text{s}}{\text{m}^2 \cdot \text{s}^2}\right)}{0.573 \frac{\frac{\text{J}}{\text{s}}}{\text{m} \cdot \text{K}}}$$

$$Pr = 9.60$$

357. In problem #355, determine the Nusselt number.

$$Nu = [0.023(Re)^{0.8}](Pr)^n; n = 0.4 \text{ for heating, } 0.3 \text{ for cooling}$$

$$Nu = [0.023(Re)^{0.8}](Pr)^n = [0.023(12,213.74)^{0.8}](9.60)^{0.3}$$

$$Nu = 84.31$$

For numbers 358-360. Consider the problem below:

Supposing that in a certain area the temperature is 34°C, relative humidity is 65 percent and barometric pressure is 101.3kPa. Assume further:

S = Saturated; W = water

Saturation pressure at 34°C = 5.318kPa

Specific heat of dry air = 1.006 kJ/kg-K

Enthalpy of saturated steam at 34°C = 2563.6 kJ/kg

Gas constant of dry air = 287 J/kg-K

Gas constant of water vapor = 461.5 J/kg-K

358. The humidity ratio in decimal is

Solution:

$$(\phi) = \frac{P_w}{P_{ws}} = 0.65 \text{ or } 65\%$$

$$P_w = 0.65 P_{ws} = 0.65(5.318 \text{ kPa})$$

$$P_w = 3.4567 \text{ kPa}$$

$$\text{Humidity Ratio} = \frac{0.622 P_w}{P_{\text{atm}} - P_w} = \frac{0.622(3.4567 \text{ kPa})}{101.325 \text{ kPa} - 3.4567 \text{ kPa}}$$

$$\text{Humidity Ratio} = 0.02197$$

359. The Enthalpy, in kJ/kg

Solution:

$$h = C_{p-\text{air}} T + (HR) h_s = \left(\frac{1.006 \text{ kJ}}{\text{kg} \cdot ^\circ\text{K}}\right) (34^\circ\text{C}) + (0.02197) \left(\frac{2,563.6 \text{ kJ}}{\text{kg}}\right)$$

$$h = \frac{90.53 \text{ kJ}}{\text{kg}}$$

360. The specific volume of the air-vapor mixture in cubic meter per kilogram.

Solution:

$$v = \left(\frac{R_a T}{P}\right) (1 + 1.608 RH)$$

$$v = \left[\frac{\left(\frac{0.287 \text{ kJ}}{\text{kg} \cdot \text{K}}\right) (34^\circ\text{C} + 273)}{101.325 \text{ kPa}} \right] (1 + 1.608(0.02197))$$

$$v = \frac{0.9\text{m}^3}{\text{kg}}$$

For numbers 361 and 362. Consider the problem below:

The outer-facing of the room is constructed from 25 cm brick, 2.5 cm mortar, 10 cm of limestone ($k = 0.186 \text{ W/m-K}$) and 1.2 cm plaster ($k = 0.096 \text{ W/m-K}$). Thermal conductivities of mortar and brick are both 0.52 W/m-K . Assume that the heat transfer coefficients on the inside (plaster side) and outside (brick side) surfaces of the wall to be 6 and $12 \text{ W/m}^2\text{-K}$, respectively.

361. Calculate the overall coefficient of heat transfer

Solution:

$$U = \frac{1}{\frac{1}{h_o} + \frac{x_1}{k_1} + \frac{x_2}{k_2} + \frac{x_3}{k_3} + \frac{x_4}{k_4} + \frac{1}{h_i}}$$

$$U = \frac{1}{\frac{1}{12\text{W}} + \frac{0.25\text{m}}{0.52\text{W}} + \frac{0.025\text{m}}{0.52\text{W}} + \frac{0.1\text{m}}{0.186\text{W}} + \frac{0.0012\text{m}}{0.096\text{W}} + \frac{1}{6\text{W}}}$$

$$U = \frac{0.6937\text{W}}{\text{m}^2 - ^\circ\text{K}}$$

362. Calculate the rate of heat transfer (W) per 10m^2 of the wall surface from the room at 18°C to the outside air at 36°C .

Solution:

$$Q = AU\Delta T = (10\text{m}^2) \left(\frac{0.6937\text{W}}{\text{m}^2 - ^\circ\text{K}} \right) (36^\circ\text{C} - 18^\circ\text{C})$$

$$Q = 124.866\text{Watts}$$

363. Twenty three hundred liters of partially frozen ice cream at -4°C enter a hardening room each day. Hardening is completed and the temperature of ice cream is lowered to -28°C in 10 hours. The average density of ice cream is 0.6 kg/L , the average latent heat per kg is 233 kJ . Assume C_p above freezing is $0.7\text{cal/g-}^\circ\text{C}$, C_p below freezing is $0.39\text{cal/g-}^\circ\text{C}$ and freezing points is -5.6°C . 1cal is equal to 4.18 J . The product load in kW is

Solution:

$$m = 2,300\text{Liter} \left(\frac{0.6\text{kg}}{\text{L}} \right) = 1,380\text{kg of ice}$$

$$\frac{0.7\text{Cal}}{\text{g} - ^\circ\text{C}} = \frac{2,926\text{J}}{\text{kg} - ^\circ\text{C}}$$

$$\frac{0.39\text{Cal}}{\text{g} - ^\circ\text{C}} = \frac{1,630.2\text{J}}{\text{kg} - ^\circ\text{C}}$$

$$Q = (1,380\text{kg}) \left[\left(\frac{2,926\text{kJ}}{\text{kg} - ^\circ\text{C}} \right) (-4^\circ\text{C} + 5.6^\circ\text{C}) + \left(\frac{1,630.2\text{kJ}}{\text{kg} - ^\circ\text{C}} \right) (-5.6^\circ\text{C} + 28^\circ\text{C}) + \frac{233\text{kJ}}{\text{kg}} \right]$$

$$Q = \frac{378,387.168\text{kJ}}{(10\text{hrs})(3,600\text{s})}$$

$$Q = 10.51\text{kW}$$

For number 364 and 365. Consider the problem below:

Water at the rate 68 kg/min is heated from 35°C to 75°C by an oil having a specific heat of 1.9 kJ/kg-°C. The fluids are used in counter-flow double-pipe heat exchanger, and the refrigerant enters the heat exchanger at 110°C and leaves at 75°C. The overall heat-transfer coefficient is 320 W/ m²-°C

364. The total heat transfer, in kW, from the energy absorbed by the water is

Solution:

$$Q = mC_p\Delta T = \left(\frac{68\text{kg}}{\text{min}}\right)\left(\frac{1\text{min}}{60\text{s}}\right)\left(\frac{4.185\text{kJ}}{\text{kg} - ^\circ\text{C}}\right)(75^\circ\text{C} - 35^\circ\text{C})$$

$$Q = 189.81\text{kW}$$

365. The log mean temperature difference, in Celsius, is calculated to be

Solution:

$$\text{LMTD} = \frac{\Delta T_A - \Delta T_B}{\ln\left(\frac{\Delta T_A}{\Delta T_B}\right)} = \frac{(75 - 35) - (110 - 75)}{\ln\left(\frac{75 - 35}{110 - 75}\right)}$$

$$\text{LMTD} = 37.44^\circ\text{C}$$

366. About 3 hp per ton of refrigeration is required to maintain a temperature of -40°C in a refrigerator. If the refrigerator works on the reverse Carnot cycle, what is the heat rejected to the sink per ton of refrigeration.

Solution:

$$3\text{hp}(0.746\text{kW}) = \frac{2.238\text{kJ}}{\text{s}}\left(\frac{1\text{kCal}}{4.185\text{kJ}}\right)\left(\frac{60\text{s}}{\text{min}}\right) = \frac{32.07\text{kCal}}{\text{min}}$$

$$1\text{TOR}\left(\frac{211\text{kJ}}{\text{min}}\right)\left(\frac{1\text{kCal}}{4.185\text{kJ}}\right) = \frac{50.39\text{kCal}}{\text{min}}$$

$$Q_A = Q_R + W = \frac{32.086\text{kCal}}{\text{min}} + \frac{50.39\text{kCal}}{\text{min}}$$

$$Q_A = \frac{82\text{kCal}}{\text{min}}$$

367. What Kelvin temperature is equal to 167 °F?

Solution:

Since no formula for direct Fahrenheit to Kelvin, convert the Fahrenheit to Celsius first.

$$T_k = \frac{5}{9}(T_F - 32) + 273 = 348\text{K}$$

368. What is the density (kg/m³) of dry air at 101kPa and 25°C? Gas constant = 287 J/kg-K for air and 462 J/kg-K for water.

Solution:

Pv = RT (The equation of state)(v = specific volume)

$$\rho_{\text{air}} = \frac{1}{v} = \frac{P}{RT} = \frac{\frac{101,325\text{N}}{\text{m}^2}}{\left(\frac{287\text{N} - \text{m}}{\text{kg} - \text{K}}\right)(25^\circ\text{C} + 273)}$$

$$\rho_{\text{air}} = \frac{1.18\text{kg}}{\text{m}^3}$$

369. What is the mass, in kg, of air contained in a room of dimensions 20ft x 20ft x 10ft if the density of air is 1.20 kg/m³?

Solution:

$$m = \rho_{\text{air}} V = \left(\frac{1.20 \text{ kg}}{\text{m}^3} \right) (4,000 \text{ ft}^3) \left(\frac{1 \text{ m}}{3.28 \text{ ft}} \right)^3 = 136 \text{ kg}$$

370. At the critical point, the quality is

Answer: indeterminate

371. Water is flowing through a pipe at the rate of 5.6 ft³/s. What is the mass flow rate in pounds per seconds?

Solution:

$$\dot{m} = \rho_w V = \left(\frac{62.4 \text{ lb}}{\text{ft}^3} \right) \left(\frac{5.6 \text{ ft}^3}{\text{s}} \right) = \frac{349.44 \text{ lb}}{\text{s}}$$

372. A rectangular tank measuring 2ft by 3ft at the base is filled to a depth of 6 in. with water having a density of 62.4lb/ft³. Determine (a) the total gravitational force, and (b) the pressure exerted by the water on the base of the tank.

Solution:

a. the total gravitational force

Solution:

$$F = ma = \rho g V = \left(\frac{62.4 \text{ lb}}{\text{ft}^3} \right) \left(\frac{32.2 \text{ ft}}{\text{s}^2} \right) (2 \text{ ft} * 3 \text{ ft} * 0.5 \text{ ft}) \left(\frac{\text{lb}_f - \text{s}^2}{32.2 \text{ ft} - \text{lb}} \right)$$

$$F = 187.2 \text{ lb}$$

b. The pressure exerted by the water on the base of the tank.

Solution:

$$P = \frac{F}{A} = \frac{187.2 \text{ lb}}{2 \text{ ft} * 3 \text{ ft}} = \frac{31.2 \text{ lb}}{\text{ft}^2} \left(\frac{1 \text{ ft}}{12 \text{ in}} \right)^2 = 0.217 \text{ psi}$$

373. A pressure gage on a refrigerant condenser reads 120 psi. What is the absolute pressure of the refrigerant in the condenser?

Solution:

$$P_{\text{abs}} = P_{\text{atm}} + P_{\text{gauge}} = 14.7 \text{ psi} + 120 \text{ psi}$$

$$P_{\text{abs}} = 134.7 \text{ psi}$$

374. Determine the total external energy per pound of water flowing at a rate of 50 fps on a raceway located 200ft above a reference datum.

Solution:

$$E_T = \frac{1}{2} m V^2 + mgh$$

$$E_T = \frac{1}{2} (1 \text{ lb}) \left(\frac{50 \text{ ft}}{\text{s}} \right)^2 + (1 \text{ lb}) \left(\frac{32.2 \text{ ft}}{\text{s}^2} \right) (200 \text{ ft})$$

$$E_T = \frac{7,690 \text{ ft} - \text{lb}_f}{32.2}$$

$$E_T = 238.33 \text{ ft} - \text{lb}$$

375. Twenty kilograms of water at 65°C are supplied with 5,000 kJ of heat energy. What mass of the water will be evaporated?

Solution:

$$Q = m C_p \Delta T + m h_f$$

$$5,000 \text{ kJ} = (20 \text{ kg}) \left(\frac{4.187 \text{ kJ}}{\text{kg} - ^\circ \text{K}} \right) (100^\circ \text{C} - 65^\circ \text{C}) + m \left(\frac{2,257 \text{ kJ}}{\text{kg}} \right)$$

$$m = 0.92 \text{ kg}$$

376. The interior wall of a furnace is maintained at a temperature of 900°C. The wall is 60 cm thick, 1 m wide, 1.5 m broad of material whose thermal conductivity is 0.26 kcal/hr.-m.-°C. The

temperature of the outside surface of the wall is 200°C. Determine the heat loss through the wall. Also determine the thermal resistance to heat flow.

Solution:

$$Q = \frac{kA\Delta T}{x} = \frac{\left(\frac{0.26\text{kcal}}{\text{hr} - \text{m} - ^\circ\text{C}}\right)(1\text{m} * 1.5\text{m})(900^\circ\text{C} - 200^\circ\text{C})}{0.6\text{m}}$$

$$Q = \frac{455\text{kCal}}{\text{hr}}$$

$$R = \frac{x}{kA} = \frac{0.6\text{m}}{\left(\frac{0.26\text{kcal}}{\text{hr} - \text{m} - ^\circ\text{C}}\right)(1.5\text{m}^2)} = \frac{1.54\text{hr} - ^\circ\text{C}}{\text{kCal}}$$

377. A cold room has one of the walls 5 m x 2.5 m made of bricks 12 cm thick insulated externally by cork slabbing 8 cm thick. Cork is protected externally by 2.5 cm wood. Estimate the heat infiltration through the wall in 24 hours, if the interior of the cold room is maintained at a temperature of 0°C and the outside temperature is 20°C. Thermal conductivities for brick, cork and wood are 0.8, 0.038 and 0.15 kcal/m-hr.-°C, respectively.

Solution:

$$Q = \frac{\Delta T}{\frac{x_1}{k_1} + \frac{x_2}{k_2} + \frac{x_3}{k_3}}$$

$$Q = \frac{(20^\circ\text{C} - 0^\circ\text{C})(5\text{m} * 2.5\text{m})}{\frac{0.12\text{m}}{0.8\text{kCal}} + \frac{0.08\text{m}}{0.038\text{kCal}} + \frac{0.025\text{m}}{0.15\text{kCal}}}$$

$$Q = \frac{103.22\text{kCal}}{\text{hr}} (24\text{hrs})$$

$$Q = 2,477.36\text{kCal}$$

378. The temperature of the outside surface of a double extra-strong bare steel pipe (ID = 6.87 in, OD = 8.625 in) is 600°F while the temperature of the inside surface is 605°F. (a) What is the heat loss per feet of pipe? (b) What will be the saving in heat if the pipe is insulated with 3 in. of 85% magnesia and the temperature of the outside surface of the insulation is reduced to 200°F.? $k_{\text{steel}} = 26.56 \text{ Btu-ft./hr.-ft}^2\text{-}^\circ\text{F}$, and $k_{\text{magnesia}} = 0.046 \text{ Btu-ft./hr.-ft}^2\text{-}^\circ\text{F}$.

Solution:

$$8.625\text{in} = 0.71875\text{ft}$$

$$6.87\text{in} = 0.5725\text{ft}$$

$$\text{OD}_3 = 14.625\text{in} = 1.21875\text{ft}$$

$$Q = \frac{605^\circ\text{F} - 600^\circ\text{F}}{\ln\left(\frac{0.71875\text{ft}}{0.5725\text{ft}}\right)} = \frac{3,667.71\text{BTU}}{\text{hr} - \text{ft}}$$

$$Q = \frac{2\pi\left(\frac{26.56\text{BTU} - \text{ft}}{\text{hr} - \text{ft}^2 - ^\circ\text{F}}\right)}{605^\circ\text{F} - 200^\circ\text{F}} + \frac{\ln\left(\frac{1.21875\text{ft}}{0.71875\text{ft}}\right)}{2\pi\left(\frac{0.046\text{BTU} - \text{ft}}{\text{hr} - \text{ft}^2 - ^\circ\text{F}}\right)}$$

$$Q = \frac{221.80\text{BTU}}{\text{hr} - \text{ft}}$$

$$\text{Savings} = \frac{3,667.71\text{BTU}}{\text{hr} - \text{ft}} - \frac{221.80\text{BTU}}{\text{hr} - \text{ft}} = \frac{3,446\text{BTU}}{\text{hr} - \text{ft}}$$

379. A flat composite plate is made of two layers of aluminum and steel of 5 cm and 2 cm thickness, respectively. The thermal conductivities of aluminum and steel are 175 kcal/m-hr.-°C and 39 kcal/m-hr.-°C, respectively. The hot surface of aluminum side is in contact with hot liquid at 200°C, the heat transfer of liquid film being 12 kcal/m²-hr.-°C. The cold surface of steel side is in contact with liquid at 25°C, the heat transfer coefficient of liquid film being 25 kcal/m²-hr.-°C. Determine the (a) overall coefficient of heat transfer, and (b) the heat transfer rate from hot liquid to cold liquid through surface area of 10m².

Solution:

$$U = \frac{1}{\frac{1}{\frac{12\text{kCal}}{\text{m}^2 - \text{hr} - ^\circ\text{C}}} + \frac{0.05\text{m}}{\frac{175\text{kCal}}{\text{m} - \text{hr} - ^\circ\text{C}}} + \frac{0.02\text{m}}{\frac{39\text{kCal}}{\text{m} - \text{hr} - ^\circ\text{C}}} + \frac{1}{\frac{25\text{kCal}}{\text{m}^2 - \text{hr} - ^\circ\text{C}}}}$$

$$U = \frac{8.056\text{kCal}}{\text{m}^2 - \text{hr} - ^\circ\text{C}}$$

$$Q = AU\Delta T = (10\text{m}^2) \left(\frac{8.056\text{kCal}}{\text{m}^2 - \text{hr} - ^\circ\text{C}} \right) (200^\circ\text{C} - 25^\circ\text{C})$$

$$Q = \frac{14,097.91\text{kCal}}{\text{hr}}$$

380. Find the sensible heat gain through a wall adjacent to another room (78°F) made from hard maple with $U = 0.22$ Btu/hr·ft²-°F. The wall measures 20'x 15. Inside conditions are 75°F dry bulb and 50% relative humidity.

Solution:

$$Q = AU\Delta T = (20\text{ft} * 15\text{ft}) \left(\frac{0.22\text{BTU}}{\text{hr} - \text{ft}^2 - ^\circ\text{F}} \right) (78^\circ\text{F} - 75^\circ\text{F})$$

$$Q = \frac{198\text{BTU}}{\text{hr}}$$

381. Find the total heat load from a window made from 3/16-in tinted glass with venetian blinds. $A = 60$ ft², $U = 0.92$ Btu/hr·ft² -°F, CLTD = 22°F, solar heat gain factor = 207 Btu/hr-ft², shading coefficient = 0.64, and cooling factor = 0.72.

Solution:

CLTD = Cooling Load Temperature Difference

$$Q = AU(\text{CLTD}) = (60\text{ft}^2) \left(\frac{0.92\text{BTU}}{\text{hr} - \text{ft}^2 - ^\circ\text{F}} \right) (22^\circ\text{F})$$

$$Q = \frac{1214.4\text{BTU}}{\text{hr}}$$

For combined convection and radiation:

CLF = Cooling Load Factor

SC = Shading coefficient (tint, drapes, rolls, blinds)

SHGF – solar heat gain factor (local altitude, date, solar time, window orientation)

$$Q = A(\text{max SHGF})(\text{SC})(\text{CLF})$$

$$Q = (60\text{ft}^2) \left(\frac{207\text{BTU}}{\text{hr} - \text{ft}^2} \right) (0.64)(0.72)$$

$$Q = \frac{5,723.136\text{BTU}}{\text{hr}}$$

Total Heat Load

$$Q = \frac{1214.4\text{BTU}}{\text{hr}} + \frac{5,723.136\text{BTU}}{\text{hr}}$$

382. The dimensions of a storage cooler in a hotel kitchen are 15ft x 20ft X 10ft, and usage is heavy (equivalent to 9.5 air changes per 24hrs with 50% additional heat). The inside is maintained at 32°F and 85% relative humidity ($h = 11.3 \text{ Btu/lb.}$), while the outside conditions are 75°F and 50% RH ($h = 28.3 \text{ Btu/lb.}$). Air density is 0.0751lb/ft³. Calculate the air change load.

Solution:

$$Q = (\text{Room Volume})(\text{air change per 24 hours})(\text{air density})(h_o - h_i)$$

$$Q = (15\text{ft} * 20\text{ft} * 10\text{ft}) \left(\frac{9.5}{24\text{hrs}} \right) \left(\frac{0.0751\text{lb}}{\text{ft}^3} \right) \left(\frac{28.3\text{BTU}}{\text{lb}} - \frac{11.3\text{BTU}}{\text{lb}} \right) (1.50)$$

$$Q = \frac{2,274.12\text{BTU}}{\text{hr}}$$

383. Seventy-five hundred pounds of fresh lean beef enter a chilling cooler at 102°F and are chilled to 45°F each day. For beef, $c = 0.75 \text{ Btu/lb.-°F}$ with chilling factor of 0.67. Compute the product load in Btu per 24 hr.

Solution:

$$Q = \frac{m\Delta T}{t(\text{CF})}$$

$$Q = \frac{(7500\text{lb}) \left(\frac{0.75\text{BTU}}{\text{lb} - \text{°F}} \right) (102\text{°F} - 45\text{°F})}{(24\text{hrs})(0.67)}$$

$$Q = \frac{19,939.37\text{BTU}}{\text{hr}}$$

384. Three thousand lug boxes of apples ($c = 0.87 \text{ Btu/lb.-°F}$) are stored at 35°F in a storage cooler. The apples enter the cooler at a temperature of 75°F and at a rate of 200 lug boxes per day for the 15-day harvest period. The average weight of apples per lug box is 59 pounds. The lug boxes have an average weight of 4.5 pounds and a specific heat of 0.6 Btu/lb.-°F. Calculate the product load if the respiration heat of apple is 0.0225 Btu/hr.-lb.

Solution

$$\text{Total Heat} = Q_{\text{apple}} + Q_{\text{respiration}} + Q_{\text{box}}$$

$$Q_{\text{apple}} = \frac{m\Delta T}{t} = \frac{(11,800) \left(\frac{0.87\text{BTU}}{\text{lb} - \text{°F}} \right) (75\text{°F} - 35)}{(24\text{hrs})} \text{ (per day only)}$$

$$Q_{\text{apple}} = \frac{17,110\text{BTU}}{\text{hr}}$$

$$Q_{\text{respiration}} = mh = (11,800\text{lb}) \left(\frac{0.0225\text{BTU}}{\text{lb} - \text{hr}} \right)$$

$$Q_{\text{respiration}} = \frac{265.5\text{BTU}}{\text{hr}}$$

$$Q_{\text{box}} = \frac{\left(\frac{4.5\text{lb}}{\text{box}} \right) (200\text{box}) \left(\frac{0.6\text{BTU}}{\text{lb} - \text{°F}} \right) (75\text{°F} - 35\text{°F})}{24\text{hrs}}$$

$$Q_{\text{box}} = \frac{900\text{BTU}}{\text{hr}}$$

$$\text{Total heat} = \frac{17,110\text{BTU}}{\text{hr}} + \frac{265.5\text{BTU}}{\text{hr}} + \frac{900\text{BTU}}{\text{hr}}$$

$$\text{Total heat} = \frac{18,275.5\text{BTU}}{\text{hr}}$$

385. A 1m³ rigid tank contains 10 kg of water (in any phase or phases) at 150°C. The pressure in the tank is

Solution:

From the table:

$$P = 475.80 \text{ kPa}$$

386. Water is boiled in a pan on a stove at sea level. During 10 min of boiling, it is observed that 200 g of water has evaporated. Then the rate of heat transfer to the water is

Solution:

$$\text{Note: @ } 101.325 \text{ kPa, } h_{fg} = \frac{2,258 \text{ kJ}}{\text{kg}}$$

$$Q = \dot{m} h_{fg} = \left(\frac{0.2 \text{ kg}}{10 \text{ min}} \right) \left(\frac{2,258 \text{ kJ}}{\text{kg}} \right)$$

$$Q = \frac{45.16 \text{ kJ}}{\text{min}}$$

387. A 2m³ rigid tank contains nitrogen gas at 500kPa and 300K. Now heat is transferred to the nitrogen in the tank and the pressure of nitrogen rises to 800kPa. The work done during this process is.

Solution:

$$W = 0 \text{ (constant volume)}$$

388. A refrigerator operates on the standard vapor compression refrigeration cycle with R-134a as the working fluid between the pressure limits of 140kPa and 800kPa. If the rate of heat removal from the refrigerated space is 25 kJ/s, the mass flow rate of the refrigerant is

Solution:

$$h_f @ 140 \text{ kPa} = \frac{387.5 \text{ kJ}}{\text{kg}}$$

$$h_f @ 800 \text{ kPa} = \frac{243.6 \text{ kJ}}{\text{kg}}$$

$$Q = \dot{m} \Delta h$$

$$\dot{m} = \frac{\frac{25 \text{ kJ}}{\text{s}}}{\frac{387.5 \text{ kJ}}{\text{kg}} - \frac{243.6 \text{ kJ}}{\text{kg}}}$$

$$\dot{m} = \frac{0.174 \text{ kg}}{\text{s}}$$

389. A typical new household refrigerator consumes about 700 kWh of electricity per year and has a coefficient of performance of 1.4. The amount of heat removed by this refrigerator from the refrigerated space per year is

Solution:

$$\text{COP} = \frac{Q_A}{W}$$

$$Q_L = (1.4) \left(\frac{700 \text{ kWh}}{\text{year}} \right) \left(\frac{3,600 \text{ s}}{\text{hr}} \right)$$

$$Q_L = \frac{3.528 \text{ MJ}}{\text{year}}$$

390. What is the ideal work for a refrigerator if the cold room temperature is -15°C, ambient temperature is 30°C, and heat rejection is 10kW?

Solution:

$$\text{COP} = \frac{(-15^\circ\text{C} + 273)}{(30^\circ\text{C} + 273) - (-15^\circ\text{C} + 273)} = 5.733$$

$$\text{COP} = \frac{Q_A}{Q_R - Q_A} = \frac{Q_A}{10 \text{ kW} - Q_A} = 5.733$$

$$Q_A = 8.515 \text{ kW}$$

$$W = Q_R - Q_A$$

$$W = 1.48 \text{ kW}$$

391. An apple with a an average mass of 0.15 kg and average specific heat of 3.65 kJ/kg-°C is cooled from 20°C to 5°C. The amount of heat transferred from the apple is

Solution:

$$Q = mC_p \Delta T = (0.15 \text{ kg}) \left(\frac{3.65 \text{ kJ}}{\text{kg} - ^\circ\text{C}} \right) (20^\circ\text{C} - 5^\circ\text{C})$$

$$Q = 8.21 \text{ kJ}$$

392. A bin is 12ft in diameter and has an eave height of 10 ft. If the bin is filled with grains leveled at the eave height, what is the volumetric capacity of the bin in bushel?

Solution:

$$V = \frac{\pi}{4} D^2 h = \frac{\pi}{4} (12 \text{ ft})^2 (10 \text{ ft}) = 1,1130.97 \text{ ft}^3$$

$$V = 1,1130.97 \text{ ft}^3 \left(\frac{1 \text{ bushel}}{1.25 \text{ ft}^3} \right)$$

$$V = 904.80 \text{ bushel}$$

393. Throughput capacity is the amount of grain which flows through a continuous-flow equipment in one hour. If the holding capacity is 6 ton, and it uses a 30-minute process, then throughput capacity is

Solution:

$$\text{Throughput Capacity} = \frac{6 \text{ ton}}{30 \text{ min} \left(\frac{hr}{60 \text{ min}} \right)} = 12 \text{ tons}$$

394. A rice miller buys 100 kg of dried and clean palay at ₱13/kg. Assuming that his cost of operation is ₱4/kg milled rice, how much should he sell (₱/kg) to break even? Neglect by-products and assume milling recovery of 64%.

Solution:

$$100 \text{ kg} (0.64 \text{ kg}) = 64 \text{ kg milled rice}$$

$$\left(\frac{₱4}{\text{kg}} \right) (64 \text{ kg}) = ₱256$$

$$₱1,300 \text{ kg} + ₱256 = ₱1556$$

$$x = \frac{₱1,556}{64 \text{ kg}} = 24.31 \text{ kg}$$

395. A Carnot engine receives 130 Btu of heat from a hot reservoir at 700°F and rejects 49 Btu of heat. Calculate the temperature of the cold reservoir.

Solution:

$$\text{COP} = \frac{Q_L}{Q_H - Q_L} = \frac{49 \text{ BTU}}{130 \text{ BTU} - 49 \text{ BTU}} = 0.605 = \frac{T_L}{T_H - T_L} = \frac{T_L}{(700^\circ\text{F} + 460) - T_L}$$

$$T_L = -22.74^\circ\text{F}$$

396. A Carnot refrigerator operates between 10°F and 80°F. It's coefficient of performance is

Solution:

$$\text{COP} = \frac{T_L}{T_H - T_L} = \frac{10^\circ\text{F} + 460}{(80^\circ\text{F} + 460) - (10^\circ\text{F} + 460)} = 6.71$$

397. The maximum thermal efficiency possible for a power cycle operating between 1200°F and 225°F is

Solution:

$$\eta_{\text{th}} = \frac{T_{\text{H}} - T_{\text{L}}}{T_{\text{L}}} = \frac{(1,200^{\circ}\text{F} + 460) - (225^{\circ}\text{F} + 460)}{(1,200^{\circ}\text{F} + 460)}$$
$$\eta_{\text{th}} = 58.73\%$$